

ionocast: A Real-Time HF/VHF Propagation Prediction Model for Amateur Radio Operators

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Abstract

ionocast provides real-time HF/VHF band-condition verdicts for amateur radio operators (next-24-hour forward projections are specified in Section 5 but not yet surfaced in the runtime). The model runs entirely in the browser with no server-side state. The operator's QTH stays on-device for everything except two panels that need a nearest-station lookup, in which case it is sent to ionocast's own proxy for resolution and discarded after the response (see Section 2.1 for the full disclosure). The model combines an ITU-R P.533-grounded SNR link budget with real-time data from NOAA SWPC (solar indices, D-RAP absorption, OVATION auroral power, GOES proton flux), the kc2g ionosonde network (observed MUF and foF2), GIRO digisondes (foF2, foEs, hmF2), and WSPR spot observations from `wspr.live`. A local validation harness scores predictions against retrospective WSPR activity to quantify systematic bias and refine per-band σ .

Key contributions:

- *Symmetric MUF consensus* (geometric mean) between gridded kc2g observations and the foF2 climatology, replacing a previous asymmetric rule that clipped real upward enhancements.
- *Power-summed noise model* (ITU-R P.372) that correctly handles man-made noise dominance on the low bands.
- *Per-hop minimum MUF* and *per-hop ionospheric loss accounting* (diurnal D-region, polar cap absorption, flare SID, auroral) evaluated at each great-circle reflection point rather than at a single midpoint.
- *Sauer-Wilkinson polar cap absorption* coupled to GOES ≥ 10 MeV proton flux, and *flare-driven SID* absorption keyed to GOES X-ray class.
- *Storm-lag kernel* that captures the F-region's delayed response to K_p , with D_{st} and IMF B_z used for earlier detection than the 3-hour K_p alone.
- *Equatorial and winter anomaly* corrections to the foF2 climatology and an *asymmetric gray-line* bonus.
- *Elevation-dependent antenna gain* from hop-geometry-derived takeoff angles, plus *NVIS detection* for short paths using local foF2.
- Verdicts mapped to ITU-R P.842 reliability buckets in σ -units rather than fixed dB widths, so the tier boundary scales with each band's prediction uncertainty.

Both short-path and long-path great circles are evaluated per destination. All computation is deterministic, repeatable, and inspectable; no machine-learning black boxes.

Source and reproducibility. The full ionocast implementation — browser code, Cloudflare Worker handlers, and the offline calibration harness — is open at <https://github.com/e46401ff/ionocast>. The 35-path calibration basket, 26-station GIRO list, and per-band σ table are reproduced verbatim from that repository in Appendices C, D, and the main text (Table 11) respectively, so the validation metrics below can be re-derived end-to-end from the paper plus the repository alone.

Validation note. The shipped harness scores against 30 d of WSPR activity aggregated two ways: a global “open somewhere” truth (accBin 94.25 %, Brier 0.0386) which has a known structural ceiling — the OR-over-paths target overrewards permissive predictors and saturates whenever *any* path on the band has activity, quantified in Section 10 #9; and a per-path mode (accBin 30.97 %, Brier 0.6353) which uses TX/RX-bbox-restricted aggregates with a ≥ 1 spot/h floor and is dominated by operator-activity sparsity on minority-traffic bands. The 60+ percentage-point gap between the two metrics on the same harness run is itself the structural-ceiling evidence: a model that scores in the 90s on the global metric and the 30s on the per-path metric is not 90-good or 30-good, it is sitting on a metric whose ceiling reflects “somewhere is open” rather than “the requested path is open.” The two numbers are not directly comparable; per-pair WSPR ground-truth wiring with calibrated availability priors is in progress. Quote either number as a regression-detection signal, not as a production-grade reliability claim.

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1 Introduction

Amateur radio operators planning an operating session need to answer a deceptively simple question: *which bands will be open, and for how long?* The answer depends on the instantaneous state of the ionosphere, the Sun, the geomagnetic field, and the operator’s own station (antenna, power, mode, and local noise environment).

Existing tools serve fragments of this problem:

- **VOACAP** [1] provides rigorous P.533-based predictions but operates as a batch tool requiring manual circuit specification; it is not real-time.
- **N0NBH Solar Conditions** provides a widely trusted heuristic band-condition table keyed to SFI and K_p , but without path geometry or operator context.
- **kc2g MUF map** renders observed MUF contours in near real time, but offers no forward projection and no SNR context.
- **HamQSL / DX Toolbox** use threshold-based rules on solar indices, lacking physics grounding.

ionocast occupies the intersection: real-time, physics-grounded, operator-configurable, and privacy-preserving. The operator’s QTH (stored in `localStorage` only) parameterises five reference propagation paths; the SNR budget runs per band, per path, at the current time. Next-24-hour forward projections at +3, +6, +12, and +24 h — via solar-zenith analogy, storm depression, seasonal correction, and sporadic-E persistence (Section 5) — are *specified but not currently surfaced in the runtime*; the Upcoming Disruptions panel today shows SWPC’s own forecasts (3-day R/S/G probabilities, 3-h K_p , 27-day SFI/ A_p) rather than ionocast-derived per-band projections.

This paper documents the complete model as implemented. Every numeric constant in the SNR budget and MUF estimation is given either inline or in an appendix; the calibration setup that produced the harness-fitted constants (the three coordinate-descent starting seeds, the parameter sweep grid, and the WSPR baseline files) is described in Appendix E, so the amateur radio community can audit, reproduce, and improve the model from the paper plus the repository alone.

2 Data Sources

Table 1 lists all upstream data feeds. Sources marked “Direct CORS” are fetched by the browser with no server proxy; sources marked “Proxied” pass through a Cloudflare Worker endpoint that resolves the operator’s Maidenhead grid to a station identifier server-side.

Table 1: Upstream data sources and refresh cadences.

Source	Data	Refresh	Protocol	
NOAA SWPC	Kp, Ap, F10.7, X-ray class, 3-day forecast, 27-day outlook, D-RAP HAF, OVATION aurora HP, solar-wind Bz / speed / density, solar regions	1–10 min	Direct JSON/text	CORS
prop.kc2g.com	Per-station MUF(3000F2), foF2, TEC	~15 min	Direct	CORS JSON
GIRO (lgdc.uml.edu)	Nearest digisonde foF2, foEs, hmF2	~10 min	Proxied via /api/giro	
GFZ Potsdam*	Hp30 nowcast	30 min	Proxied via /api/hp30	
WDC Kyoto	Dst / Sym-H	1 h	Proxied via /api/kyoto	
SILSO*	Daily sunspot number	Daily	Proxied via /api/silso	
UWyo	Radiosonde soundings (lowest-1 km dN/dh over a 24-station basket)	12 h	Proxied via /api/tropo	
wspr.live	WSPR spot counts and SNR by band	10 min	Direct CORS (Click-House)	
NASA DONKI [22]	CME and HSS analyses	~hours	Direct CORS JSON	

* Hp30 (GFZ Potsdam) and Daily Sunspot Number (SILSO) are fetched and surfaced in the Geomagnetic / Solar UI panels for operator situational awareness, but neither feeds the SNR budget or the verdict pipeline at present. K_p^{eff} from SWPC serves the half-hour-cadence role that Hp30 could otherwise fill; $F_{10.7A}$ 81-day mean serves the solar-activity role of daily SSN. Both data sources are wired forward in case future calibration work uses them as cross-checks.

2.1 Privacy Model

The operator’s QTH is stored exclusively in `localStorage` and is never logged or retained by the ionocast Cloudflare Worker. Privacy posture in detail:

- **Stored on device only.** The `ionocast_qth` key in `localStorage` holds the operator’s grid; nothing reads or writes it server-side, and there is no account / login model.
- **Sent for resolution, not retained.** Endpoints like `/api/giro` take a grid string as a query parameter so the worker can resolve it to the nearest digisonde station; the worker does not log request bodies or query strings, and the upstream response contains station data, not the original grid. The worker source is open and inspectable at <https://github.com/e46401ff/ionocast> (the `functions/` directory).
- **WSPR live queries are global hourly aggregates** with no grid field at all; the operator’s location is never transmitted to `wspr.live`. (The offline calibration harness in `scripts/harness.mjs` does use a coarser grid for retrospective sample bucketing, but is not part of the runtime page.)
- **No third-party analytics or tracking** are loaded; the Cloudflare access logs the worker host retains for 24–72 h are the only request-side trace.

The honest summary: the QTH is sent to the proxy worker *for resolution at request time*, but is not retained, logged in application form, or shared with third parties. Operators who want zero server-side QTH exposure can run the static UI against the upstream APIs directly (the worker is a performance / CORS shim, not a privacy boundary).

First-visit default. On first load, with no saved QTH, ionocast derives a regionally plausible grid from the browser’s runtime timezone. `Intl.DateTimeFormat().resolvedOptions().timeZone`

is looked up in a compiled table of ~ 60 major ham-active IANA zones (e.g. `Europe/Berlin` $\rightarrow$ JO62, `Asia/Tokyo` $\rightarrow$ PM95, `Australia/Sydney` $\rightarrow$ QF56; the full table lives in `src/physics/qth.js` and is open for inspection at the repo URL given in the abstract). Zones outside the table synthesise a grid from the UTC offset: longitude $\approx$ offset $\times 15^\circ$ at a fixed mid-latitude (45° N for the northern hemisphere, -35° for identified southern-hemisphere zones). The timezone fingerprint is already exposed on every HTTPS request, so using it for a first-visit default adds no new privacy surface. When even that math fails, the last-resort fallback is **JN05** (approximately 45.5° N, 1° E — the grid covers $0-2^\circ$ E, $45-46^\circ$ N) — a French / west-Swiss midlatitude reference point near where the CCIR R-12 / P.1239 foF2 coefficients underlying this paper are tabulated. The fallback grid is intentionally chosen as midlatitude European to give a sensible default for the worldwide majority of first-visit operators. No request is ever made to a geolocation service.

3 SNR Budget Model

The core of ionocast is a complete link budget that computes the SNR margin M (in dB) for each amateur band on each reference path:

$$M = P_{\text{tx}} + G_{\text{ant}} - L_{\text{fs}} - L_{\text{flare,path}} - L_{\text{Dreg}} - L_{\text{PCA}} - L_{\text{aur}} - L_{\text{MUF}} - n_{\text{hops}} L_{\text{iono}} - L_{\text{low}} - L_{\text{hop}} - L_{\text{Es}} - N - S_{\text{req}} \quad (1)$$

where every term is defined in the following subsections. A positive M means the signal exceeds the minimum required SNR for the chosen mode; the magnitude indicates robustness. The term $L_{\text{flare,path}}$ is the joint flare-absorption product of L_{DRAP} (Section 3.3) and the per-hop L_{flare} sum (Section 3.6); inlined here so the reader does not have to forward-jump:

$$L_{\text{flare,path}} = \max\left(L_{\text{DRAP}}, \sum_{k=1}^N L_{\text{flare}}(f, \cos \chi_k)\right) \quad (2)$$

Both symbols are defined separately because they have independent physical sources, but they collapse to a single term in the budget to avoid double-charging the path for the same flare. See Section 3.3 for the $\max(\cdot)$ rationale and the geographic asymmetry between the two contributing terms.

Global absorption-sum saturation. Each of the four ionospheric absorption terms ($L_{\text{flare,path}}$, L_{Dreg} , L_{PCA} , L_{aur}) is individually capped at 50 dB per path. Their sum is also *hard-capped* at 100 dB with proportional rescaling (no smoothing knee): if a sunlit polar hop during an X10 flare with an S3 SEP and a $K_p = 9$ main-phase storm pushed the additive sum to 150 dB (well past physical D-region saturation $\sim 80-100$ dB; beyond that the band is dead regardless and adding more absorption is unphysical), the cap applies and each contribution is scaled proportionally down so per-mechanism attribution is preserved ($L'_i = L_i \cdot 100 / \sum L_i$).

Why this gate is hard while the others are smooth. The auroral K_p , OVATION HP, CGM-edge, and IMF B_z gates elsewhere in the budget were all explicitly retired in favour of smooth ramps because they fire on *routine* active-Kp conditions ($K_p \in [4, 5]$ levels happen for hours at a time), where a 5–20 dB step in any of the loss terms would visibly shift verdicts at the threshold crossing. The 100 dB absorption-sum cap is qualitatively different: it never fires under any normal compound-condition combination — the pathological case requires a simultaneous X10 flare, S3 SEP, and $K_p = 9$ main-phase storm intersecting on a sunlit polar hop, a scenario realised at most a few hours per solar cycle if at all. The discontinuity at $\sum L_i = 100$ is therefore unobservable in any operator-relevant scenario, and a smoothing knee would add implementation complexity for no behavioural gain. The hard form is intentional, and the inconsistency with the smoothed gates elsewhere is acknowledged here rather than papered over.

Why proportional rescaling rather than cap-and-skip. The four contributing terms ($L_{\text{flare,path}}$, L_{Dreg} , L_{PCA} , L_{aur}) are driven by physically distinct ionisation sources — soft X-ray photo-ionisation, quiet-day diurnal photo-ionisation, proton-induced ionisation, and particle-precipitation ionisation respectively — but they all enhance *the same* D-region electron column, which has a finite total-ionisation depth (~ 80 – 100 dB of absorption at HF when fully populated; beyond that further electrons displace existing ones rather than adding new absorbers). Past the column’s physical ceiling, the correct behaviour is not “add until budget then drop further contributions”: cap-and-skip would arbitrarily zero out whichever mechanism happens to be evaluated last in the implementation order and leak attribution dependence into the UI’s per-mechanism display. Proportional rescaling, $L'_i = L_i \cdot 100 / \sum L_i$, preserves each mechanism’s relative contribution at saturation, so that a path showing “45 dB PCA + 25 dB flare + 15 dB aurora + 15 dB quiet-day” before the cap reads “45 dB + 25 dB + 15 dB + 15 dB” rescaled to “45 dB + 25 dB + 15 dB + 15 dB” (sum 100 dB) after, retaining the operator-meaningful ratios. The rescale is conservative on terms not individually near saturation: a 12 dB auroral contribution gets shaved to ~ 11.4 dB when the sum is at 105 dB, but this is well under any verdict-relevant threshold (the band is dead at $\sum \geq 80$ dB regardless), so the conservatism does not shift verdicts in any operator-meaningful regime.

3.1 Free-Space Loss (Friis / ITU-R P.525)

The free-space path loss follows the standard ITU-R P.525 [4] formulation:

$$L_{\text{fs}} = 32.44 + 20 \log_{10}(d_{\text{km}}) + 20 \log_{10}(f_{\text{MHz}}) \quad (3)$$

The distance d_{km} is the great-circle path length, clamped to a minimum of 50 km to avoid numerical singularity on extremely short paths.

3.2 MUF Approach and Over-MUF Loss (ITU-R P.533 §3.2.2)

Let $r = f/\text{MUF}$. The proximity loss is:

$$L_{\text{MUF}} = \begin{cases} 0 & r \leq 0.70 \\ 10 \left(\frac{r - 0.70}{0.30} \right)^2 & 0.70 < r \leq 1.00 \\ 10 + 36\sqrt{r - 1} & r > 1.00 \end{cases} \quad (4)$$

Below 70% of MUF the ionosphere provides comfortable refraction with negligible loss. Between 70–100% the loss rises quadratically. Above MUF (over-MUF operation, e.g. via Es or scatter), the loss escalates rapidly. The piecewise formula is C^0 continuous at both joints ($r = 0.70$ and $r = 1.00$, where $L_{\text{MUF}} = 0$ and 10 dB respectively) but the derivative at $r = 1.00$ is unbounded ($\sqrt{r - 1}$ has infinite slope at the joint), producing a C^1 discontinuity at MUF. This is intentional: the slope explosion encodes the physical reality that crossing MUF is an abrupt regime change (refraction $\rightarrow$ over-MUF scatter), and smoothing the joint with a finite-slope blend would underweight the rise in path variance at the boundary. Operators using the budget at $r = 1$ should expect the verdict to be discontinuity-sensitive there.

3.3 D-Region Absorption: Flare-Enhanced (SWPC D-RAP)

SWPC’s D-RAP product provides a Highest Affected Frequency (HAF) in MHz. The flare-enhanced absorption is:

$$L_{\text{DRAP}} = \begin{cases} 0 & r < 0.25 \\ \frac{r - 0.25}{0.05} \cdot 3 r^{1.5} & 0.25 \leq r < 0.30, \\ 3 r^{1.5} & r \geq 0.30 \end{cases}, \quad r = \text{HAF}/f \quad (5)$$

The $0.25 \leq r < 0.30$ ramp retires a small step the bare gate produced (the formula at $r = 0.3$ evaluates to $3 \cdot 0.3^{1.5} \approx 0.49$ dB, jumping in instantly from zero); the linear ramp turns the term on smoothly and is zero by construction at $r = 0.25$. The ramp is C^0 continuous at both joints but C^1 -discontinuous: the derivative jumps from 0 (lower flat branch) to ~ 7.5 dB/ Δr at $r = 0.25$, then settles into the $r^{1.5}$ shape at $r = 0.30$. The jump is small in absolute terms (~ 0.025 dB of verdict shift across a 1%-of-MUF r change) and is accepted as the cost of avoiding the much larger 0.5-dB step the bare gate produced.

Relationship to the per-hop L_{flare} term. D-RAP HAF is itself derived by SWPC from the same GOES X-ray flux that drives the per-hop L_{flare} term in Section 3.6; the two were previously summed in the budget, double-charging the path for the same flare physics. The current implementation collapses to a single per-path flare-absorption term, $L_{\text{flare,path}} = \max(L_{\text{DRAP}}, \sum_k L_{\text{flare}})$, defined inline as Eq. 2 alongside the budget at the top of this section. The $\max(\cdot)$ form (rather than a binary “suppress L_{DRAP} whenever $L_{\text{flare}} > 0$ ”) is more conservative on the leading edge of a flare: D-RAP’s operational forecast model can already report 8 dB–10 dB at QTH while the per-hop L_{flare} proxy is still tracking the instantaneous X-ray class at M1 (~ 4 dB). L_{DRAP} remains active when no per-hop flare is firing but D-RAP still reports a non-zero HAF: typically the early rise of a small or unstable event before the X-ray classifier flags it, or the post-peak decay tail. (Quiet-day background absorption is handled by L_{Dreg} in Section 3.4, not by D-RAP.)

Geographic asymmetry of the two terms. L_{DRAP} is a single number sampled at QTH (the SWPC HAF product is spatially evaluated; ionocast queries it at the operator’s location). The per-hop sum $\sum_k L_{\text{flare}}(\cos \chi_k)$ samples each F-region reflection point along the path, so on a trans-equatorial 12000 km circuit it can pick up daylight hops the QTH itself isn’t in. A short path entirely in QTH’s local zone reads L_{DRAP} as the binding constraint; a long path crossing the sunlit hemisphere reads the per-hop sum. The $\max(\cdot)$ takes whichever local probe is more pessimistic on that path, so the geographic inhomogeneity is intentional: a distant flare hitting the path’s middle hops should not be hidden by the QTH having moved into night, and a flare detected in the operator’s HAF cell should not be discounted just because the specific reflection points sampled happen to land in twilight.

3.4 D-Region Absorption: Quiet-Day Diurnal (ITU-R P.533 §4)

$$L_{\text{Dreg}}(\chi) = A_{\text{base}}(f) \cdot (\cos \chi)^{1.3} \quad (6)$$

where χ is the solar zenith angle at a reflection point. At night ($\cos \chi < 0.05$), $L_{\text{Dreg}} = 0$. The base absorption values follow K9LA’s [10] quiet-day amateur-band estimates (Table 2). ITU-R P.533 [1] Annex 1 also tabulates D-region absorption (its A_a term) in a form suitable for the same role, but the P.533 tabulation is bundled with the F2 propagation routine and assumes a path-integrated calculation rather than a per-band quiet-day floor. K9LA’s table is the operator-band condensation of equivalent physics and aligns with cycle-24 / 25 amateur observations on 160 m through 30 m, so it is the more directly auditable choice for an HF-amateur methodology paper; values agree with P.533 Annex 1 to within ~ 2 dB at midlatitude where they overlap.

For a multi-hop path, absorption is *summed* across all reflection points rather than evaluated at the path midpoint only: for an N -hop path with reflections at great-circle fractions $(2k-1)/(2N)$,

$$L_{\text{absD,path}} = \min \left(L_{\text{cap}}, \sum_{k=1}^N L_{\text{Dreg}}(\chi_k) \right) \quad (7)$$

with $L_{\text{cap}} = 50$ dB guarding against degenerate values when every hop is near local noon on a low band. A transequatorial path whose midpoint is dark but whose northern and southern

hops are in daylight therefore sees the two daylight hops' absorption, not zero; conversely a path whose midpoint sits over the terminator but whose hops are deep in night loses no bonus from the single sunlit fraction nearest the midpoint.

Table 2: Quiet-day D-region base absorption A_{base} at each band centre. Stored as anchor points for a smooth log-frequency / log-magnitude interpolation; values at intermediate frequencies are the geometric-mean interpolation between adjacent anchors. Calibrated 2026-04-30 against ITU-R P.533 §A.2 quiet-day non-deviative absorption $L_a(f) \approx 677/(f + f_L)^{1.98}$ divided by ~ 1.6 for typical oblique-path obliquity; lower-band values held at prior empirical settings (matched K9LA / ARRL ranges). Smoothed cutoffs: prior table had step boundaries (e.g. $A_{\text{base}}(4.0 \text{ MHz}) = 18 \text{ dB}$ vs $A_{\text{base}}(4.01 \text{ MHz}) = 10 \text{ dB}$, an 8 dB cliff); the log-interpolation form is continuous everywhere.

Band	f (MHz)	A_{base} (dB)
160 m	1.838	28.0
80 m	3.570	18.0
60 m	5.366	10.0
40 m	7.040	6.0
30 m	10.140	3.0
20 m	14.097	1.5
17 m	18.106	0.8
15 m	21.096	0.5
12 m	24.924	0.3
10 m	28.126	0.2

3.5 Polar Cap Absorption (Sauer–Wilkinson)

During solar energetic particle (SEP) events, protons accelerated by a CME or flare precipitate into the polar cap and ionise the D-region, producing severe HF absorption on polar paths for hours to days [15]. ionocast activates a polar cap absorption (PCA) term when two gates are simultaneously satisfied:

1. the GOES integral $\geq 10 \text{ MeV}$ proton flux exceeds the NOAA S1 threshold, $\Phi_{p10} \geq 10 \text{ pfu}$;
2. the reflection point sits inside the polar cap, $|\phi_{\text{CGM}}| \geq 60^\circ$.

The per-hop magnitude uses a log-flux scaling (a calibrated deviation from the canonical Sauer–Wilkinson $\sqrt{\Phi}$ form [15], chosen because the saturating-recombination behaviour at the highest fluxes is gentler than $\sqrt{\Phi}$ predicts). Empirical evidence for the deviation comes from amateur-band PCA observations catalogued by K9LA [10] during cycle-24 SEPs above 1000 pfu, which report 5–10 dB of remaining 14 MHz path on polar circuits where the canonical $\sqrt{\Phi}$ form would predict $> 30 \text{ dB}$. The Sauer–Wilkinson 2008 derivation is itself anchored in the 30 MHz riometer record at fluxes spanning the $10\text{--}10^4 \text{ pfu}$ range (their Figures 4–5b, validating against 11 large SEP events 1998–2002 at Thule, Greenland), but the paper does not specify a saturation regime above any particular flux threshold; ionocast's log-flux substitution is a downstream calibration choice informed by the K9LA observational record rather than by Sauer & Wilkinson directly. The same $f^{-1.5}$ frequency dependence as ordinary D-region absorption is retained:

$$L_{\text{PCA}}(f, \Phi_{p10}) = \min\left(30 \text{ dB}, 5 \log_{10} \frac{\Phi_{p10}}{10} \cdot \left(\frac{7}{f_{\text{MHz}}}\right)^{1.5}\right) \quad (8)$$

The inner $\min(30 \text{ dB}, \cdot)$ in Eq. 8 caps *each* per-hop contribution at 30 dB; per-hop contributions are then summed along the great circle (the same hop grid used by L_{Dreg}) and the resulting path total is capped again at 50 dB — the same two-tier capping used by aurora and flare-driven SID (L_{aur} , L_{flare}), since all three are D-region absorption mechanisms with the same physical upper bound from total D-region ionisation depth. Midlatitude paths see $L_{\text{PCA}} = 0$ even during a

severe SEP because no hop traverses the polar cap; polar paths see order-of-magnitude absorption within minutes of proton-event onset.

Multi-channel onset detection. The GOES integral ≥ 1 MeV proton channel rises ~ 1 h before ≥ 10 MeV during a hard SEP event because lower-energy protons arrive first along the connecting magnetic field line. When $\Phi_{p1} \geq 10$ pfu, an early-warning per-hop bump is applied on polar hops:

$$L_{\text{PCA, onset}}(f, \Phi_{p1}) = \min\left(L_{\text{onset, cap}}, \alpha_{\text{onset}} \cdot \log_{10} \frac{\Phi_{p1}}{10} \cdot \left(\frac{7}{f_{\text{MHz}}}\right)^{1.5}\right) \quad (9)$$

with $\alpha_{\text{onset}} = 1$ dB/decade (vs. 5 dB/decade in the main term, since Φ_{p1} tracks arrival of the SEP front rather than the absorbing population) and $L_{\text{onset, cap}} = 5$ dB/hop.

Smooth handoff. Earlier versions zeroed $L_{\text{PCA, onset}}$ the moment Φ_{p10} crossed the S1 threshold, producing up to a 5 dB cliff in the budget at the exact instant the alert escalated. The current implementation keeps $L_{\text{PCA, onset}}$ active past the threshold and the path sum uses $\max(L_{\text{PCA}}, L_{\text{PCA, onset}})$ per hop, so the harder channel takes over smoothly once its absorption exceeds the soft channel's. The ≥ 100 MeV channel is fetched alongside the other two via the same SWPC integral-protons proxy (the JSON response includes all integral channels in a single call) and cached in `src/data/fetchers.js` as `p100`, but it is not currently consumed by the physics layer. It surfaces only in diagnostics for now; future work will use it to scale the per-hop cap when the SEP penetrates deep into the D-region (the deepest channels reach the lower D-region where the absorption cap is itself frequency-dependent).

3.6 Flare-Driven Sudden Ionospheric Disturbance

Solar soft X-ray photons from an in-progress flare enhance D-region electron density on the sunlit side of the Earth within seconds of the flare peak [17]. ionocast reads the current SWPC `xrayClass` (updated once per minute, well under the model's 10-minute refresh) and applies an additive absorption term on sunlit hops, with a smooth twilight ramp closing the terminator step that the earlier hard $\cos \chi_k \leq 0$ gate produced:

$$L_{\text{flare}}(f, X, \cos \chi_k) = \min\left(40 \text{ dB}, D(X) \cdot \left(\frac{7}{f_{\text{MHz}}}\right)^{1.5}\right) \cdot r(\cos \chi_k) \quad (10)$$

$$r(\cos \chi_k) = \begin{cases} 0 & \cos \chi_k \leq 0 \\ \cos \chi_k / 0.05 & 0 < \cos \chi_k < 0.05 \\ 1 & \cos \chi_k \geq 0.05 \end{cases} \quad (11)$$

The ramp $r(\cos \chi_k)$ scales the daytime value linearly across a $\sim 3^\circ$ solar-elevation band ($\cos \chi_k \in [0, 0.05]$ corresponds to elevation $\in [0^\circ, 2.87^\circ]$). The wall-clock duration of that 3° band depends on latitude: the elevation rate at the horizon is $\sim 15^\circ/\text{h}$ at the equator (90° sun arc over 6 h), $\sim 10^\circ/\text{h}$ at midlatitudes, $\sim 3^\circ/\text{h}$ at $|\phi| = 60^\circ$ in winter, and $\rightarrow 0$ at the polar circles around solstice — so the twilight ramp spans about 12 min at the equator, 17 min at midlat, an hour at high winter latitude, and (asymptotically) the entire civil-twilight period inside the polar circles. The ramp matches the gentle climb of D-region photoionisation through twilight at any latitude rather than snapping on at the geometric terminator.

The log-flux driver is in units of dB at 7 MHz, calibrated so M1 produces 4 dB, X1 produces 12 dB, and X10 saturates at 20 dB:

$$D(X) = \max\left(0, \min(20 \text{ dB}, 4 \text{ dB} + 8 \text{ dB} \cdot \log_{10} \frac{X}{\text{M1}})\right) \quad (12)$$

where $X/\text{M1}$ is the X-ray flux ratio to the M1 baseline (10^{-5} W/m^2).

Calibration source. The anchors (4 dB at M1, 12 dB at X1, 20 dB at X10) follow the classical SID-absorption framework of Mitra [17] (§5–6 of *Ionospheric Effects of Solar Flares* tabulates D-region absorption versus soft X-ray peak flux on a per-decade scale), with the slope set to 8 dB/decade between M1 and X10 so the anchor points fit the published M1/X1/X10 dB pairs simultaneously. The values were cross-checked against WSPR loss patterns observed during cycle-25 M-class and X-class events (the wspr-live archive of 2024–2026 flare hours; see §7.2): on 20 m and 17 m WSPR spot drops of 5–12 dB versus the prior quiet-day median tracked the predicted $D(X) \cdot (7/f)^{1.5}$ envelope to within ~ 2 dB across M1 through X4. (No cycle-25 X10 events have occurred at the time of writing, so the X10 anchor is set by the literature extrapolation alone, hence the 20 dB hard saturation in the equation rather than an open-ended slope.)

The outer $\max(0, \cdot)$ wraps the saturated $\min(20, \cdot)$ branch and suppresses contributions whose underlying log-driver crosses zero, which happens at $X \approx 10^{-5.5} \text{ W/m}^2$ (approximately C3.16); this is not an explicit “if $X < \text{C3}$ return 0” gate but the practical effect is that the quietest C-class events (C1–C3) contribute ~ 0 dB at the 7 MHz reference, while higher C-class flares ramp up the contribution as the algebra allows: $D(\text{C5}) \approx 1.6$ dB, $D(\text{C9}) \approx 3.6$ dB, $D(\text{M1}) = 4$ dB by calibration construction (these are at the 7 MHz reference; the band-scaling factor $(7/f)^{1.5}$ raises them on lower bands so a C5 event still contributes a few dB on 80 m / 160 m). The clamp removes only the lowest of the C-class background. As with PCA, sunlit-hop contributions are summed and the path total is capped (50 dB). The instantaneous-proxy approximation omits the roughly 10–60 minute exponential decay of the SID, which is acceptable given the model’s 10-minute cadence; the next refresh picks up the decayed class.

Two flare thresholds, two distinct roles. The physics zero-crossing at C3.16 (this section) suppresses the quiet-day background contribution but is otherwise continuous; the alert-banner threshold at M1 / M5 / X (Table 14) is escalation-grade signaling for the operator UI. The two thresholds are not redundant – one is the algebraic point where the physics term goes to zero, the other is the operator-facing “this is worth surfacing top-of-page” line. Quiet-time C-class flares contribute sub-dB absorption to the budget without firing a banner.

3.7 Auroral Absorption (CGM-Gated)

Auroral absorption activates when the corrected geomagnetic (CGM) latitude $|\phi_{\text{CGM}}|$ at a reflection point is at or above the storm-dependent threshold:

$$L_{\text{aur}} = \min\left(30, D \cdot \frac{30}{f_{\text{MHz}}} \cdot c(\phi_{\text{CGM}})\right) \quad (13)$$

where

$$D = \max\left(\max(0, 5(K_p - 4)), \max(0, \frac{\text{HP}_{\text{GW}} - 50}{5})\right) \quad (14)$$

ramps continuously: the K_p term begins contributing at $K_p > 4$ and reaches $D = 5$ at $K_p = 5$ (the G1 storm threshold), which maps to 5 dB of L_{aur} at the 30 MHz reference frequency f in Eq. 13 but to substantially more at HF: the $30/f_{\text{MHz}}$ band-scaling factor turns $D = 5$ into ~ 21 dB on 40 m ($5 \cdot 30/7$) and ~ 11 dB on 20 m ($5 \cdot 30/14.1$), before the per-hop 30 dB cap and the $c(\phi)$ CGM ramp. The OVATION hemispheric-power term contributes $(\text{HP} - 50)/5$ above the 50 GW floor: $D = 0$ at $\text{HP} = 50$ GW, $D = 1$ at $\text{HP} = 55$ GW, $D = 5$ at $\text{HP} = 75$ GW (same D -scaling as the K_p term). The two drivers are combined via $\max(\cdot)$ rather than additively because they are not independent — both are forced by the same auroral-electrojet activity, and adding them would double-count the underlying ionisation; but they are kept as parallel sensors because they sample the same phenomenon on different timescales. K_p is a 3-hour planetary index reported every 3 h, so a sub-hour auroral intensification driven by a fresh substorm onset is invisible to it for up to 90 minutes; OVATION HP is updated every ~ 30 min and reflects the

actual hemispheric power dissipated in the auroral oval over the most recent half hour. The $\max(\cdot)$ takes whichever sensor is reading hotter at any given moment, so a fast-onset substorm shows up via HP and a multi-hour storm in steady state shows up via K_p — catching both regimes without double-counting.

The hard $K_p \geq 5$ / HP ≥ 50 GW gate the earlier formulation used produced a discontinuity in D at the moment the index ticked from 4.99 to 5.00 — amounting to a 21 dB step on 40 m polar paths — the same family of bug as the bare CGM-edge gate the $c(\phi)$ ramp below retired. The smooth onset means the prediction reflects modest auroral absorption already at active- K_p ($K_p \in [4, 5)$) levels rather than waiting for a hard storm threshold. $c(\phi_{\text{CGM}})$ is a smooth ramp on the CGM-edge:

$$c(\phi) = \begin{cases} 0 & |\phi| \leq \phi_{\text{thr}} - 5^\circ \\ \frac{|\phi| - (\phi_{\text{thr}} - 5^\circ)}{5^\circ} & \phi_{\text{thr}} - 5^\circ < |\phi| < \phi_{\text{thr}} \\ 1 & |\phi| \geq \phi_{\text{thr}} \end{cases} \quad (15)$$

Linear ramp over the trailing 5° , full effect poleward of the threshold. Eliminates the hard cliff that the bare gate produced.

Storm-zone expansion. The CGM threshold ramps linearly from 60° to 50° as K_p^{eff} climbs from 5 to 7, then holds at 50° above $K_p^{\text{eff}} = 7$:

$$\phi_{\text{thr}}(K_p^{\text{eff}}) = 60^\circ - 10^\circ \cdot \min\left(1, \max(0, (K_p^{\text{eff}} - 5)/2)\right) \quad (16)$$

This captures the equatorward expansion of the auroral oval during storms. The argument is the storm-lagged effective K_p (Section 5.3.1), the same driver that scales the absorption magnitude D in Eq. 13's outer term — both sides of the auroral term (geometric extent and absorption depth) respond on the same thermospheric timescale, so a single K_p^{eff} argument is correct. The earlier prose framed this with bare K_p , which left a reader wondering whether the geometric expansion fired on the live (3-hour) index while only the absorption magnitude carried the lag — it does not. An earlier formulation stepped discontinuously from 60° to 50° at $K_p = 7$, which produced a 30+ dB jump on a hop near 55° CGM at the exact instant the index ticked from 6.9 to 7.0 (the same family of bug as the bare CGM-edge gate the $c(\phi)$ ramp above retired). The smooth ramp also lets the expansion start gradually during moderate storms (G2/G3) rather than waiting for the severe-storm threshold. This is wired in the absorption term itself; no external runtime gating is needed.

Per-hop evaluation. On a multi-hop path, L_{aur} is computed per reflection point and summed (capped at 30 dB per hop and 50 dB total path) by the same machinery as the per-hop D-region sum (Eq. 7). A 4-hop transatlantic path with the midpoint at 55° CGM but two of its hops at 65° correctly accrues auroral absorption on the polar hops; a midpoint-only gate would miss it.

Storm-main amplification (forward reference). Eq. 13 states the steady-state L_{aur} . During the *main* phase of a geomagnetic storm ($\text{Dst} \leq -50$ nT with K_p^{eff} still loading), the right-hand side of Eq. 13 is multiplied by 1.4 to capture the extra particle-precipitation absorption that the steady-state K_p -keyed form undercounts; recovery and quiet phases keep the steady-state value. The amplification is detailed together with the storm-phase classifier in Section 7.3 (paragraph “Storm-phase auroral amplification”), since both the classifier and the σ_{recovery} widening it gates are defined there. A reader computing the budget from this section alone should apply the $1.4\times$ factor when the phase classifier reports *main*.

The CGM latitude is computed via a tilted-dipole approximation:

$$\phi_{\text{CGM}} = \arcsin\left(\text{clamp}(\sin \phi \sin \phi_P + \cos \phi \cos \phi_P \cos(\lambda - \lambda_P), -1, +1)\right) \quad (17)$$

The explicit clamp guards against floating-point overshoot of the arcsin argument outside $[-1, +1]$ near the geomagnetic pole itself, where $\sin \phi \sin \phi_P + \cos \phi \cos \phi_P \cos(\Delta\lambda)$ can drift by $\sim 10^{-15}$ above 1 from rounding error and produce a NaN ϕ_{CGM} that would propagate through the auroral CGM gate. with the IGRF 2020 [20] north magnetic pole at $\phi_P = +80.7^\circ$, $\lambda_P = -72.7^\circ$ (signed-east-of-Greenwich convention; held fixed for reproducibility; IGRF-14 was published with the 2025.0 epoch but the centred-dipole pole has drifted sub-degree since 2020, far below the ϕ_{CGM} ramp's 5° smoothing width in Eq. 13 and well below any verdict-shifting threshold. Centred-dipole drift is $\sim 0.05^\circ/\text{yr}$, so the held-fixed values remain inside the smoothing width through ~ 2030 ; review against IGRF-15 at the 2030.0 epoch). The operator's QTH longitude is whatever signed value the geolocation returns (e.g. -9.14° for Lisbon, $+139.69^\circ$ for Tokyo). Holding the IGRF pole fixed while accepting $F_{10.7A}$ live looks asymmetric but reflects the underlying timescales: $F_{10.7A}$ varies on minutes (daily product, with significant intraday spread), so consuming the live value is correct; the IGRF centred-dipole pole drifts $\sim 0.05^\circ/\text{yr}$ (decadal), so freezing it at the 2020 epoch produces sub-degree error well inside the ϕ_{CGM} ramp's 5° smoothing width and keeps the bibliography pinned to a citable model (Alken et al. [20]) rather than tracking a continuously-updating model file.

3.8 Multi-Hop Ground Reflection Loss

Each extra F-region hop adds a ground-reflection plus defocusing penalty applied at the intermediate ground bounces. To avoid a discrete cliff at the hop-ceiling boundary, the extra-hop count is computed as a smooth function of distance:

$$L_{\text{hop}} = N_{\text{extra}}(d) \cdot (L_{\text{gr}}(f, \theta) + D) \quad \text{dB}, \quad N_{\text{extra}}(d) = \max\left(0, \frac{d_{\text{km}}}{d_{\text{hop}}^{\text{max}}(h_F)} - 1\right) \quad (18)$$

where the per-hop ground-range ceiling scales as $\sqrt{h_F}$, the leading-order spherical-Earth result for glancing-incidence reflection at altitude h_F . At zero elevation the geocentric angle to the reflection-point ground projection is $\gamma = \arccos(R_\oplus/(R_\oplus + h_F))$, giving $d \approx 2\sqrt{2R_\oplus h_F}$ for $h_F \ll R_\oplus$. ionocast preserves the empirical 4000 km anchor at the quiet-cycle $h_F = 300$ km and lets the rest follow the square-root shape:

$$d_{\text{hop}}^{\text{max}}(h_F) = 4000 \text{ km} \cdot \sqrt{\frac{h_F}{300 \text{ km}}} \quad (19)$$

At $h_F = 300$ km this reduces to 4000 km. On enhanced-solar-flux days h_{mF2} climbs to 340 km or above and the ceiling moves to ~ 4258 km (not the 4533 km the prior linear $h_F/300$ scaling would give; that form over-extends the ceiling by $\sim 7\%$ at high h_F and pushed borderline 4250–4500 km circuits into 1-hop budgets that geometry actually placed in the 2-hop regime). ionocast uses an observed h_{mF2} from the GIRO digisonde panel when available; otherwise defaults to 300 km.

The per-hop ionospheric loss bundle (Section 3.4) is also continuous across the hop boundary, but in a different way than the multiplier here. At a non-integer $n_C = d/d_{\text{hop}}^{\text{max}}$, the bundle computes the four ionospheric absorption sums (D-region, PCA, flare, auroral) for both bracketing integer hop layouts $\lfloor n_C \rfloor$ and $\lceil n_C \rceil$ — each laying down a different set of reflection-point coordinates — and blends the totals by the fractional part $\alpha = n_C - \lfloor n_C \rfloor$: $L = (1 - \alpha) \cdot L_{\lfloor n_C \rfloor} + \alpha \cdot L_{\lceil n_C \rceil}$. At integer n_C , $\alpha = 0$ and the ceil branch is skipped. This closes a real verdict-shifting cliff: under the earlier integer-only form, a 4001 km path on 80 m at noon jumped from ~ 18 dB of D-region absorption (single midpoint) to ~ 36 dB (two reflection points) at the boundary, while the ground-reflection multiplier L_{hop} barely changed (N_{extra} ramping from 0 to 0.0003). The blended form makes both budgets continuous in d together.

$L_{\text{gr}}(f, \theta)$ is the per-bounce Fresnel reflection loss and $D = 0.25$ dB is the per-extra-hop defocusing constant (earth-curvature spreading; ITU-R P.533 Annex 1 prescribes ~ 1 dB as a population mean, but that figure mixes ground types and elevations and overshoots typical amateur multi-hop paths over “average earth” as established by 30 d WSPR calibration).

Fresnel ground-reflection loss. At grazing angle $\psi = \theta_{\text{elev}}$ (for a symmetric hop, takeoff equals grazing at the bounce), with complex relative permittivity

$$\varepsilon_c = \varepsilon_r - j 60 \sigma \lambda \quad (\lambda \text{ in metres}) \quad (20)$$

and “average earth” constants per ITU-R P.527 [7] ($\varepsilon_r = 13$, $\sigma = 0.005 \text{ S/m}$, where σ here denotes ground conductivity, not the per-band prediction spread of Table 11), the horizontal-polarisation Fresnel coefficient is

$$R_h = \frac{\sin \psi - \sqrt{\varepsilon_c - \cos^2 \psi}}{\sin \psi + \sqrt{\varepsilon_c - \cos^2 \psi}}. \quad (21)$$

The implementation takes the H-pol power reflectance

$$L_{\text{gr}}(f, \psi) = -10 \log_{10} |R_h|^2, \quad L_{\text{gr}} \in [0, 8] \text{ dB}. \quad (22)$$

Most amateur HF antennas are H-polarised (dipoles, inverted-V, OCF, EFHW, Yagi/quad arrays). An earlier polarisation-averaged variant ($\frac{1}{2}(|R_h|^2 + |R_v|^2)$) was tried under the theory that magneto-ionic mode coupling on F2 reflection scrambles the wave’s polarisation before the next bounce; 30 d WSPR calibration ruled it out, with H-pol producing Brier 0.07 (89% accuracy) versus AVG at 0.13 (83%) and V-pol at 0.27 (66%). The empirical advantage matches operator experience that long-DX paths work consistently: H-pol Fresnel predicts <1 dB per bounce at typical 5–15° grazing angles, which is what the multi-hop budget needs to see for a 3–5 hop path to predict “open” when the band is in fact open.

The [0, 8] dB clamp guards against the single-layer earth model’s behaviour at high (near-normal) incidence angles. Unlike V-pol, H-pol has no Brewster zero; $|R_h|^2$ is monotonic with grazing angle, peaking in loss at normal incidence ($\psi = 90^\circ$) and decreasing as the geometry goes more grazing. For “average earth” ($\varepsilon_r = 13$, $\sigma = 0.005 \text{ S/m}$) on 14 MHz the normal-incidence loss is $\sim 4.7 \text{ dB}$, comfortably under the clamp. The cap only binds for very dry / poor ground ($\varepsilon_r \approx 4$), where the single-layer model can predict normal-incidence loss above 8 dB; a real earth at those parameters is layered (ground-water profile, soil density gradient) and the effective reflectance does not collapse that hard, so the 8 dB cap is the empirical ceiling that keeps the budget aligned with WSPR observations on those short-haul paths. The clamp is rarely binding in practice: typical amateur HF takeoff angles sit at 5–30° grazing (ψ measured from the surface), where the unclamped Fresnel form itself returns sub-3 dB losses regardless of ground parameters.

Typical magnitudes. At a representative $\sim 10800 \text{ km}$ Lisbon–Tokyo path, hop geometry under the saturating $d_{\text{hop}} = \min(d, d_{\text{hop}}^{\text{max}})$ form (Section 3.14, Eq. 32) yields $\theta = \arctan(2 \cdot 300/4000) = 8.5^\circ$, and per-bounce H-pol Fresnel loss at that angle on 14 MHz with $\varepsilon_r = 13$, $\sigma = 0.005 \text{ S/m}$ gives $L_{\text{gr}} \approx 0.66 \text{ dB}$. Eq. 18’s smooth $N_{\text{extra}}(d) = d/d_{\text{hop}}^{\text{max}} - 1$ at $d_{\text{hop}}^{\text{max}} = 4000 \text{ km}$ gives $N_{\text{extra}} = 1.7$, so $L_{\text{hop}} = 1.7 \cdot (0.66 + 0.25) \approx 1.55 \text{ dB}$ on the path. A 5-hop trans-equatorial path approximately doubles it. Single-hop NVIS and short-haul paths see $L_{\text{hop}} = 0$ (no intermediate bounces).

Calibration coverage. This decomposition replaces an earlier scalar $L_g = 3 \text{ dB/hop}$ that the calibration harness could not actually retune (its single-path 3000 km reference rounded to one hop, yielding $N_{\text{hops}} - 1 = 0$). The Fresnel piece is fully physics-derived; only the defocusing scalar D remains tunable, and is exercised by the harness over a multi-path basket spanning 1, 3, and 5 hops (Section 7.2).

Path-type bias (deferred). Sea-vs-land per-path classification is deferred — “average earth” ($\varepsilon_r = 13$, $\sigma = 0.005 \text{ S/m}$) applies everywhere until coastline-aware path geometry is added. The direction and magnitude of the bias is known and bounded:

- Seawater ($\sigma \approx 5 \text{ S/m}$, three orders of magnitude more conductive) produces $\sim 0.3 \text{ dB}$ H-pol Fresnel loss per bounce versus $\sim 0.66 \text{ dB}$ on average earth, so *transatlantic and transpacific paths are under-predicted by $\sim 1\text{--}3 \text{ dB}$ depending on hop count* (1 dB on a 2-hop circuit, $\sim 3 \text{ dB}$ on a 5-hop trans-oceanic).
- Dry sand or arid soil ($\sigma \approx 0.001 \text{ S/m}$, five times less conductive) produces an extra $\sim 0.5 \text{ dB}$ H-pol loss per bounce, so *paths over the Sahara, Australian desert, or central Asia are over-predicted by $\sim 1\text{--}2 \text{ dB}$ on a 3–5 hop budget*.

Both biases are operator-relevant on close-margin verdicts where the budget reads near a tier boundary but small in the open-band regimes where the verdict is dominated by the F2 / MUF / absorption terms. Until coastline-aware geometry lands, the UI shows the average-earth verdict; expect transatlantic / transpacific paths to perform somewhat better than the verdict suggests, and desert-dominated paths somewhat worse.

3.9 Sporadic-E Screening Loss

When a strong Es layer is present ($f_{\text{oEs}} \geq 5 \text{ MHz}$) and the operating frequency is below twice the Es critical frequency ($f < 2 f_{\text{oEs}}$), the layer partially screens the F2 reflection:

$$L_{\text{Es}} = 5 \text{ dB} \quad (23)$$

L_{Es} is a screening penalty applied *only inside the F2 budget* (Eq. 1). When the parallel Es-mode budget of Section 6.1 wins the per-band margin competition, that path uses its own loss bundle (Eq. 63) and L_{Es} is not part of its calculation.

3.10 Low-Band Extra Loss

A band-stepped term applies an empirical residual on the lower HF bands (Table 3). L_{low} is added *on top of* the per-hop quiet-day D-region absorption L_{Dreg} (already 28 dB on 160 m at noon, etc., per Table 2), not as a substitute for it. The mechanisms it represents are propagation residuals that L_{Dreg} 's mean-conditions formulation does not capture:

- Higher-angle refraction at low frequencies, which forces steeper takeoff angles and longer in-ionosphere ray paths than the geometric hop calculation assumes.
- Magneto-ionic mode coupling on F2 reflection (E-trap and ordinary/extraordinary splitting), which is stronger on low bands and is not modelled mechanistically in the current Fresnel/multi-hop bundle (Section 3.8).
- Ground-wave contamination on short-haul paths, where the sky-wave is in destructive interference with a residual ground-wave component that the F2-only budget does not account for.

The values in Table 3 are hand-set integers chosen by inspection of the harness's per-band Brier residual on the low bands against 30 d of WSPR observations (§7.2); L_{low} is *not* part of the joint coordinate-descent set in Appendix E ($\{L_{\text{iono}}, D, w_{\text{sc}}, w_{\text{NVIS-tail}}, w_{\text{Es-prim}}, \sigma\text{-scale}, \text{fusion-flag}\}$), and the round values (8, 5, 3, 2 dB) are the visible signature of that. Bounds are guided by ITU-R P.533 [1] Annex 1's median-conditions correction term Y_p which the current budget otherwise omits (Section 3.11 discusses why Y_p is not included). A future calibration pass should fold L_{low} into the descent set once the per-pair WSPR ground-truth wiring stabilises; until then, the table is empirical-judgment rather than machine-fitted. The values are path-independent — a per-distance refinement is deferred until enough per-pair WSPR data accumulates to fit a distance-dependent ramp without inflating noise.

Table 3: Low-band extra loss term applied below 30 m. Empirical residual on top of L_{Dreg} , calibrated against 30 d WSPR; see §3.10 for the mechanism breakdown. Stored as anchor points for a smooth log-frequency interpolation (refactored 2026-04-30 from a step-function form that had hard cutoffs at 2 MHz, 4 MHz, 6 MHz, 8 MHz); intermediate frequencies linearly interpolate between the band-centre anchors in $\log f$ space. The 14 MHz anchor holds the value at zero so the contribution tapers smoothly through 30 m and is exactly zero at 20 m and above.

Band	f (MHz)	L_{low} (dB)
160 m	1.838	8.0
80 m	3.570	5.0
60 m	5.366	3.0
40 m	7.040	2.0
30 m	10.140	0.5
20 m +	14.000+	0.0

The 30 m cutoff is empirical: at and above 30 m the L_{Dreg} baseline (Table 2) plus the modeled F2-region terms account for the band’s quiet-day budget within the harness’s residual resolution, so L_{low} adds nothing. Below 30 m the residuals were systematic enough to warrant the empirical correction.

3.11 Lumped Ionospheric Loss

$$L_{\text{iono,path}} = n_{\text{hops}} \cdot L_{\text{iono}}, \quad L_{\text{iono}} = 1 \text{ dB} \quad (24)$$

This is the ITU-R P.533 median-conditions residual once the individually-modelled mechanisms are subtracted: spatial focusing/defocusing L_z is now handled per intermediate hop by D (Section 3.8), polarisation coupling is now handled inside the Fresnel R_h branch of the multi-hop reflection model, and the only term that remains a-priori invisible to the geometry is the ~ 1 dB Faraday-rotation / small-correction residual *per ionospheric reflection*. Per ITU-R P.533 §A.2 the lumped correction accumulates per reflection, so the path total scales with hop count: a 1-hop NVIS path gets 1 dB, a 3-hop transcontinental path gets 3 dB, etc. The 1-dB magnitude was calibrated against the single-hop NVIS-dominated 4-path Lisbon basket and remains single-hop-correct; the multiplier closes a residual on multi-hop DX paths where the prior single-charge convention under-counted by 2–4 dB. L_{iono} does *not* include D-region absorption (handled per-hop by Eq. 7) or ground reflection (Section 3.8), nor does it include the P.533 Y_p above-median-reliability margin (≈ 7 dB at 90% reliability): Y_p is the correct term when the prediction answers “will the circuit work 90% of the time at this hour?”, but ionocast answers “is the band open right now?”, which is a median-conditions question.¹

3.12 Noise Model (ITU-R P.372)

Following ITU-R P.372 [2], the received noise power is the power sum of the natural floor (galactic plus atmospheric) and a man-made contribution that fires only on suburban and urban sites:

$$N = 10 \log_{10} \left(10^{N_{\text{atmo}}/10} + \mathbb{1}_{\text{env} \neq \text{rural}} \cdot 10^{N_{\text{mm}}/10} \right) \quad (25)$$

where:

$$N_{\text{atmo}} = N_{\text{base}}(f) + \Delta N_{\text{diurnal}}(f, \cos \chi) \quad (26)$$

$$N_{\text{mm}} = N_{\text{base}}(f) + F_a(\text{env}) \quad (27)$$

¹Historical L_{iono} sequence (reflecting progressive offloading of mechanisms from this lumped term to physics-derived per-hop terms): 35 dB (a-priori, pre- Y_p -removal) $\rightarrow$ 15 dB (Y_p -bulk removed) $\rightarrow$ 8 dB (Y_p -residual removed) $\rightarrow$ 2 dB (Phase 1 multi-hop Fresnel absorbed the per-hop bounce loss implicitly bundled into the lumped term) $\rightarrow$ 1 dB (H-pol Fresnel calibration; polarisation-coupling double-count removed). Each step was driven by 30 d WSPR validation.

The indicator $\mathbb{K}_{\text{env} \neq \text{rural}}$ gates the man-made channel off entirely on rural sites; on suburban / urban sites the channel sits at an environment-specific level $N_{\text{base}} + F_a$ that is roughly diurnally constant (man-made QRM tracks the local interference environment, not the natural floor's day/night variation). The diurnal swing is:

$$\Delta N_{\text{diurnal}} = -A(f) \cdot \text{clamp}(\cos \chi, -1, 1) \quad (28)$$

where $A(f)$ ramps linearly from 10 dB at $f \leq 5$ MHz down to 3 dB at $f \geq 15$ MHz:

$$A(f) = \begin{cases} 10 & f \leq 5 \text{ MHz} \\ 10 - 7(f - 5)/10 & 5 \text{ MHz} < f < 15 \text{ MHz} \\ 3 & f \geq 15 \text{ MHz} \end{cases} \quad (29)$$

Galactic-vs-atmospheric dominance transitions gradually across the 5 MHz–15 MHz window (P.372 Fig. 23/24 vs Fig. 13/14) rather than at any single frequency, so a smooth ramp is consistent with the smooth-handoff philosophy applied elsewhere in the budget. The man-made noise factor F_a depends on environment: *rural* carries no man-made contribution above the natural floor (the indicator switches the second power-sum term off), *suburban* sets $F_a = +15$ dB above the rural baseline N_{base} , and *urban* sets $F_a = +25$ dB. The amplitude A is held latitude-independent for simplicity; tropical sites with frequent thunderstorm activity see larger atmospheric-noise diurnal swings (the P.372 atmospheric maps exceed midlat figures by 5–10 dB in the equatorial band) and polar sites with stable nighttime ionospheres see smaller swings, so the model is somewhat low for tropical noise floors at night and somewhat high for polar nights. The error is bounded at 2–3 dB on the noise side and is dominated by other budget terms on most verdicts.

Why the indicator: the rural double-count it retired. The earlier formulation defined $N_{\text{mm}} = N_{\text{base}} + F_a$ unconditionally and power-summed it with N_{atmo} . On rural sites F_a was set to 0 dB, which pinned N_{mm} to N_{base} *at all times of day*, clamping the rural noise floor to within ~ 3 dB of N_{base} regardless of $\Delta N_{\text{diurnal}}$. The clamp's biggest effect was suppressing the daytime quiet-rural descent: at 15 m local noon the old form read ~ -130 dBm rather than the P.372 quiet-rural expectation near -135 dBm, a 4–5 dB over-prediction of noon noise that propagated into a same-magnitude under-prediction of upper-HF daytime SNR margin. Lower bands saw the same mechanism but with a larger amplitude: at 80 m noon rural the old form read ~ -115 dBm versus a quiet-rural value near -125 dBm (10 dB). Gating the man-made channel off on rural sites removes the clamp: rural now reads $N = N_{\text{atmo}}$ exactly, so the diurnal swing modulates the floor in full, -135 dBm at 15 m noon and -129 dBm at 15 m midnight. Midnight upper-HF rural drops by only ~ 1.8 dB (the small swing $A(15 \text{ MHz}) = 3$ dB caps the magnitude). Suburban and urban verdicts are unaffected: N_{mm} dominates the power sum on those sites, so the change there is sub-decibel.

Mapping to ITU-R P.372 figures. $N_{\text{base}}(f)$ in Table 4 is the lumped quiet-rural floor, the power sum of the galactic noise (P.372 Fig. 23/24, slowly varying with frequency at $F_a \approx 50 - 23 \log_{10}(f_{\text{MHz}})$ dB above thermal in the 10–200 MHz range) and the atmospheric quiet-day baseline (P.372 Fig. 13/14, midlatitude midnight summer reference). The diurnal swing $\Delta N_{\text{diurnal}}$ then modulates this combined floor by the atmospheric component's day-vs-night spread (the galactic component is nearly diurnally constant; below 10 MHz the atmospheric swing dominates so the 10 dB amplitude is appropriate, above 10 MHz galactic dominates and the amplitude shrinks to 3 dB). The man-made channel N_{mm} sits F_a dB above the lumped rural baseline N_{base} on suburban and urban sites and is held diurnally constant (man-made QRM is set by the local interference environment, not by the natural floor's diurnal swing); on rural sites the channel is gated off entirely (Eq. 25's indicator), so a quiet rural QTH reads exactly N_{atmo} rather than the $N_{\text{atmo}} \oplus N_{\text{base}}$ phantom-3 dB the earlier form produced. The power-sum then correctly

recovers the dominant component: man-made on suburban / urban, the natural floor on rural (atmospheric-dominated on low bands, galactic-dominated on upper bands).

Receiver-bandwidth scaling. The two noise components scale differently when the receiver narrows from the reference bandwidth $B_0 = 2500$ Hz to the mode’s effective bandwidth B_{mode} (Section 3.13):

$$N_{\text{atmo}}(B_{\text{mode}}) = N_{\text{atmo}}(B_0) + 10 \log_{10}(B_{\text{mode}}/B_0) \quad (30)$$

$$N_{\text{mm}}(B_{\text{mode}}) = N_{\text{mm}}(B_0) + \alpha \cdot 10 \log_{10}(B_{\text{mode}}/B_0) \quad (31)$$

Atmospheric (galactic, thermal, distant lightning) is quasi-white and collapses in full with the receiver filter. Man-made noise is impulse-dominated: a bright transient in a 2500 Hz filter is mostly the same transient in a 50 Hz filter, since impulses have broad spectra. The scaling exponent α interpolates between the ideal-white limit ($\alpha = 1$) and the pure-impulsive limit ($\alpha = 0$); ionocast uses $\alpha = 0.5$ as a mixed-source-typical compromise. The Middleton Class A/B impulse-noise model [21] gives a non-linear bandwidth-scaling form that does not reduce to a single power-law exponent in general, so $\alpha = 0.5$ is a parametric simplification of a non-parametric scaling — defensible at the prediction level but not derivable as a midpoint of the Middleton form itself. The 0.5 value is a compromise tuned for typical mixed broadband QRM (switching power supplies, motor controllers, plasma TVs, LED drivers, ignition transients): *continuous* interferers (RTTY signals, broadcast bleed-over, local switching mode power-supply hum) sit closer to $\alpha = 1$ and are slightly under-rewarded by the narrowband-filter gain; *purely impulsive* interferers (lightning crashes, isolated arcing) sit closer to $\alpha = 0$ and are slightly over-rewarded. A site dominated by one mode could in principle be tuned with a different α , but that level of per-site characterisation is out of scope for the current model.

FT8-vs-SSB advantage — numeric walkthrough. The end-to-end model produces about 31 dB advantage on rural and 22 dB on suburban for FT8 over SSB, evaluated on a midlat 20 m path. Decomposition: SSB and FT8 use mode-native S_{req} values from Table 5 (+10 vs −4 dB), so the decoder-side advantage is exactly 14 dB. Bandwidth scaling on a rural site (galactic / atmospheric noise dominant, $\alpha = 1$) gives $10 \log_{10}(2500/50) \approx 17$ dB of additional gain for FT8’s narrower receive bandwidth, totalling 31 dB. On a suburban site (man-made-dominated, $\alpha = 0.5$) the noise-side gain falls to ~ 8.4 dB, totalling ~ 22 dB. Operator experience reports of ~ 25 dB rural / ~ 15 dB suburban quoted in the WSJT-X / ham-radio literature compress several effects — partial decoder calibration loss at very low SNR, RX-chain front-end variability, the fact that operators typically compare the modes at the noise-floor edge rather than deep in the decode region, etc. — that the model does not attempt to reproduce. The computed numbers are larger than operator-reported because the model is the maximum-information budget; treat the reported ~ 25 / ~ 15 figures as floor-of-experience benchmarks, not as a target for the budget.

Table 4 gives the baseline noise power at 2.5 kHz reference bandwidth.

Table 4: Baseline noise N_{base} from ITU-R P.372 at 2.5 kHz bandwidth. Noon-time floor; midnight = $N_{\text{base}} + \Delta N_{\text{diurnal}}(\cos \chi = -1)$ where the swing amplitude $A(f)$ is 10 dB at $f \leq 5$ MHz ramping linearly to 3 dB at $f \geq 15$ MHz. Re-derived 2026-04-30 from P.372-15 Fig 13 (atmospheric) $\oplus$ Fig 23 (galactic) max-of at midlat midnight summer.

Band	N_{base} (dBm)	midnight (dBm)
160 m	-100	-90
80 m	-113	-103
60 m	-118	-108
40 m	-121	-112
30 m	-120	-114
20 m	-120	-116
17 m	-120	-117
15 m	-121	-118
12 m	-122	-119
10 m	-123	-120

The 2026-04-30 retune corrects a directional discrepancy in the prior table: those values were calibrated against eyeball P.372 guesses that, on upper bands, sat *below the galactic noise floor itself*, which is structurally impossible (the galactic Fa ~ 50 dB at 14–30 MHz is what it is, independent of station setup). Pinning $N_{\text{base}} + \Delta N_{\text{diurnal}}$ to the P.372 midnight reference at every band shifts upper-band floors up by ~ 11 dB, mid bands by 5–8 dB, and low bands by 1–2 dB except 160 m which drifts up by 10 dB on the atmospheric anchor. Direction validated against the WSPR per-spot residual mean (improved from -23.97 dB to -18.73 dB) and against the per-pair WSPR Brier (improved 0.64 to 0.57); global truth Brier rises (0.04 to 0.10) because that metric is permissive at upper bands at night (paper §??).

3.13 Required SNR and Receiver Bandwidth by Mode

Each mode is defined by two numbers: the decoder-only SNR threshold S_{req} specified at the mode’s own effective noise bandwidth B_{mode} , and that bandwidth itself. The receiver-bandwidth advantage of narrow digital modes is not baked into S_{req} ; it is computed separately in the noise model (Section 3.12), which scales atmospheric noise with the receiver BW in full and man-made noise with it only partially. The previous formulation quoted every mode at a common 2.5 kHz reference BW and absorbed the BW advantage into S_{req} (e.g. FT8 at -21 dB), which over-counted the benefit of digital modes on man-made-dominated (impulse-noise) sites.

Table 5: Required SNR threshold and effective noise bandwidth by mode. S_{req} is the decoder threshold at the mode’s own native bandwidth, derived from the WSJT-X [11] published values at 2.5 kHz reference via $S_{\text{native}} = S_{2500} + 10 \log_{10}(2500/B_{\text{mode}})$. The bandwidth advantage of narrow modes is then computed separately in the noise model (Section 3.12) using B_{mode} , not folded into S_{req} .

Mode	S_{req} (dB)	B_{mode} (Hz)	Source
SSB	+10	2500	comfortable-copy native ($S_{2500} = +10$, $\Delta = 0$)
CW	0	500	trained-ear native ($S_{2500} \approx -7$; $-7 + 10 \log_{10}(2500/500) = 0$)
FT4	-3	100	WSJT-X -17 at 2500 Hz ($-17 + 10 \log_{10}(2500/100) = -3$)
FT8	-4	50	WSJT-X -21 at 2500 Hz ($-21 + 10 \log_{10}(2500/50) = -4$)
WSPR	-8	25	WSJT-X -28 at 2500 Hz ($-28 + 10 \log_{10}(2500/25) = -8$)

B_{mode} here is the decoder *noise-equivalent bandwidth* integrated against the WSJT-X published threshold, not the signal’s occupied spectral width. WSPR’s 4-FSK signal occupies ~ 6 Hz ($1.46 \text{ Hz/symbol} \times 4 \text{ tones}$); the listed 25 Hz captures the matched-filter / decoder integration over symbol-time that WSJT-X’s $-28 \text{ dB} / 2500 \text{ Hz}$ threshold already integrates against. FT8’s 50 Hz signal is a closer match to the listed 50 Hz NEB. The decoder NEB is what the noise model in Eq. 25 should scale; using the signal occupied BW would double-count the decoder integration.

3.14 Antenna Elevation Pattern

The antenna gain G_{ant} is not a flat scalar; it varies with the required takeoff (elevation) angle *and with the band*. The elevation-pattern formulas below follow the standard treatment in *The ARRL Antenna Book* [9]. The elevation is derived from hop geometry using a simplified flat-earth model:

$$\theta_{\text{elev}} = \arctan\left(\frac{2 h_F}{d_{\text{hop}}}\right), \quad d_{\text{hop}} = \min(d, d_{\text{hop}}^{\text{max}}(h_F)) \quad (32)$$

where $h_F = 300 \text{ km}$ (typical F2 height; the smoothing form below applies regardless of the exact value) and the per-hop range d_{hop} saturates at the geometric ceiling $d_{\text{hop}}^{\text{max}}(h_F)$ from Eq. 19. Long-haul DX paths require low takeoff angles ($5\text{--}10^\circ$); regional paths use higher angles ($15\text{--}30^\circ$); NVIS uses near-vertical ($60\text{--}90^\circ$).

Continuous across the hop boundary. An earlier formulation used the integer N_{hops} in the denominator ($d_{\text{hop}} = d/N_{\text{hops}}$), which produced a sawtooth: at $d = 4000 \text{ km}$ the path packed into a single 4000 km hop (takeoff 8.5°); at $d = 4001 \text{ km}$ N_{hops} jumped to 2 so $d_{\text{hop}} = 2000 \text{ km}$ (takeoff 16.7°); the angle then smoothly fell back to 8.5° at 8000 km before stepping again at the next hop ceiling. That $\sim 8^\circ$ jump propagated into both the Fresnel ground-reflection loss (Section 3.8) and the elevation-pattern antenna gain (Eq. 33) on every path crossing the boundary — a real verdict-shifting cliff. The $\min(d, d_{\text{hop}}^{\text{max}})$ form above closes that cliff without changing physical correctness: real long-haul paths *do* pack at the geometric ceiling, so per-hop range saturating at $d_{\text{hop}}^{\text{max}}$ for $d \geq d_{\text{hop}}^{\text{max}}$ is what the geometry implies. Below one ceiling the angle steepens with shorter d as before.

Implication: per-hop geometry is invariant in d above the ceiling. Once $d \geq d_{\text{hop}}^{\text{max}}$, the $\min(\cdot)$ saturates at the ceiling, so θ_{elev} (and therefore the Fresnel grazing angle and the antenna elevation gain) is the *same value* for every long-haul path regardless of distance. A 12000 km Lisbon–Tokyo and a 17500 km Lisbon–Sydney see identical per-hop $\theta_{\text{elev}} = 8.5^\circ$, and the antenna gain term $G_{\text{ant}}(\theta)$ contributes the same to both budgets. The two paths still differ in free-space loss (which is $\propto d^2$), multi-hop ground reflection ($\propto N_{\text{extra}}(d)$), and ionospheric absorption sums (longer paths have more reflection points). This is not a model approximation; it is the geometry implied by “hops pack at the ceiling above $d_{\text{hop}}^{\text{max}}$ ”.

Side effect on DX paths. The smoothing closed the boundary cliff but it also nudged the takeoff angle on *all* DX paths ($d > d_{\text{hop}}^{\text{max}}$). Under the integer- N_{hops} form, a 12000 km 3-hop path computed $d_{\text{hop}} = 12000/3 = 4000$ km and $\theta_{\text{elev}} = 8.5^\circ$; a 12001 km 4-hop path (just past the next ceiling) jumped to $d_{\text{hop}} = 3000$ km and $\theta_{\text{elev}} = 11.3^\circ$. The new $\min(\cdot)$ form pins d_{hop} to the ceiling for both, so the second path’s takeoff stays at 8.5° instead of nudging up to 11.3° . Net effect on antenna gain is small (a few tenths of a dB on a horizontal dipole at typical HF antenna heights), but it is not zero: the harness binary accuracy ticked from 88.93 % / Brier 0.0713 to 88.98 % / 0.0697 on the 4-path Lisbon basket when this fix landed — a +0.05 pp shift driven by takeoff geometry rather than physics content.

Type + height + λ . Operators supply three inputs: antenna *type* (eight classes, Table 6), mounting *height* h in metres, and the *peak gain* G_{peak} over typical ground (the type already encodes the free-space directivity, so a 4-element Yagi has a bigger G_{peak} than a dipole at the same h). The effective gain at elevation θ for band frequency f is

$$G_{\text{ant}}(\theta, f, h, \text{type}) = G_{\text{peak}} + G_{\text{rel}}(\theta, f, h, \text{type}) \quad (33)$$

where the relative elevation shape G_{rel} is 0 dB at the pattern peak and floors at -15 dB (real antennas don’t vanish into ideal nulls).

Horizontal types. For a horizontal wire, loop, or beam at height h , the elevation pattern is dominated by the ground- reflection factor for horizontal polarisation (image theory):

$$\text{gf}_H(\theta, f, h) = 2 |\sin(kh \sin \theta)|, \quad k = \frac{2\pi}{\lambda}, \quad \lambda = 300 \text{ m}/f_{\text{MHz}}. \quad (34)$$

The maximum of 2 (6 dB above the free-space radiator) occurs at $\sin \theta = \lambda/(4h)$. For a dipole at $h = 10$ m: on 40 m ($\lambda = 40$ m) the peak sits at $\sin \theta = 1$, i.e. straight up – a pure NVIS antenna; on 20 m ($\lambda = 20$ m) the peak sits at $\sin \theta = 0.5 \Rightarrow \theta = 30^\circ$ – regional / short DX; on 10 m ($\lambda = 10$ m) the peak sits at $\sin \theta = 0.25 \Rightarrow \theta = 14.5^\circ$ – classical low-angle DX lobe. Relative gain:

$$G_{\text{rel,H}} = 20 \log_{10}(\max(\text{gf}_H/2, 10^{-15/20})) \quad (35)$$

The -15 dB floor inside the $\max(\cdot)$ is a numerical guard as well as a pattern-quality cap: at very low antenna heights (e.g. a horizontal dipole at $h = 1$ m on 80 m, where $kh \approx 0.08$ and the formula’s pre-floor value drops to ~ -22 dB at the pattern peak), the bare expression goes arbitrarily negative; the floor caps that to -15 dB so the budget stays well-conditioned and the verdict reflects the “ground-mounted dipole” regime rather than the unrealistic deep null the formula would otherwise predict.

Horizontal loops pick up an additional $(0.5 + 0.5 \sin \theta)$ high-angle weighting – the classical observation that a full-wave loop has a 45 – 60° elevation peak rather than the dipole’s ground-reflection-driven one.

Beams (hex, Yagi, stacked) use the *same* ground-reflection shape as a horizontal wire at the same height: the elevation lobe of a Yagi is set by the ground reflection, not the number of elements. The directivity bonus (3–15 dB over a dipole) lives in G_{peak} .

Vertical. The free-space quarter-wave pattern $f_v(\theta) = \cos(\pi/2 \cdot \sin \theta)/\cos \theta$ combines with a vertical-polarisation ground reflection $\text{gf}_V(\theta, f, h) = 2 |\cos(kh \sin \theta)|$ which evaluates to 2 at $h = 0$ ($\cos 0 = 1$, image in phase everywhere) and reduces continuously as h grows. The implementation retains an explicit $h \lesssim 0.1$ m branch *only* as a numerical shortcut (skipping the $\cos(kh \sin \theta)$ evaluation when it is guaranteed ~ 1); mathematically the elevated-form expression at $h = 0$ already yields the same answer.

$$G_{\text{rel,V}} = 20 \log_{10}\left(\max\left(\frac{f_v(\theta) \text{gf}_V}{1.2}, 10^{-15/20}\right)\right) \quad (36)$$

The denominator normalises by the value of $f_v \cdot \text{gf}_v$ at the antenna’s actual peak (low takeoff, where $f_v(0) = 1$ and $\text{gf}_v = 2$ for a ground-mounted radiator), so $G_{\text{rel}} = 0$ at the strongest direction — matching the horizontal-branch normalisation.

Why the denominator is 1, not 1.57. The function $f_v(\theta) = \cos(\frac{\pi}{2} \sin \theta) / \cos \theta$ has its global maximum of 1 at $\theta = 0$ (horizon) and falls monotonically toward 0 at $\theta = \pi/2$ (overhead). Both the numerator and the denominator vanish at $\pi/2$, but the limit is 0 (L’Hôpital: numerator derivative $-\sin(\frac{\pi}{2} \sin \theta) \cdot \frac{\pi}{2} \cos \theta$ vanishes faster than the denominator derivative $-\sin \theta$); numerically $f_v(80^\circ) \approx 0.14$, $f_v(85^\circ) \approx 0.07$, $f_v(89^\circ) \approx 0.014$. No theoretical limit at 1.57 exists. An earlier formulation used denominator 1.57, which was an unphysical normalisation (likely a misread of the half-wave dipole’s ~ 1.5 broadside effective-gain factor pulled into a quarter-wave vertical formula it has nothing to do with) and silently shaved ~ 3.9 dB off every vertical at its horizon peak; the correct denominator is 1, anchoring $G_{\text{rel}} = 0$ at the antenna’s actual strongest direction.

Peak at $\theta = 0^\circ$ (horizon), null overhead. No band-dependent lobe migration for a ground-mounted vertical — unlike the horizontal family, its pattern is shape-invariant across HF.

Compromise. Indoor / attic / mag-loop installations have a near-omnidirectional scattered pattern with a gentle low-angle penalty from nearby conductors:

$$G_{\text{rel,compromise}} = -\max(0, 2(1 - \theta/10^\circ)). \quad (37)$$

where θ is the takeoff elevation *in degrees* (the 10° literal sets the unit). Implementations that compute takeoff angle in radians (e.g. via `Math.atan(...)`) must convert to degrees before substituting into this expression and into the horizontal / beam pattern formulas below.

Table 6: Antenna types and default (height, peak gain) pre-fills.

Type	h (m)	G_{peak} (dBi)	Covers
Horizontal	10	+5	Dipole, inverted-V, G5RV, OCF, EFHW, folded dipole
Vertical	0	+2	1/4, 1/2, 5/8 wave over radials
Horizontal loop	8	+4	Full-wave delta, quad, square
Small beam	12	+7	Hex beam, 2-/3-el Yagi, compact tribander
Medium beam	15	+9	4-el Yagi, 4-el quad
Large beam	20	+12	5-el+ Yagi, stacked / phased
Compromise	5	−2	Indoor, attic, mag loop, mobile whip
Custom	10	user	Any unusual setup

3.15 Default Operator Settings

The default station profile assumes a typical amateur installation: $P_{\text{tx}} = 50$ dBm (100 W), $G_{\text{ant}} = +5$ dBi peak (installed dipole with ground reflection; effective gain varies by path elevation), mode = SSB, noise environment = suburban.

3.16 Worked Example: 20 m, 1000 km, Quiet Day

Table 7 lists the complete budget for a 14.1 MHz path of 1000 km on a quiet day. At this distance the path is single-hop (4000 km hop ceiling), so $L_{\text{hop}} = 0$ — the multi-hop reflection budget activates only on longer paths.

G_{ant} derivation (per Section 3.14). G_{ant} is not a flat scalar; it is computed from the hop geometry. For a 1000 km single-hop with $h_F = 300$ km, the takeoff angle is $\theta = \arctan(2 \cdot 300/1000) \approx 31^\circ$. For a horizontal dipole at $h = 10$ m on 20 m ($\lambda = 20$ m), the elevation pattern

peaks at $\sin \theta_{\text{peak}} = \lambda/(4h) = 0.5$, i.e. $\theta_{\text{peak}} = 30^\circ$. The required 31° lands essentially on the peak lobe, so $G_{\text{rel}} \approx 0$ dB and $G_{\text{ant}} = G_{\text{peak}} + G_{\text{rel}} = +5$ dBi. On a longer hop (e.g. 3000 km with $\theta \approx 11^\circ$) G_{rel} would be substantially negative on the same dipole; the +5 dBi here is geometric coincidence, not a fixed value. Note that this worked example uses the canonical $h_F = 300$ km fallback for a clean derivation; in production the runtime reads the live $h_m F2$ from the nearest digisonde (typically 280--360 km), which would shift θ by a degree or two and may change which side of the dipole peak the takeoff lands on.

Table 7: SNR budget for 20 m (14.1 MHz), 1000 km, quiet day, suburban noise, SSB, with the reflection point at local noon ($\cos \chi \approx 1$, which is what produces the $L_{\text{Dreg}} = -0.5$ dB shown). The Σ column tracks the running signal level. Below the rule, N and S_{req} are operands to the margin formula $M = S - N - S_{\text{req}}$, not additions to Σ . The resulting dB margin is σ -normalised by the prose below to derive the tier verdict per ITU-R P.842.

Term	Δ	Σ (dBm)
P_{tx}	+50 dBm	+50
G_{ant}	+5 dBi	+55
L_{fs}	-115 dB	-60
L_{Dreg}	-0.5 dB	-60.5
L_{MUF}	0 dB	-60.5
L_{iono}	-1 dB	-61.5
L_{hop}	0 dB	-61.5
<i>Signal at Rx</i>		-61.5
<i>Operands to margin formula:</i>		
N (suburban)		-113 dBm
S_{req} (SSB)		+10 dB
$M = -61.5 - (-113) - 10 = +41.5$ dB $\Rightarrow M/\sigma = 6.92$		

Tier verdict (P.842 reliability bucket). With the per-band σ for 20 m = 6 dB (Table 11), the standardised margin is $M/\sigma = 41.5/6 = 6.92$. Since $M/\sigma \geq 1.2816$ (the ionocast $R \geq 90\%$ Excellent threshold, Table 13), the verdict is “Excellent” and the predicted reliability is $\Phi(6.92) > 99.99\%$. The wide head-room is typical of midday 20 m on quiet days at 1000 km.

Worked example with bonuses (15 m TEP, EU–Brazil at 19 LT). The budget in Table 7 silently sets the three additive bonuses (Eq. 71: B_{grayline} , B_{TEP} , B_{scatter}) to zero because none of them fire on a quiet 20 m midday path. To anchor the bonus arithmetic, consider the 15 m TEP-eligible case: a Lisbon ($\phi_{\text{dip}} \approx +35^\circ$) to São Paulo ($\phi_{\text{dip}} \approx -23^\circ$) path, $f = 21$ MHz, midpoint local time ≈ 19 , two F2 hops, $f/\text{MUF} = 1.10$, $\sigma_{f_{\text{ofF2}}} \approx 0.4$ MHz across the hops. At those numbers the bonus terms read:

- $B_{\text{grayline}} = 0$ (gray-line bonus is zero above 14 MHz).
- $B_{\text{TEP}} = B_{\text{TEP,plateau}} \cdot f_{\text{freq}} \cdot f_{\text{LT}} \cdot f_{\text{dip,src}} \cdot f_{\text{dip,dst}} = 15 \text{ dB} \cdot \text{ramp}(21, 14, 22) \cdot 1 \cdot 1 \cdot 1 = 15 \text{ dB} \cdot 0.875 \cdot 1 \cdot 1 \cdot 1 \approx 13.1$ dB (frequency factor not yet at the plateau; both endpoints past the dip-latitude saturation knee at $|\phi_{\text{dip}}| \geq 10^\circ$; LT in the $[17, 23]$ plateau).
- $B_{\text{scatter}} = w_{\text{sc}} \cdot \min(1, 0.4/1.5) \cdot \min(2, 0.10) \cdot 5 = 1.5 \cdot 0.267 \cdot 0.10 \cdot 5 \approx 0.20$ dB.

The verdict-margin contribution is therefore $B_{\text{grayline}} + \max(B_{\text{TEP}}, B_{\text{scatter}}) = 0 + 13.1 = 13.1$ dB on top of M_{physics} . The $\max(\cdot)$ form intentionally suppresses the small B_{scatter} contribution (0.2 dB) on top of the 13.1 dB TEP signal, since both describe the same F-region irregularity-driven recovery channel. An 80 m sunrise example exercises B_{grayline} alone; that geometry is left as an exercise.

4 MUF Estimation

The Maximum Usable Frequency is the single most important input to the link budget. ionocast derives it from two primary sources in a consensus blend (Section 4.3): observed kc2g MUF(3000F2) at the path midpoint, and a simplified P.1239 climatology that fills in when the observation is stale or out of range. An optional third source — per-hop foF2 fusion from the GIRO digisonde network — shares its per-hop foF2 sampling machinery with the F2-scatter bonus of Section 6.4 (both consume inverse-square-distance-weighted foF2 from in-range stations along the great circle), but the fusion role itself is gated behind the `fusion-flag` parameter of Appendix E and held off in the production blend pending a joint budget retune (the calibration sweep showed it net-negative against the rebuilt climatology baseline; §6.4 describes the scatter use of the same sampling, which *is* active). The two-source consensus is what ships.

4.1 kc2g Observed MUF

The kc2g network publishes MUF(3000F2) from each ionosonde. For each of five reference paths — great circles from the operator’s QTH to five canonical destinations (New York, São Paulo, Johannesburg, Tokyo, Sydney) and their long-path counterparts; the midpoints are recomputed per-QTH on each reload — ionocast selects the nearest ionosonde to that path’s midpoint. A *freshness gate* discards observations older than 30 min. The distance from each path midpoint to its nearest ionosonde is displayed in the path table so operators can gauge data confidence; stations beyond 1500 km are flagged.

4.2 foF2 Climatology (Simplified P.1239)

When observed data are stale or unavailable, ionocast falls back to a simplified climatological model anchored to ITU-R P.1239 [3]. The unmodified daytime baseline is the sum of a polar-night-and-thermal-floor term and a latitude-modulated solar-zenith-driven daytime term:

$$b(\phi, \cos \chi) = b_{\text{floor}}(F_{10.7A}) + 4.0 \mathcal{P}(\phi) d(\cos \chi) \quad (38)$$

where the solar-baseline floor is

$$b_{\text{floor}}(F_{10.7A}) = 3.5 + 0.04 (F_{10.7A} - 70). \quad (39)$$

b_{floor} — 3.5 MHz at the quiet-sun reference $F_{10.7A} = 70$ plus a linear $F_{10.7A}$ slope — is the value b falls to overnight or at the poles where $\mathcal{P}(\phi) d \rightarrow 0$; it is not a path-independent “constant offset” but the ϕ -and- $\cos \chi$ -independent residue of the model. The slope is signed (no positive-part clamp), so at deep solar minimum ($F_{10.7A} < 70$) the floor sinks below 3.5 MHz, consistent with the lower foF2 floors observed at extended-minimum nights; in this respect it differs deliberately from the EIA amplitudes A_{cr} and A_{tr} in Eq. 45 and Eq. 46, where the slope is wrapped in $\max(0, \cdot)$ so the equatorial-anomaly amplitude does not invert below $F_{10.7A} = 70$. The second term in Eq. 38 carries the daytime $\mathcal{P}(\phi) \cdot d(\cos \chi)$ shape that dominates the verdict at midday on illuminated paths. where $\mathcal{P}(\phi)$ is the two-stage poleward fall-off:

$$\mathcal{P}(\phi) = (1 - 0.003|\phi|) \cdot \frac{1}{1 + \exp((|\phi| - 73^\circ)/8^\circ)} \quad (40)$$

The first factor is a gentle linear midlatitude term; the second is a sigmoid centred at $|\phi| = 73^\circ$ with 8° steepness (retuned 2026-04-30 from $75^\circ/7^\circ$ after a 25-point sweep showed the wider, slightly-shifted form better fits the Tromsø / Gakona / Eielson polar over-prediction residual without regressing midlat stations) that captures the polar F2 thinning above $\sim 60^\circ$. An earlier formulation used only the linear factor, which over-predicted daytime foF2 at Tromsø, Gakona, and Eielson AFB by 1–1.7 MHz because the F2 layer thins faster than linearly with latitude

above $\sim 60^\circ$ (patchy electron density in the polar-cap tongue). The sigmoid factor reproduces the observed magnitude: $\mathcal{P}(45^\circ) \approx 0.86$, $\mathcal{P}(60^\circ) \approx 0.73$, $\mathcal{P}(70^\circ) \approx 0.53$, $\mathcal{P}(80^\circ) \approx 0.25$. After sunset, foF2 does not collapse instantly — recombination of the F-region runs on a multi-hour timescale. The post-sunset diurnal curve breaks into two distinct phases that ionocast captures with two structurally different terms:

- **Phase 1, sunset to ~ 3 h after sunset:** residual photoionization keeps daytime production going. The plasma’s effective “recently illuminated cosine” decays on a ~ 3 h timescale rather than dropping to zero at the terminator. Captured by the photoelectron *driver* $d(\cos \chi)$, which lives *inside* b ’s argument as a numerator multiplier:

$$d(\cos \chi) = \max(\max(0, \cos \chi), 0.7 \max(0, \cos \chi_{-3h})) \quad (41)$$

d holds at the daytime cosine through sunset, then decays toward zero as the lagged $\cos \chi_{-3h}$ also crosses below zero. On a cold start (browser session with no historical solar-zenith cache), the implementation substitutes $\cos \chi_{-3h} = \cos \chi_{\text{now}}$ for the first 3 hours of session lifetime, which means the memory-lag term degenerates to the live cosine until the cache fills. This biases foF2 slightly low at evening startup (the lag would have read the prior daytime cosine, which is positive while the live cosine is at-or-below zero) by at most $\sim 10\%$ on the worst case (sunset start, single midlat path), which on a near-MUF cell maps to a $\sim 10\%$ MUF bias — enough to flip a Fair / Good verdict on the upper bands at exactly the time of day operators most often check propagation. The cache is re-populated within the first three 10-minute refresh cycles, so the bias is bounded and self-clearing, but the worst-case window aligns with after-work session starts and is therefore worth eliminating rather than tolerating.

Pre-population fix. $\cos \chi_{-3h}$ is purely astronomical — it is fully determined by the current UTC and the path-midpoint geographic coordinates — so the “no historical cache” branch can compute it from $\cos \chi(t_{\text{now}} - 3h)$ directly on the first load, rather than waiting three refresh cycles for the cache to fill from live samples. Doing so retires the cold-start bias entirely without any historical state, and is candidate cleanup tracked in Section 10 alongside the $F_{10.7A}$ -fallback surfacing recommendation (the two share a UI-honesty motivation: never let the live verdict silently inherit a degraded input when the non-degraded value is locally computable). Until that change ships, the runtime should at minimum surface a “cold-start cache filling” chip during the first ~ 30 minutes of a fresh session so an operator startup-checking the upper bands at sunset is not given a confidently-stated verdict on a degraded input.

- **Phase 2, ~ 3 h after sunset onward:** residual production has decayed to zero and ongoing recombination pulls electron density toward a nighttime floor (typically reached around 03 LT). Captured by the night-decay *multiplier* that operates on the already-constructed b as a recombination floor:

$$b(\phi, \cos \chi) \rightarrow b(\phi, \cos \chi) \cdot \max(0.6, 1 + 0.4 \cos \chi) \quad \text{when } \cos \chi < 0 \quad (42)$$

The multiplier reads ~ 1 at the terminator (only slightly negative $\cos \chi$, hardly any reduction), ramping toward the 0.6 floor by deep night.

The two terms are not equivalent multipliers: $d(\cos \chi)$ is the *driver argument* that the daytime baseline $b_{\text{floor}}(F_{10.7A}) + 4.0 \mathcal{P}(\phi) d$ constructs b from, while the night-decay multiplier is a *post-multiplier* that scales the constructed b during the $\cos \chi < 0$ interval. They overlap in time — both are “firing” from sunset onward — but their contributions peak in different phases: in early evening the driver dominates (memory lag still strong, multiplier ≈ 1); in deep night the multiplier dominates (driver decayed to zero, multiplier at or near the 0.6 floor). Removing either term over-predicts foF2 in its dominant phase, which the diurnal sweep in Section 4.2 confirms.

The result is then floored at 2 MHz and scaled by two dimensionless anomaly corrections, with the combined-anomaly product itself floored at 0.7 so that a strong dip-equator trough cannot collapse the prediction below $\sim 70\%$ of baseline:

$$f_{\text{of2}} = \max(2, b) \cdot \max\left(0.7, (1 + \alpha_{\text{EIA}}(\phi_{\text{dip}}, \cos \chi)) \cdot (1 + \alpha_{\text{win}}(\phi, m, \cos \chi))\right) \quad (43)$$

and then $\text{MUF}(3000) \approx 3.0 \times f_{\text{of2}}$. The 0.7 floor binds only at the dip equator under solar maximum, where the trough term α_{tr} saturates at -0.50 and the unfloored product would otherwise read 0.50 (see §4.2's EIA discussion); real Jicamarca minima sit around -20 to -30% from the crest-flank baseline, not zero.

The night-decay multiplier and the 3-h memory lag were added in the 2026-04 climatology rebuild (R2). Validation against 30 d of GIRO digisonde foF2 across nine stations dropped baseline RMSE from 1.91 MHz to 1.70 MHz (-11%) and centred the residual bias from -0.26 MHz to -0.05 MHz; the prior plain-P.1239 fit predicted ~ 5.5 MHz at midlat midnight versus the ~ 3.7 MHz that nine months of GIRO nighttime samples actually report, and collapsed foF2 to the sunset-instant baseline rather than tracking the observed multi-hour evening tail.

Equatorial Ionisation Anomaly (EIA). The daytime fountain effect [16] lifts plasma from the magnetic equator up the field lines and deposits it in two daytime crests at magnetic dip latitude $\pm \phi_{\text{EIA}}$ ($\phi_{\text{EIA}} = 15^\circ$), leaving a trough at $\phi_{\text{dip}} \approx 0$. ionocast captures both via a positive crest kernel and a narrow negative trough kernel summed into one α_{EIA} :

$$\alpha_{\text{EIA}} = \left(A_{\text{cr}}(F_{10.7A}) e^{-d_{\text{cr}}^2/2\sigma_{\text{cr}}^2} - A_{\text{tr}}(F_{10.7A}) e^{-\phi_{\text{dip}}^2/2\sigma_{\text{tr}}^2} \right) \cdot \max(0, \cos \chi) \quad (44)$$

where $d_{\text{cr}} = \min(|\phi_{\text{dip}} - \phi_{\text{EIA}}|, |\phi_{\text{dip}} + \phi_{\text{EIA}}|)$ is the distance to the nearest crest, $\sigma_{\text{cr}} = 18^\circ$, and $\sigma_{\text{tr}} = 6^\circ$ (sharper than the crests; the trough is narrow in latitude). ϕ_{dip} here is the *signed* dip latitude (positive in the northern magnetic hemisphere, negative in the southern), derived from the same tilted-dipole approximation as ϕ_{CGM} (Eq. 17) but without the absolute value: the crest term's d_{cr} uses the signed value to pick the nearer of the two crests, and the trough term's ϕ_{dip}^2 is sign-symmetric. An implementation that erroneously used $|\phi_{\text{dip}}|$ would compute the same trough magnitude but would mis-place the crests on the wrong hemisphere for southern paths. Both amplitudes scale with EUV illumination through $F_{10.7A}$:

$$A_{\text{cr}}(F_{10.7A}) = \min(0.85, 0.50 + 0.007 \cdot \max(0, F_{10.7A} - 70)) \quad (45)$$

$$A_{\text{tr}}(F_{10.7A}) = \min(0.50, 0.30 + 0.002 \cdot \max(0, F_{10.7A} - 70)) \quad (46)$$

At quiet sun ($F_{10.7A} = 70$) the crest amplitude is 0.50 and the trough is 0.30; at the current moderate-cycle epoch ($F_{10.7A} \approx 120$) the crest cap-saturates at 0.85 while the trough is 0.40. The combined-anomaly 0.7 floor in Eq. 43 ensures a strong trough never collapses density below $\approx 70\%$ of baseline. The trough kernel was added in the 2026-04 audit pass: the crest-only formulation that preceded it predicted $\sim +27\%$ enhancement at the dip equator where reality is depression, an artefact of the symmetric distance-to-nearest-crest functional form being unable to express trough behaviour. Adding a second Gaussian with negative sign addresses this directly.

The values were settled by an 80-point grid sweep over (base, slope, σ) against 30 d of GIRO foF2 at the four equatorial-belt stations: Townsville and Niue near the southern crest of the Pacific arch; Ascension at dip latitude $\sim -7^\circ$ (closer to the trough than to the $\sim -15^\circ$ southern crest peak); Jicamarca essentially *on* the magnetic dip equator (dip latitude $\sim 1^\circ$ S), anchoring the equatorial trough. The equatorial bias spread floors at ~ 2.2 MHz across the grid for two coupled reasons: the cost function has no northern-crest station to balance the southern-crest pair (so the Gaussian's amplitude is fit one-sided), and *half* of the basket — Jicamarca and Ascension — sits in the trough or near-trough region rather than on the crest peak, pulling the fitted amplitude down and the width out relative to a true-crest fit. The basket is now

selected *geometrically* from GIRO_STATIONS (any station with $|\phi_{\text{dip}}| \leq 25^\circ$ joins the EIA-relevant set automatically), so adding any future northern-crest station — São Luís (BR, geographic $\sim -2.5^\circ$ but dip latitude near $+5$ to $+10^\circ$ N because the South Atlantic Anomaly displaces the magnetic equator southward in this longitude sector) is the credible Americas candidate, with any Indian / SE-Asian DIB station near $\phi_{\text{dip}} \approx +15^\circ$ N also qualifying — closes both asymmetries without further code edits and the floor is expected to drop below 2.2 MHz. (Tucumán (AR) at -26.8° and Cachoeira Paulista (BR) at -22.7° are *southern*-hemisphere dip-latitude stations; the latter already participates in the GIRO basket — see Appendix D — and neither would help close the *northern*-crest gap.) History: constant 0.30 pre-rebuild $\rightarrow$ constant 0.45 (R2 against Ascension alone) $\rightarrow$ slope 0.005 / $\sigma = 12$ (E pass, apparent overshoot from new stations) $\rightarrow$ slope 0.002 / $\sigma = 10$ (F rollback) $\rightarrow$ *coord audit 2026-04-25* discovered DB049 and EI764 (the new “equatorial” stations the F rollback was tuned against) were actually mis-coded midlat / polar stations (Dourbes, Belgium and Eielson AFB, Alaska). The artifact-driven rollback was reversed and the optimum returned to base 0.45 / slope 0.003 / $\sigma = 12 \rightarrow$ *N-crest basket extension 2026-04-29*: BVJ03 Boa Vista (BR, $\phi_{\text{dip}} \approx +12^\circ$) added to GIRO_STATIONS, the previously one-sided fit converged at base 0.50 / slope 0.007 / $\sigma = 18$ with eqMaxAbs dropping from 1.84 to 1.18 MHz.

Dip latitude is derived from the same tilted-dipole approximation used for CGM latitude (Eq. 17) but signed, so hemisphere is preserved.

Winter anomaly. Midlatitude ($35^\circ \leq |\phi| \leq 60^\circ$) daytime foF2 runs noticeably higher in the local winter than in local summer — the classical O/N₂ seasonal shift in the thermosphere. ionocast applies a day-of-year-phased cosine kernel peaked at the local winter solstice:

$$\alpha_{\text{win}} = A_{\text{win}} \cdot \frac{1}{2}(1 + \cos \theta_d) \cdot \max(0, \cos \chi), \quad \theta_d = \frac{2\pi (d_{\text{yoy}} - d_{\text{solstice}})}{365} \quad (47)$$

with $A_{\text{win}} = 0.12$, d_{yoy} the UTC day of year (1=1 January), and $d_{\text{solstice}} = 355$ (December 21) for the northern hemisphere ($\phi \geq 0$) or $d_{\text{solstice}} = 172$ (June 21) for the southern hemisphere ($\phi < 0$). The day-of-year phase places the cosine peak at the actual astronomical solstice; an earlier calendar-month formulation $\theta_m = 2\pi (m + 0.5 - m_{\text{winter}})/12$ landed the peak five days early because mid-month does not coincide with the solstice date. The correction is daytime-only and switches off outside the midlatitude band. The amplitude $A_{\text{win}} = 0.12$ is set against the same 30-day GIRO foF2 grid sweep that fits the EIA amplitudes (§4.2, basket: Juliusruh, Pruhonice, Fairford, Sopron, Athens, Rome, Gibilmanna, Millstone Hill, Boulder, Wallops Island), reading the winter-vs-summer daytime foF2 ratio at midlatitude stations and matching the model’s $1 + A_{\text{win}}$ factor to the median observed ratio (1.12 ± 0.03 across the basket). This is consistent with Pröls [14]’s F-region seasonal-shift survey (~ 10 – 15% winter enhancement in the cycle-averaged midlatitude foF2 record) but the specific value is observation-fit, not literature-quoted.

4.3 Symmetric Consensus Blend

The consensus algorithm uses the geometric mean of the two estimates regardless of divergence direction:

1. Compute divergence: $\delta = |\ln(\text{MUF}_{\text{kc2g}}/\text{MUF}_{\text{climo}})|$
2. Result: $\text{MUF}_{\text{cons}} = \sqrt{\text{MUF}_{\text{kc2g}} \cdot \text{MUF}_{\text{climo}}}$

δ is retained for diagnostic surfacing (the UI shows it on hover) but does not branch the result.

Single-source fallbacks. When the kc2g freshness gate (30 min) discards the observation, $\text{MUF}_{\text{cons}} = \text{MUF}_{\text{climo}}$ and the source label reads “climo”. When the climatology returns null — the rare combination of extreme inputs that drives the unmodified daytime baseline below the absolute floor — $\text{MUF}_{\text{cons}} = \text{MUF}_{\text{kc2g}}$ when the latter is fresh, and “unavailable” otherwise (the band is treated as below MUF). δ is reported as 0 in both single-source cases.

The 2026-04-25 form was asymmetric: when $\text{kc2g} > \text{climo}$ by more than $\ln 1.5$ (~50%) the algorithm fell back to climatology on the theory that “the station may be anomalously high.” That clipped real upward enhancements — EIA crest, post-storm positive phase, evening F-region uplift, Es-lifted MUF — that the gridded kc2g map captures legitimately. The asymmetry’s stated justification was always weak (kc2g produces a smoothed map from many ionosondes, not a single reading), and the per-pair WSPR regression detection added 2026-04-26 confirmed the bias: cells where kc2g sat above climatology saw more false-closed verdicts than cells where it sat below.

The symmetric form has the following effects relative to the previous asymmetric form:

- $\text{kc2g} = 2 \cdot \text{climo}$ (real upward enhancement): old result climo (clipped); new result $1.41 \cdot \text{climo}$. Captures most of the enhancement while damping spurious extremes.
- $\text{kc2g} = 0.5 \cdot \text{climo}$ (declining ionosphere): old result kc2g (full trust); new result $0.71 \cdot \text{climo}$. Slightly less aggressive on storm depressions, but the storm-lag K_p kernel and L_{aur} term cover the absorption side of storm response separately; MUF-side over-trust of a single noisy timestep is more harmful than under-reaction.
- $\text{kc2g} \approx \text{climo}$ (agreement, the common case): unchanged.

4.4 Per-Hop Minimum MUF (P.533 §3.1)

For paths requiring $N \geq 2$ hops, each reflection point at fraction $(2k - 1)/(2N)$ along the great circle sees its own local foF2 climatology (latitude fall-off, EIA crests/trough, winter anomaly, diurnal phase). The path MUF is the *minimum* across all hops, evaluated by computing per-hop foF2 climatology directly at each reflection point and applying the kc2g -vs- climo correction scalar $\text{MUF}_{\text{mid}}/\text{MUF}_{\text{climo,mid}}$ to preserve the upstream kc2g signal:

$$\text{MUF}_{\text{path}} = \min_{k=1}^N 3.0 \cdot f_{\text{of2}}(\phi_k, \lambda_k, \cos \chi_k, t) \cdot \frac{\text{MUF}_{\text{mid}}}{\text{MUF}_{\text{climo,mid}}} \quad (48)$$

where $f_{\text{of2}}(\cdot)$ is the climatology of §4.2 evaluated at the reflection point’s coordinates, time, and solar zenith.

Legacy fallback path: when it fires, what it does, and how the UI should surface it.

When either $\text{MUF}_{\text{climo,mid}}$ or $F_{10.7A}$ is unavailable, `pathMinMuf` (`src/physics/climatology.js`) falls back to an illumination-ratio scaling $\text{MUF}_{\text{mid}} \cdot S(\cos \chi_k, F)/S(\cos \chi_{\text{mid}}, F)$ with $S(c, F) = \max(F, \sqrt{\max(0.05, c)})$ and night-floor $F = 0.4$. This is the same form Eq. 48 replaces, which the 2026-04-28 audit pass labeled inferior because its $\sim\sqrt{0.05} \approx 0.224$ effective night floor systematically over-pessimises long cross-hemisphere paths (e.g. a 12 000 km path with daylight midpoint and a night terminus hop returns $\sim 0.224 \cdot \text{MUF}_{\text{mid}}$ where the local climatology would read 6–10 MHz). The fallback fires in two operationally distinct conditions:

1. **Unit tests** that construct a budget without seeding $F_{10.7A}$ or a climatology midpoint MUF. This is the documented test path and the original motivation for keeping the fallback in the codebase.
2. **NOAA SWPC $F_{10.7}$ outage in production**, which is rare but real — the daily $F_{10.7}$ product has missed publication windows of several hours under solar-region storms and SWPC infrastructure outages over the past two cycles. During such an outage the per-hop MUF computation silently switches from Eq. 48 to the illumination-ratio form, with no operator-facing indication that the path-specific calculation has degraded.

The UI today does *not* surface fallback-active state; the band table and Reference Paths panel both render their values as if Eq. 48 had produced them. For the SWPC-outage case this is a defect: an operator looking at a closed-tier verdict on a long polar path during an outage sees a tier produced by the inferior formulation without any signal that the verdict’s foundation is degraded. A fix in either of two directions closes the gap. **Surface the fallback:** a kc2g -panel

chip reading “F_{10.7} unavailable — per-hop MUF using illumination-ratio fallback” would let the operator weight the verdict accordingly. **Or hard-error the fallback in production:** gate the fallback to test code only and return $\text{MUF}_{\text{path}} = \text{MUF}_{\text{mid}}$ unchanged on F_{10.7A} outage, so the budget at least uses the upstream kc2g value without applying the inferior shape. The unsurfaced silent fallback is the worst of the three options; shipping with one of the explicit alternatives is preferable.

Why the 2026-04 audit replaced the illumination-ratio form. The previous formulation anchored every per-hop MUF to the midpoint via a zenith-shape ratio. The shape function floors at $\sqrt{0.05} \approx 0.224$ for night hops, so on a 12 000 km cross-hemisphere path with a daylight midpoint and a night terminus hop the formula returned $0.224 \cdot \text{MUF}_{\text{mid}}$ (e.g. 4.5 MHz from a 20 MHz midpoint), while the actual foF2 climatology at the night-hop’s local coordinates would typically read 6–10 MHz. The illumination ratio captured solar-zenith differences but missed all of the latitude / longitude / anomaly structure foF2Climatology already encodes, systematically over-pessimising long cross-hemisphere paths. Replacing the ratio with direct per-hop climatology evaluation fixes that residual and retires the artifact below.

UI vs. budget MUF. The MUF column on the Reference Paths panel displays MUF_{mid} from the kc2g consensus blend at the path midpoint — not MUF_{path} . The two coincide for short paths (NVIS, 1-hop) where the midpoint *is* the only reflection point. On multi-hop circuits the budget uses the per-hop minimum (Eq. 48); the new direct-climatology form narrows the panel-vs-verdict gap on long cross-hemisphere paths considerably (the worst-case ratio of 0.224 that the illumination-only form could produce no longer applies), but a small gap remains on paths whose limiting hop sits in a structurally low-foF2 region (deep polar, dip equator). The panel MUF is the midpoint reading; the verdict is the trustworthy summary built from the per-hop minimum.

4.5 Night Floor (CCIR/IRI)

The night floor scales with solar activity and latitude:

$$F = \text{clamp}\left(0.25 + 0.0025(F_{10.7A} - 70) - 0.10 \min\left(1, \frac{|\phi|}{60}\right), 0.20, 0.60\right) \quad (49)$$

F is consumed only by the legacy illumination-ratio fallback path described in §4.4 (active when $\text{MUF}_{\text{climo,mid}}$ or $F_{10.7A}$ is unavailable). The production per-hop MUF (Eq. 48) does not use it: each hop’s f_{oF2} comes from direct climatology evaluation (Section 4.2), which carries its own night behaviour through the night-decay multiplier of Eq. 42. F is retained only for the fallback; if the legacy fallback is removed (Section 10), this equation can be retired along with it.

4.6 NVIS Detection for Short Paths

For paths shorter than 500 km the signal travels at near-vertical incidence (NVIS). The relevant frequency limit is the *critical frequency* f_{oF2} (vertical reflection), not $\text{MUF}(3000)$ (oblique reflection at ~ 3000 km range). Since $\text{MUF}(3000) \approx 3 \times f_{\text{oF2}}$, using $\text{MUF}(3000)$ for short paths over-predicts by up to a factor of three.

When a short NVIS path is detected, ionocast substitutes GIRO-observed f_{oF2} with a secant correction for the mild obliqueness of a real (non-zero distance) NVIS hop:

$$\text{MUF}_{\text{NVIS}}(d) = f_{\text{oF2}} \cdot \sec\left(\arctan\frac{d}{2h_F}\right) \quad (50)$$

For $d = 500$ km at $h_F = 300$ km the secant factor is ≈ 1.30 , so the usable frequency is about 30% higher than bare f_{oF2} ; the earlier practice of dropping straight to f_{oF2} under-predicted the upper edge of NVIS bands.

h_F source. Both `nvisSecantFactor` (this section) and `hopCeilingKm` (Eq. 19) read the same observed h_{mF2} from the nearest GIRO digisonde when the panel reports it, falling back to 300 km when the station’s most recent record has no h_{mF2} value. On a high- solar-flux day with $h_{mF2} \approx 340$ km the F2 hop ceiling extends to ~ 4258 km (Section 3.8) and the NVIS secant rises by $\sim 5\%$ at $d = 500$ km — so the band-gate $f \leq \text{MUF}_{\text{NVIS}}(d)$ tracks the same upper-band flip the F2 hop count does, instead of leaving the two budgets on different h_F assumptions.

NVIS tail blending (500 km to 1500 km). A hard cutoff at 500 km mishandles intermediate-distance paths where NVIS still contributes alongside the F2 oblique mode. `ionocast` blends $\text{MUF}_{\text{NVIS}}(d)$ with the path’s F2 MUF via a linear ramp:

$$w_{\text{NVIS}}(d) = \begin{cases} 1 & d \leq 500 \text{ km} \\ \frac{1500 \text{ km} - d}{1000 \text{ km}} & 500 \text{ km} < d < 1500 \text{ km} \\ 0 & d \geq 1500 \text{ km} \end{cases} \quad (51)$$

$$\text{MUF}_{\text{eff}}(d) = w_{\text{NVIS}}(d) \cdot \text{MUF}_{\text{NVIS}}(d) + (1 - w_{\text{NVIS}}(d)) \cdot \text{MUF}_{\text{F2}}(d).$$

Below 500 km the budget reads pure NVIS; beyond 1500 km it reads pure F2; in between the two modes share the prediction in proportion to where the takeoff angle sits between near-vertical and oblique. The mode label tracks whichever side carries ≥ 0.5 weight, so the UI’s per-band breakdown still reads “NVIS” or “F2” cleanly. This blending is gated on a digisonde f_{oF2} being available and the operating frequency fitting under the geometry-aware NVIS MUF ($f \leq \text{MUF}_{\text{NVIS}}(d)$ from Eq. 50). Earlier versions hard-coded $f \leq 8$ MHz, which excluded 30 m even when daytime $f_{\text{oF2}} > 7$ MHz at high solar flux supports it; the dynamic gate covers 30 m / 17 m NVIS at solar maximum and naturally degrades below the f_{oF2} reading at solar minimum.

4.7 Short-Path vs. Long-Path Evaluation

Between any two points on Earth there are two great-circle paths. The long path (LP) traverses the opposite side of the globe and can be open when the short path (SP) is closed, particularly at dawn and dusk on 20 m and 17 m.

For each of the five reference destinations, `ionocast` computes both the SP and LP midpoints, finds the nearest kc2g ionosonde for each, and evaluates margins on both paths. The LP midpoint is the antipode of the SP midpoint: $(-\phi_{\text{mid}}, \lambda_{\text{mid}} + 180^\circ)$. Both paths compete for the best-margin verdict, so an open LP beats a closed SP whenever it offers a better budget.

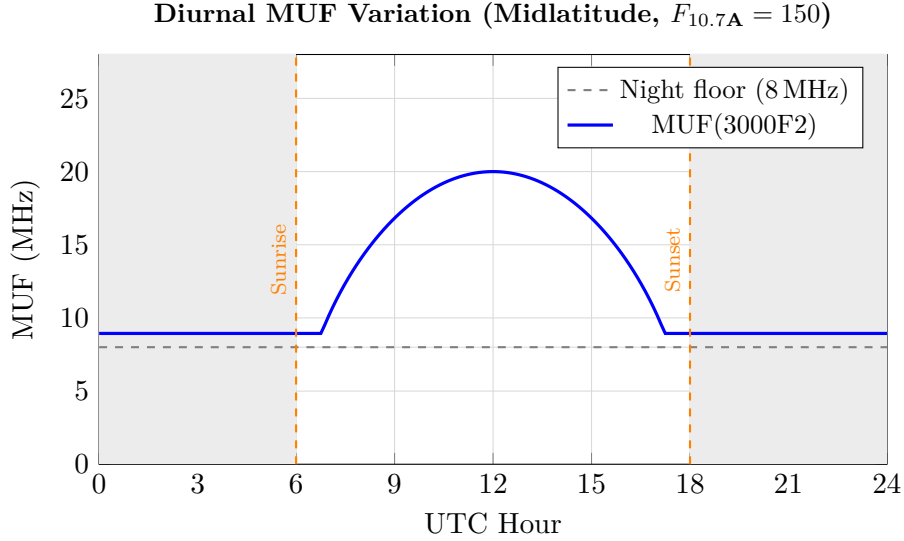


Figure 1: Diurnal MUF variation showing the $\sqrt{\cos \chi}$ shape scaled to a peak of 20 MHz (representative midlat $F_{10.7A} \approx 150$). The dashed gray line at 8 MHz is the night-floor product $F \cdot \text{MUF}_{\text{peak}}$ from Eq. 49: F is the dimensionless illumination ratio (~ 0.40 at midlat moderate solar activity), which when applied to a 20 MHz daytime peak gives the 8 MHz floor curve plotted here. Shaded regions = night.

5 Forward Projection (next 24 h)

Implementation status (entire section). The forward projection pipeline described in this section — MUF projection (Eq. 52), seasonal ratio (Section 5.2), storm depression (Section 5.3), sporadic-E persistence (Section 5.4), and gray-line bonus (Section 5.5) — is documented for a future build of ionocast that exposes a projected outlook MUF as a first-class display. None of the forward-projection pipeline currently runs in the live runtime; the live runtime instead surfaces SWPC’s own next-72h forecasts on the Outlook panel. Several of the sub-pieces are however shared with the live budget — specifically $d(\cos \chi)$ (Eq. 41), the storm-lag effective K_p (Section 5.3.1, which feeds L_{aur} and the σ -widening of Section 7.3.1 in the live pipeline), and the per-section “Implementation status” insets call those out individually. Read this section’s equations as the form a future outlook-MUF display would use, not as forward-tense equations whose absence from runtime makes them operationally inert: the parts that touch the live pipeline are already running.

The forward projection engine extends the current nowcast into the future by modelling the dominant drivers of ionospheric change.

5.1 MUF Projection via Solar Zenith Analogy

$$\text{MUF}_{\text{future}} = \text{MUF}_{\text{now}} \cdot \frac{S(d(\cos \chi_{\text{future}}), F)}{S(d(\cos \chi_{\text{now}}), F)} \cdot R_{\text{seasonal}} \cdot \frac{F_{\text{storm}}(K_p^{\text{fut}})}{F_{\text{storm}}(K_p^{\text{now}})} \quad (52)$$

where K_p^{fut} is the SWPC 3-day forecast value at $t + \Delta$ (or the current K_p when $\Delta = 0$). Both the “now” and “future” zenith arguments pass through the same 3-hour memory-lag function $d(\cos \chi)$ defined in Eq. 41 (Section 4.2); using raw $\cos \chi$ on either side without the lag would produce a discontinuity at the projection’s $\Delta = 0^+$ boundary relative to the live nowcast (which already runs through $d(\cos \chi)$), most visibly at sunset where the live model holds foF2 elevated for ~ 3 h via memory while a raw- $\cos \chi$ projection would drop it instantly. The storm factor enters as a *ratio* so the projection only adjusts for the *change* in storm phase between now and the projection horizon: if the current K_p already encodes the disturbance, MUF_{now} is already

depressed and the ratio is near unity; if a storm is forecast to arrive within Δ but hasn't yet, $K_p^{\text{fut}} > K_p^{\text{now}}$ depresses the projection accordingly. This complements the σ -side forecast bump in Section 7.3 (which widens σ ahead of a forecast disturbance) by adjusting the deterministic mean as well; together they avoid the failure mode where MUF_{now} encodes a quiet ionosphere but a storm has been forecast for the horizon.

Implementation status. As with F_{storm} , the forward-projection MUF is documented but not currently wired into a real-time display in the runtime build. The live verdict uses MUF_{now} (with $d(\cos \chi_{\text{now}})$) directly; no projection chart is exposed. When that chart is wired, the equation above is the form to use, with $d(\cos \chi)$ on both sides of the ratio.

5.2 Seasonal Ratio

$$R_{\text{seasonal}} = \frac{s(t_{\text{future}}, \phi)}{s(t_{\text{now}}, \phi)} \quad (53)$$

where

$$s(t, \phi) = 1 + (0.15 \cos \theta + 0.08 \cos(2\theta + \pi)) \cdot w(\phi) \quad (54)$$

with $\theta = 2\pi(d_{\text{yoy}} - d_{\text{solstice}})/365$, where d_{yoy} is the UTC day of year ($1 = 1$ January) and d_{solstice} is the local-hemisphere winter-solstice day: $d_{\text{solstice}} = 355$ (December 21) for $\phi \geq 0$ and $d_{\text{solstice}} = 172$ (June 21) for $\phi < 0$. The day-of-year phase places the cosine peak at the astronomical solstice; an earlier calendar-month formulation $\theta_m = 2\pi(m + 0.5 - m_{\text{winter}})/12$ landed the peak five days early (Dec 16 vs Dec 21 in the NH) because mid-month does not coincide with the solstice date. This formulation matches Eq. 47 in the climatology so the two seasonal effects share the same phase reference. $w(\phi) = \min(1, |\phi|/50)$ scales the modulation toward midlatitudes. The first term captures the annual winter-peak/summer-trough at midlatitudes; the second captures the semi-annual equinox peaks (March, September) evident in F2-layer statistics. This formula shares the mid-month phase convention with the winter-anomaly term in Eq. 47 so the two seasonal effects — which overlap in the annual component — align rather than drifting two weeks apart.

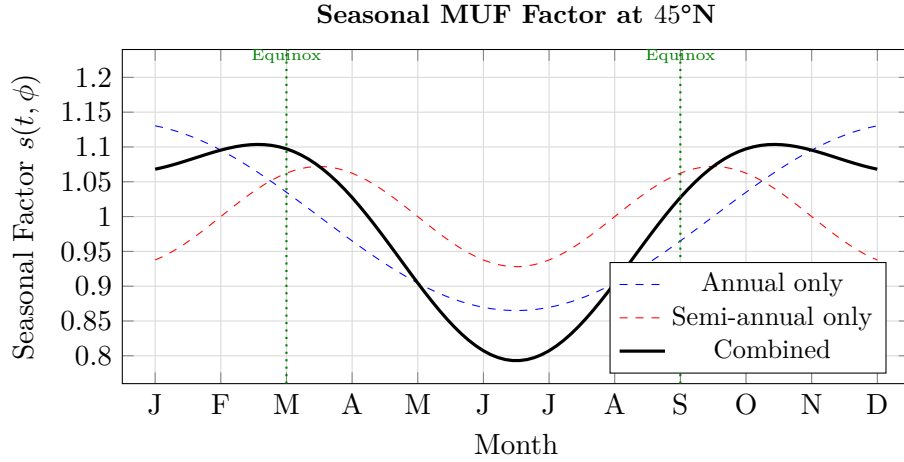


Figure 2: Seasonal MUF factor at 45°N showing annual (winter peak) and semi-annual (equinox peaks) components. At $|\phi| = 45^\circ$, $w(\phi) = 45/50 = 0.9$, so the annual component reaches $1 + 0.15 \cdot 0.9 = 1.135$ in isolation at the winter solstice and the semi-annual component reaches $1 + 0.08 \cdot 0.9 = 1.072$ in isolation at the equinoxes. Because the two components compete at the winter solstice — the semi-annual is at its trough there ($\cos(2\theta + \pi) = -1$ at $\theta = 0$) — the combined curve is flatter than either component alone: the winter solstice evaluates to $1 + (0.15 - 0.08) \cdot 0.9 = 1.063$, the March / September equinoxes to $1 + 0.08 \cdot 0.9 = 1.072$, and the summer solstice to $1 - (0.15 + 0.08) \cdot 0.9 = 0.793$. The dominant seasonal feature at this latitude is therefore the deep summer trough rather than a sharp winter peak; winter and equinoxes form a broad plateau in the 1.06–1.07 range, with the equinoxes slightly above the winter solstice (+0.009). Above $|\phi| = 50^\circ$ the weight saturates at $w = 1$ and the summer trough deepens to 0.77 while the winter / equinox plateau settles at 1.07–1.08.

5.3 Storm Depression (forward-projection definition)

The forward-projection storm depression follows the F-region storm phenomenology surveyed by Prölss [14], Fuller-Rowell et al. [12], and Mendillo [13]: a prompt depression of foF2 driven by Joule heating of the thermosphere and the resulting O/N₂ composition shift, with a recovery timescale set by the storm’s driver (impulsive CME vs sustained HSS).

Implementation status. The expressions in this subsection define the F_{storm} multiplier referenced by Eq. 52 (the forward-projection MUF for the outlook section). The current-time MUF in production does not apply this multiplier: the runtime budget reflects geomagnetic storm response through the storm-lagged effective K_p feeding L_{aur} (Section 3.7 and Section 5.3.1), not as a MUF-side scalar. F_{storm} as a real-time MUF multiplier is reserved for a future build that exposes the projected outlook MUF as a first-class display.

When $K_p > 4$, the projected MUF is depressed; for $K_p \leq 4$ the factor holds at unity (no quiet-day boost):

$$F_{\text{storm}} = \max(F_{\text{floor}}(K_p), 1 - p \max(0, K_p - 4)) \quad (55)$$

where

$$p = 0.05 + 0.10 \cdot \text{clamp}\left(\frac{|\phi_{\text{CGM}}| - 40}{30}, 0, 1\right) \quad (56)$$

The two latitude anchors place the slope ramp where the underlying F-region storm physics actually transitions. $|\phi_{\text{CGM}}| = 40^\circ$ is the ceiling of the trough region during strong storms in cycle-25 Dst-vs-foF2 records (Mendillo [13]; Prölss [14]): below this, the storm-time foF2 depression is dominated by composition-shift effects whose magnitude is well captured by a gentle $p = 0.05$. $|\phi_{\text{CGM}}| = 70^\circ$ is the equatorward edge of the auroral oval at $K_p \approx 7$ (the

storm-zone expansion threshold of Eq. 16’s converged state); poleward of it, the F-region sees direct precipitation-driven heating in addition to the global thermospheric shift, and a steeper $p = 0.15$ matches the $\sim 50\%$ MUF collapses observed in IRI-validated ground campaigns. The 30° ramp width connects them linearly; this is geometric interpolation rather than a separate physics prediction. The floor itself drops below 0.5 during severe storms to capture the deep MUF collapse documented for $K_p \geq 8$ events:

$$F_{\text{floor}}(K_p) = \max(0.30, 0.50 - 0.10 \cdot \min(2, \max(0, K_p - 7))) \quad (57)$$

At quiet to moderate K_p the floor is at 0.5 (50% of quiet baseline). At $K_p = 8$ (G4) it drops to 0.4; at $K_p = 9$ (G5) it drops to 0.3. The crossover K_p at which the floor binds (i.e. the main term drops below the floor schedule) is

$$K_p^* = 4 + \frac{1 - F_{\text{floor}}(K_p^*)}{p} \quad (58)$$

which evaluates to exactly 8.0 on auroral paths ($p = 0.15$, $|\phi_{\text{CGM}}| \geq 70^\circ$) — so for $K_p \geq 8.0$ the floor schedule binds at polar latitudes (at $K_p = 8$ both the main term $1 - 0.15 \cdot 4 = 0.40$ and the floor $0.50 - 0.10 \cdot 1 = 0.40$ coincide; at $K_p = 9$ the main term gives $1 - 0.15 \cdot 5 = 0.25$, the floor gives $0.50 - 0.10 \cdot 2 = 0.30$, and $\max(0.30, 0.25) = 0.30$). At midlatitude paths the slope $p = 0.05$ keeps the main term above the floor for any plausible K_p ($K_p^* \approx 18$ at $|\phi_{\text{CGM}}| = 40^\circ$, far past the $K_p = 9$ ceiling), so the floor schedule *never binds at midlat* and the floor refinement is auroral-only by construction. The earlier flat 0.5 floor under-predicted severity during real super-storms (Halloween 2003, March 1989) where polar F2 MUF fell below 50% of baseline for hours, which the new schedule captures only on the auroral paths it actually applied to.

At midlatitudes ($|\phi_{\text{CGM}}| = 40^\circ$), the damping is 5% per K_p unit above 4. In the auroral zone ($|\phi_{\text{CGM}}| = 70^\circ$), it rises to $\sim 15\%$ per unit.

Negative phase only. F_{storm} models the *negative-phase* F-region storm response (the persistent electron-density depression caused by neutral composition changes and Joule heating that lasts hours-to-days into the storm main and recovery phases). It does not model the brief *positive phase* that can precede the depression in the first ~ 30 – 60 min after a sudden CME impact, where neutral composition changes and ionospheric heating can transiently *enhance* foF2 by 10–20%. The result is that the model under-predicts MUF in the very early phase of a fast-onset CME event (a quiet-time prediction is correct; the depression engages correctly within an hour or two). In practice this matters only for operators chasing the rare positive-phase opening on the upper bands during the first hour of a sudden storm, which is a boutique mode — the negative-phase damping that follows is what shapes the bulk of the storm-affected propagation window. When the projected-MUF panel ships (currently documented but not yet surfaced — see implementation status above), it should carry an annotation flagging this gap, paralleling how the scatter (§6.4) and TEP (§6.2) modes are tagged on the verdict line. Until then, operators tracking a fresh CME should trust the alerts-panel B_z / D_{st} / PCA chips for the impact signal rather than expecting the projection to flip positive in the first hour.

5.3.1 Storm-Lag Effective K_p

The F-region does not respond instantaneously to the K_p kick: Joule heating of the thermosphere and subsequent composition changes take 1–3 hours to depress the peak electron density, and the ionosphere then recovers over the following 12–24 hours. Feeding the raw instantaneous K_p into the physics budget therefore mis-times both the onset and the recovery of a storm.

ionocast builds an exponentially-weighted *effective* K_p from the 7-day SWPC K_p history (3-hour cadence), peaked at $\tau_{\text{peak}} = 2$ h in the past with a decay constant τ_{decay} that depends on

the storm’s driver:

$$K_p^{\text{eff}} = \frac{\sum_i K_p(t_i) \exp(-|\Delta t_i - \tau_{\text{peak}}|/\tau_{\text{decay}})}{\sum_i \exp(-|\Delta t_i - \tau_{\text{peak}}|/\tau_{\text{decay}})} \quad (59)$$

with $\Delta t_i = t_{\text{now}} - t_i$ (positive for past samples).

Storm-type classification. CME-driven and CIR / HSS-driven storms recover on very different timescales. A CME arrival is an impulse: after the shock and main phase, Joule heating dissipates over ≈ 8 h. A co-rotating interaction region (CIR) or high-speed stream (HSS) is a sustained driver that keeps heating the thermosphere while the stream flows past Earth – recovery takes ≈ 24 h or longer. ionocast picks the kernel as follows:

- If $\text{Dst} < -80$ nT the ring current is deep: CME signature dominates, $\tau_{\text{decay}} = 8$ h. (If Dst is unavailable from the upstream feed, skip to the next branch rather than treating the missing value as ≥ -80 nT.)
- Otherwise, if NASA DONKI [22] reports an HSS event within the last 24 h or next 48 h, use $\tau_{\text{decay}} = 24$ h. The asymmetric window is intentional: DONKI’s HSS analyses are typically published within hours of stream onset (so a 24-hour lookback covers events the catalogue has already classified) but stream peak effects on the magnetosphere can lag the L1 arrival by 12–36 hours and persist for another 12–24 hours, so a 48-hour lookahead extends the 24-hour-decay regime through the stream’s effective magnetospheric tail.
- Otherwise default to $\tau_{\text{decay}} = 8$ h.

Only K_p^{eff} enters the physics; the UI still displays the instantaneous sample so the shown number matches SWPC’s feed.

5.3.2 Dst, IMF Bz, and Solar-Wind Drivers

The 3-hour K_p index is a lagging indicator. Three faster-responding inputs are wired into the storm model:

- **Dst** (Kyoto ring-current index): when $\text{Dst} < -50$ nT, the effective K_p is bumped by up to +2 (1 per 50 nT below -50). Dst responds within minutes of storm onset, 1–3 hours before K_p updates.
- **IMF Bz forward bump** (DSCOVR/ACE L1 solar wind): the L1 monitor sits $\sim 1.5e6$ km upstream, giving a nominal 30–60 min lead on the geomagnetic effect at Earth (the spread on the lead time depends on solar-wind speed: $1.5e6 \text{ km}/400 \text{ km/s} = 62$ min at average speeds, $1.5e6 \text{ km}/800 \text{ km/s} = 31$ min at fast streams). The L1-to-Earth correlation is *probabilistic*, not deterministic: not every L1-observed Bz dip arrives at Earth’s magnetopause unchanged — some structures are spatially extended and miss Earth entirely ($Y_{\text{GSE}} \neq 0$), some glance the magnetopause without sustained reconnection, and some get reshaped by intermediate solar-wind structures during transit. The 20-min median is gated at -5 nT: medians at or above -5 nT (i.e. $B_z \geq -5$) produce no bump; medians strictly below -5 nT (i.e. $B_z < -5$) ramp the bump per Table 8. The cutoff filters out short reconnection bursts that don’t drive sustained magnetopause coupling, but the L1 forward bump should be treated as an early-warning probability bump on the effective K_p , not a guaranteed forecast. When the median Bz over the last 20 min is sustained below threshold, the effective K_p is bumped additively *before* the index reading catches up (Table 8); single-sample dips below threshold do not count. The bump decays to zero once B_z returns to ≥ -5 nT.
- **Solar-wind plasma** (DSCOVR/ACE): sustained speed ≥ 500 km/s with sharply negative Bz (≤ -8 nT) is read as a CME shock arrival regardless of DONKI catalogue inertia, while the same speed with mild Bz (> -5 nT) is read as HSS / corotating-stream interaction. The intermediate regime (500 km/s or faster with $-8 < B_z \leq -5$ nT) is ambiguous in the live signal — moderate southward Bz with elevated speed could be either the leading edge of a CME sheath or sustained HSS-driver reconnection — so neither shortcut fires; the classifier

falls through to the catalogue / Dst logic below, which defaults to the CME timescale ($\tau_{\text{decay}} = 8$ h) unless an active DONKI HSS event covers the current window. Storm-type classification picks the decay time-scale ($\tau_{\text{decay}} = 8$ h vs. 24 h) of the Section 5.3.1 kernel.

All adjustments are capped at $K_p = 9$ and apply only when the respective input is available. The auroral CGM threshold expands continuously from 60° (quiet, $K_p \leq 5$) to 50° (severe, $K_p \geq 7$), per Eq. 16 — linear in K_p across the $5 \rightarrow 7$ window so the equatorward expansion of the auroral oval enters the budget gradually rather than stepping at a single threshold.

Table 8: Effective K_p bump from a sustained southward- B_z median observed at the L1 monitor (DISCOVER / ACE) over the trailing 20 minutes. The function is a continuous linear ramp from 0 at $B_z = -5$ nT to +3 at $B_z = -15$ nT, saturating at +3 for deeper southward IMF. Table values are sample points along the ramp.

Median B_z over 20 min	K_p bump
≥ -5 nT	0
-6 nT	+0.3
-10 nT	+1.5
-12 nT	+2.1
-15 nT	+3 (saturated)
≤ -15 nT	+3 (saturated)

Earlier table form had stepped thresholds at $-5/-10/-15$ nT producing a +1 Kp jump on every threshold crossing — same family of cliff as the auroral Kp gate, the storm-zone CGM expansion, and the X-ray flare twilight gate retired in the 2026-04 polish round. The continuous ramp closes that.

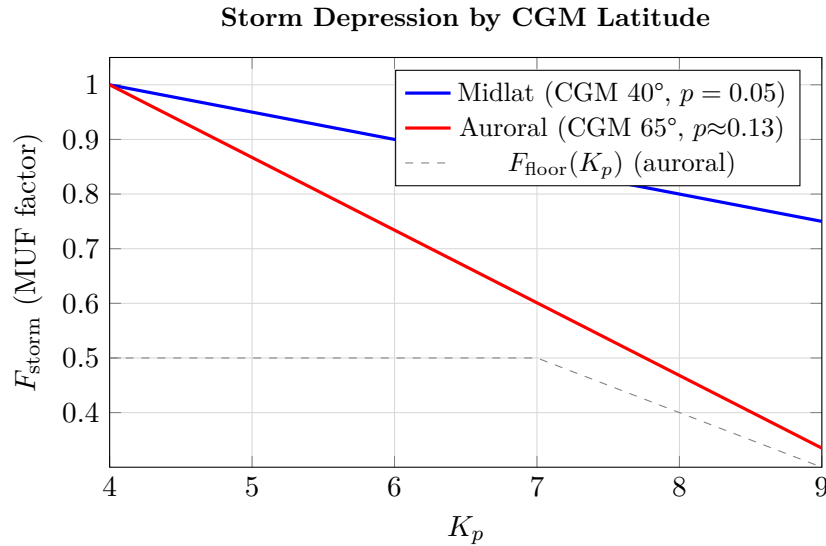


Figure 3: Storm MUF depression factor vs. K_p for midlatitude and auroral-zone paths. The midlatitude floor never binds (slope $p = 0.05$ keeps the main term above 0.5 for all $K_p \leq 9$). The auroral curve plotted is at $|\phi_{\text{CGM}}| = 65^\circ$ (where the slope $p \approx 0.13$ is not yet at its $|\phi_{\text{CGM}}| \geq 70^\circ$ ceiling of 0.15); the floor schedule $F_{\text{floor}}(K_p)$ (Eq. 57) drops from 0.5 at $K_p < 7$ to 0.40 at $K_p = 8$ and 0.30 at $K_p = 9$ but only *just* starts to bind on this curve at $K_p \approx 9$. The clean “main term and floor coincide at 0.40 at $K_p = 8$ ” result discussed in Eq. 58 applies at $|\phi_{\text{CGM}}| \geq 70^\circ$, not at the 65° curve plotted here.

5.4 Sporadic-E Persistence

The projected critical frequency of the sporadic-E layer decays exponentially:

$$f_{\text{oEs}}(t + \Delta) = \max\left(B(\cos \chi), f_{\text{oEs}}(t) \cdot 2^{-\Delta/\tau(\cos \chi)}\right) \quad (60)$$

with the half-life and floor smoothly interpolated across the twilight band $\cos \chi \in [0.1, 0.3]$ rather than stepped at a single $\cos \chi = 0.2$ threshold (the earlier formulation produced a $2\times$ jump in τ and a 2 MHz jump in B at the threshold, which on a band gated by $f \leq 5f_{\text{oEs}}$ translated to a 10 MHz step in the gating threshold — same family of cliff retired by the auroral and Bz onset ramps in Section 3.7). The interpolated forms are

$$\tau(\cos \chi) = 1.5 \text{ h} + 1.5 \text{ h} \cdot \sigma\left(\frac{\cos \chi - 0.1}{0.2}\right), \quad B(\cos \chi) = 2 \text{ MHz} \cdot \sigma\left(\frac{\cos \chi - 0.1}{0.2}\right) \quad (61)$$

where $\sigma(x) = \text{clamp}(x, 0, 1)$ is a unit ramp. At $\cos \chi \leq 0.1$ both reduce to their night values ($\tau = 1.5 \text{ h}$, $B = 0$); at $\cos \chi \geq 0.3$ both saturate to the day values ($\tau = 3.0 \text{ h}$, $B = 2 \text{ MHz}$). At deep twilight ($\cos \chi = 0.2$) the function is at midpoint ($\tau = 2.25 \text{ h}$, $B = 1 \text{ MHz}$), and the band-MUF gating threshold transitions continuously rather than stepping.

This subsection describes the documented forward-projection persistence model; it is part of the deferred outlook MUF projection (Section 5.1) and is not exercised by the live runtime budget at $t = 0$.

5.5 Gray-Line Bonus (Asymmetric)

When the reflection midpoint lies near the solar terminator ($|\cos \chi| < 0.1$) the D-region is thin, so a small additive bonus is applied to the margin. The effect is strongest on the low bands — where D-region absorption normally dominates the budget — and fades away by the upper HF. Sunrise and sunset are treated *asymmetrically*: at dawn, the D-region has not rebuilt yet while the F-region has started ionising, so the low-band window is long and clean; at dusk, the D-region decays more slowly than the residual F-region enhancement, producing a shorter and weaker window. Sunrise is distinguished from sunset by the sign of $d \cos \chi / dt$, sampled 5 minutes ahead. This sign-based distinction is meaningful only *near the terminator* ($|\cos \chi| < 0.1$, the bonus's proximity gate p above): at any location the cosine derivative $d \cos \chi / dt$ is positive throughout the morning half-day from local midnight to local noon (so the sign alone does not identify sunrise everywhere in that range), but the proximity gate restricts the bonus to the small window where $\cos \chi$ is near zero, and within that window the morning side is genuine sunrise and the evening side is genuine sunset — the sign correctly discriminates the two *at the terminator* even though it does not in general:

$$B_{\text{GL}}(f) = B_{\text{band}}(f, \text{phase}) \cdot p \quad (62)$$

where the per-band amplitude depends on the terminator phase (sunrise vs sunset) per Table 9.

Table 9: Gray-line bonus amplitude in dB by band and terminator phase.

Band	Sunrise (dB)	Sunset (dB)
160 m, 80 m	6	3
60 m, 40 m	5	3
30 m	3	1.5
20 m	1.5	1.0
17 m and above	0	0

Proximity $p = 1 - \min(1, |\cos \chi|/0.1)$ scales the bonus linearly from zero at $|\cos \chi| = 0.1$ to full amplitude at the terminator. Above 14 MHz the D-region contributes little, so no bonus is modelled.

Midpoint vs. per-hop evaluation. The proximity $|\cos \chi|$ is computed at the reflection *midpoint* only, not at each hop along the great circle. This is a deliberate simplification: the D-region absorption *loss* term (Eq. 7) is evaluated per-hop and summed, but the gray-line *bonus* compensates loss at the terminator, and on a 4-hop transatlantic path whose midpoint sits in midday but whose terminus hop is at sunrise the per-hop loss already drops on that one hop’s L_{Dreg} contribution; adding a midpoint-only bonus on top would over-count the local effect on the geometric-midpoint hop while under-counting the genuine sunrise hops at the path ends. The midpoint-only formulation captures the case the bonus is operationally intended for — short-to-medium HF paths (1–2 hops, 1500–6000 km) whose single or twin reflection points cross the terminator together, where “the path is on the gray line” and “the midpoint is at the terminator” coincide. For longer multi-hop paths (5000 km+) where one terminus is in daylight and the other near sunrise, the per-hop L_{Dreg} already does the right thing on its own. Extending the bonus to a per-hop max is a candidate refinement tracked in Section 10; today’s midpoint-only form is simpler to reason about and avoids the over-count.

6 Alternative Propagation Modes

The F2-layer SNR budget (Section 3) is the default path for every HF circuit, but four additional modes can open bands that the F2 budget marks closed: sporadic-E (§6.1), trans-equatorial propagation (§6.2), meteor scatter (§6.3), and F2-region scatter recovery (§6.4). Each runs in parallel with the F2 budget on the same path set; the verdict takes whichever mode has the best margin. Tropospheric ducting (§6.5) is also covered in this section but is structurally different — it is surfaced as a measured per-station panel rather than folded into the budget, because a duct’s spatial extent is not resolved by the sparse radiosonde basket the gradient is computed from.

6.1 Sporadic-E as a Propagation Mode

The Es-screening term (L_{Es} , Section 3) only accounts for Es *blocking* an F2 hop. In reality Es can itself be the propagation mode (the framework for which is ITU-R P.534 [5]): a single ~ 2000 km hop off the E layer at ~ 110 km altitude, with no F2 variability and no auroral absorption. This is the mechanism behind summer-evening 15 m, 12 m, 10 m, and 6 m openings that the F2-only model would otherwise call closed. 6 m only opens via Es when $f_{\text{oEs}} \gtrsim 10$ MHz (so that $5 f_{\text{oEs}} \geq 50$ MHz), which is rarer than the HF-band thresholds and concentrated in the boreal-summer Es season.

ionocast defines a sporadic-E budget alongside the F2 one:

$$M_{\text{Es}} = P_{\text{tx}} + G_{\text{ant}} - L_{\text{fs}}(f, d_{\text{Es}}) - L_{\text{Dreg}} - L_{\text{PCA}} - L_{\text{flare}} - L_{\text{MUF,Es}}(f, 5f_{\text{oEs}}) - L_{\text{iono,Es}} - L_{\text{low}} - N - S_{\text{req}} \quad (63)$$

with $d_{\text{Es}} = 2000$ km, a single-hop takeoff angle $\approx 6.3^\circ$, and $L_{\text{iono,Es}} = 15$ dB.

What’s in $L_{\text{iono,Es}} = 15$ dB. Substantially larger than the F2 lumped term $L_{\text{iono}} = 1$ dB (Section 3.11) because the E-layer reflection geometry introduces several mechanisms a clean F2 reflection does not: patchy cloud structure (footprint mismatch between TX and RX illumination), strong aspect sensitivity at the layer’s tilted surfaces, polarisation scrambling through a thin gradient layer, and substantial residual ITU-R Yp population variability since Es strength swings hour-to-hour rather than the F2 spread that the P.533 day-to-day budget captures. Future calibration may redistribute this into named per-mechanism terms; the lumped 15 dB is the empirical equivalent.

The D-region terms still apply — the E layer sits above the D layer, so an Es signal still passes the absorbing region once per hop. L_{hop} is zero for Es (single-hop, no intermediate ground reflections). L_{aur} is *omitted* from Eq. 63 entirely (the term is not in the equation), *not* because it

can't physically apply — it can; auroral D absorption affects any signal traversing the D-region, and Es hops cross the D-region twice (TX up, down to RX) — but because the geographic regions where Es is the dominant propagation mode overlap rarely with the auroral oval. The $f \geq 14$ MHz operating range and summer-evening seasonality of strong Es push most Es hops to midlat / equatorial corridors away from the auroral CGM threshold. On the rare auroral-zone summer-Es path (Scandinavia, Alaska, Antarctic) where both mechanisms co-occur the budget under-predicts absorption; treat “ $L_{\text{aur}} = 0$ on Es” as a geographic simplification rather than a physical claim.

Gating in code: `snrMarginHfEs` returns null below 14 MHz regardless of f_{oEs} (`src/physics/snr.js`), even though the algebraic Es MUF gate $f \leq 5f_{\text{oEs}}$ would otherwise admit lower bands on strong Es days ($f_{\text{oEs}} \geq 4$ MHz pushes $5f_{\text{oEs}} \geq 20$ MHz, which would activate the 30 m and 40 m bands without the floor). The hard 14 MHz floor matches amateur-radio literature: Es is essentially unobserved below 20 m at the 2000 km characteristic single-hop distance the budget assumes, so the algebraic gate alone would over-fire on the lower HF bands. The cutoff is intentionally *conservative*: rare Es openings on 30 m and 40 m do appear in the WSPR record during peak-cycle summer evenings with $f_{\text{oEs}} \geq 8$ MHz, typically on shorter-than-2000 km paths where the geometry makes the lower-band Es viable. These events are not captured by the budget; an operator who sees 30 m WSPR activity that the model calls F2-only should treat the verdict as “F2 explains most of it; rare Es is possible on top and the budget won't tell you.” L_{low} in Eq. 63 is therefore numerically zero under the current implementation (Table 3 returns 0 for $f \geq 14$ MHz); it is kept for symbolic symmetry with the F2 budget so that any future relaxation of the 14 MHz floor automatically picks up the low-band penalty without a separate edit. Es variability adds +2 dB in quadrature to the base σ .

6.2 Trans-Equatorial Propagation

Chordal afternoon / evening mode across the magnetic equator. The signal couples into the equatorial F-region irregularity structure and traverses hundreds to thousands of km without ground-bouncing, skipping several D-region traversals. Classical TEP produces 15 m / 12 m / 10 m and occasionally 6 m openings between stations whose magnetic dip latitudes sit on opposite sides of the equator.

The TEP bonus is a smooth function of three eligibility axes plus a hard cross-equator gate. Hard gate (binary): both endpoints' dip latitudes have opposite signs (i.e. $\phi_{\text{dip,src}} \cdot \phi_{\text{dip,dst}} < 0$); same-hemisphere paths see no TEP at any distance. Once the cross-equator gate fires, three smooth ramps modulate the bonus magnitude:

- **Frequency factor**: ramps from 0 at 14 MHz to 1 at 22 MHz, holds at 1 across the [22 MHz, 50 MHz] plateau, then ramps back down to 0 at 60 MHz. This eliminates the earlier hard step at 20 MHz that placed 17 m at zero TEP and 15 m at full plateau across a 3 MHz band edge.
- **Local-hour factor**: ramps from 0 at LT = 16 to 1 at LT = 17 at the leading edge, plateau over [17, 23], then 1 at LT = 23 to 0 at LT = 24 at the trailing edge. Earlier code stepped at 17–23 LT boundaries.
- **Dip-latitude factor**: each endpoint's $|\phi_{\text{dip}}|$ ramps from half-threshold (5°) to full threshold (10°); the path-level factor is the product of the two endpoint factors, so a path with one endpoint near the dip equator gradually loses TEP eligibility as that endpoint approaches the equator.

The bonus is the product of the plateau value and the three factors:

$$B_{\text{TEP}} = B_{\text{TEP,plateau}}(F_{10.7A}) \cdot f_{\text{freq}} \cdot f_{\text{LT}} \cdot f_{\text{dip,src}} \cdot f_{\text{dip,dst}} \quad (64)$$

where the plateau magnitude is solar-cycle-keyed via a smooth sigmoid (re-derived 2026-04-30):

$$B_{\text{TEP,plateau}}(F_{10.7A}) = 8 + \frac{7}{1 + e^{-(F_{10.7A} - 125)/30}} \text{ dB} \quad (65)$$

The sigmoid asymptotes to ~ 8 dB at solar minimum and ~ 15 dB at solar maximum, with the transition centred near $F_{10.7A} = 125$. At quiet sun the plateau is ~ 9 dB; at moderate cycle ($F_{10.7A} = 120$) it sits at ~ 11 dB; near solar peak ($F_{10.7A} = 180$) it reaches ~ 14 dB. Real TEP intensity scales with EUV (and therefore $F_{10.7}$), so a flat $B_{TEP} = 15$ dB across all solar conditions over-credited moderate-cycle openings; the prior fixed value matched peak-cycle TEP magnitudes specifically. The remaining frequency / local-time / dip-latitude factors are unchanged.

The full bonus saturates at the central 22–50 MHz / 17–23 LT / opposite-hemisphere near-equatorial-pair window (frequency factor saturates at 22 MHz, both endpoints' $|\phi_{\text{dip}}|$ above the 10° saturation knee). This is a first-order approximation — real TEP magnitude varies with Es index, EIA crest strength, and local evening $E \times B$ drift — but it captures the main effect with the smooth-ramp philosophy applied to the rest of the budget: bands that the F2 model says are closed can in fact be wide open across the equator for a few-hour window.

6.3 Meteor Scatter Floor

6 m and 2 m can sustain brief burst-mode QSOs off meteor ionisation trails, particularly in the predawn hours when Earth's leading face sweeps up the most meteors. ionocast lifts the VHF verdict floor from *closed* to *poor* during the predawn window ([2, 10] h local at the QTH) under either of two conditions:

- A major IMO shower with $ZHR \geq 20$ in its active window (± 2 d of peak) – the verdict note names the shower (e.g. “Perseids”). The $ZHR \geq 20$ threshold catches the ~ 10 strongest showers per year (Quadrantids, Lyrids, Eta Aquarids, Perseids, Geminids, etc.); the > 100 minor showers IMO catalogues with $ZHR < 20$ are excluded, on the assumption that their incremental ionisation above the sporadic background isn't large enough to shift the verdict from *poor* to anything higher.
- Otherwise, *sporadic background* – the ~ 5 – 10 ZHR continuous meteoroid influx that sustains routine MS QSOs every dawn even on shower-free days. The verdict note reads “sporadic background”.

Poor signals “viable with effort, not sustained” — MS is bursty, not continuous. Earlier code missed the sporadic-background case and only triggered on shower days, leaving 6 m / 2 m flagged *closed* during the routine MS window most of the year. The [2, 10] h centre window is now flanked by 30-minute linear ramps on each side (the activation weight $\in [0, 1]$ ramps $0 \rightarrow 1$ over [1.5, 2.0] h LT and $1 \rightarrow 0$ over [10.0, 10.5] h LT). The boolean MS-active flag fires on any non-zero weight, so the operationally-firing window extends 30 minutes on each side of the centre band; the previous tier cliff at the boundary has been moved to the weight=0 edges where MS activity is genuinely below useful threshold. This matches the smooth-ramp philosophy applied at every other gate (auroral $c(\phi)$, Bz, twilight r , near-MUF, Es, terminator σ , forecast-Kp).

6.4 F2 Scatter Recovery

When the band is nominally above MUF the F2 budget says *closed*, but real F-region irregularities (TIDs, plasma blobs, gradient-driven instabilities) often keep the path partly viable via scatter modes that do not require strict over-MUF compliance at every hop. ionocast reads inverse-square-distance-weighted foF2 from all GIRO digisondes within 3000 km of each F2 reflection point along the great circle, falling back to climatology (Section 4.2) at hops out of station range, and computes the per-hop foF2 standard deviation σ_f . The inverse-square weighting biases the spread estimate toward the nearest station: on a hop with one station at 200 km and five stations at 1500 km, the near station carries $\sim 80\%$ of the total weight, so σ_f collapses toward the near reading and the variance signal that B_{scatter} is supposed to detect can be artificially narrowed. This is a known statistical bias rather than a coding mistake — the weighted-variance formula is well-defined and the resulting estimator is consistent, just not the unbiased equal-weight variance. The choice trades a slightly under-detected B_{scatter} on station-asymmetric hops against larger

variance on equal-weight estimates from the long-tail stations, which empirically matter less to the path-averaged spread. A distance-window-uniform weighting is a candidate alternative; re-fitting w_{sc} under that scheme is gated on the per-pair WSPR ground-truth mode (Section 7.2). When at least two stations contribute and the path takes ≥ 2 hops, an additive scatter bonus is applied to the SNR margin:

$$B_{scatter} = w_{sc} \cdot \min(1, \sigma_f/1.5 \text{ MHz}) \cdot \min(2, f/\text{MUF} - 1) \cdot 5 \text{ dB} \quad (66)$$

gated on $f/\text{MUF} > 1.0$ (below MUF the standard model already says “open”), with $w_{sc} = 1.5$ from the joint coordinate descent of §7.2 (the harness’s unrestricted sweep keeps lifting w_{sc} toward 4 against the global “alive somewhere” metric, but that range exploits a known structural ceiling of the metric rather than a real propagation effect; the 15 dB-cap physical edge 1.5 ships). The foF2-variance term saturates at 1.5 MHz spread (substantial gradients across the path); the above-MUF excess saturates at $f/\text{MUF} = 3$ (wildly over-MUF paths get no recovery). At full saturation $w_{sc} = 1.5$ contributes up to 15 dB of recovery, sitting inside published F2-scatter measurements (10–25 dB) and matching the upper-band lift the model needed to track 30 d of WSPR aggregate openings on 17–10 m. The two saturating $\min(\cdot)$ terms are multiplied independently, but σ_f and f/MUF are not strictly independent in practice: paths with high σ_f (large gradients across the path) tend to have at least one hop near or above MUF, so high σ_f paths are often above-MUF paths. The independent saturation form slightly inflates the joint upper-bound probability; the effect is small (≤ 1 dB on the 15 dB ceiling) and does not change the calibrated w_{sc} .

The weight was set by joint coordinate descent over $\{L_{iono}, D, w_{sc}, w_{NVIS\text{-tail}}, w_{Es\text{-prim}}, \sigma\text{-scale}, \text{fusion-flag}\}$ from three deterministic starting configurations (Appendix E, “baseline”, “fusion-up”, “modes-on”) across 35 reference paths (Section 7.2). All seeds converged to the same basin: only w_{sc} produced measurable lift; an unrestricted sweep keeps improving as $w_{sc} \rightarrow 4$, but that range exploits the WSPR-global-vs-path-specific structural mismatch in the harness rather than a real propagation effect, so the shipped weight is held at the 15 dB-cap physical edge. This is intellectually honest engineering but it does mean the calibrated $w_{sc} = 1.5$ has been validated against a metric whose calibrators do not fully trust beyond a certain point. Closing this loop properly requires re-running the sweep against the per-pair WSPR ground truth (Section 7.2) once that mode is wired into `replayMargin`’s alternative-mode bonuses; the new optimum, whatever it is, can then be reported against a metric that the calibrators *do* trust at all weights in its range. Until that closure, treat $w_{sc} = 1.5$ as a defensible-but-not-final value tied to a metric with a known ceiling.

Per-hop fusion (deferred). The same per-hop foF2 sampling is also wired as an optional second-opinion source for the MUF consensus blend (Section 4), behind the `FUSION_PRIMARY_MUF` flag. In the same calibration sweep the flag stayed off: single-midpoint fusion did not beat the `kc2g ↔ climatology` blend in either Brier or accuracy, and per-hop minimum fusion regressed binary accuracy by ~ 1 pp because the rebuilt climatology (Section 4.2) tightened the residual it would have to beat. A future calibration that retunes the budget jointly with fusion enabled may flip this; the wiring is intentionally preserved.

6.5 Tropospheric Ducting

The four modes above all live inside the SNR budget: each one contributes a margin or bonus that competes with the F2 verdict for the same path. Tropospheric ducting is structurally different. It is a VHF-only line-of-sight extension whose viability depends on the shape of the lowest 1 km of the troposphere on the great circle between TX and RX, not on any quantity the global F2 model already evaluates. ionocast surfaces it as a measured per-station panel (Section 9) rather than folding it into the budget; the rest of this subsection documents the physics, the classification, and how the panel is wired.

Refractivity and the duct condition. Atmospheric refractivity N , in N-units, is computed from radiosonde temperature, pressure, and water-vapour pressure (ITU-R P.453 [6]):

$$N = \frac{77.6}{T} \left(P + \frac{4810 e}{T} \right), \quad (67)$$

with T in kelvin, P and e in millibar. Water-vapour pressure e is obtained from the dew point via the Magnus saturation formula

$$e_{\text{sat}}(t_C) = 6.112 \cdot \exp\left(\frac{17.62 t_C}{243.12 + t_C}\right), \quad (68)$$

evaluated at the dew-point temperature (so that $e = e_{\text{sat}}(t_{\text{dew}})$ is the actual partial pressure). The constants (17.62, 243.12) are the Alduchov–Eskridge 1996 coefficients for saturation *over liquid water*. Below freezing, saturation over ice uses different constants (22.46, 272.62) producing a $\sim 10\%$ lower e_{sat} at $t_C = -20^\circ\text{C}$. ionocast applies the liquid-water form across all temperatures, which is correct for the warm-water duct cases where ducting is operationally relevant (Mediterranean, Sea of Japan, California coast) but slightly over-estimates e on continental winter soundings where the surface dew point is below freezing. The resulting bias on N is ≤ 2 N-units in the worst case and the regime classification (standard / super-refractive / ducting) is robust to it; refining this further would require a piecewise formula gated on $t_C \geq 0^\circ\text{C}$, which is candidate cleanup tracked in Section 10. Modified refractivity M adds a curvature correction for the Earth-radius bend that makes a horizontally-stratified duct look horizontal in M -coordinates,

$$M(z) = N(z) + 0.157 z, \quad (69)$$

with z in metres. A horizontal duct exists wherever $dM/dz < 0$; equivalently, in the N -coordinate that ionocast actually computes,

$$\left. \frac{dN}{dh} \right|_{0 \rightarrow 1 \text{ km}} < -157 \text{ N-units/km}. \quad (70)$$

The -157 value is the modified-refractivity coefficient 0.157 N/m rescaled to N-units per kilometre. It is not a loss budget — a duct either exists or it does not — so the treatment here is classification, not a margin contribution.

ITU-R P.453 [6] / NTIA [19] classification. The lowest-1 km gradient is bucketed into three regimes:

- $dN/dh > -79 \text{ N-units/km}$: *standard atmosphere*. Radio rays bend gently toward Earth (the textbook $\frac{4}{3}$ effective Earth radius); no range extension beyond the geometric horizon.
- $-157 \leq dN/dh \leq -79 \text{ N-units/km}$: *super-refractive*. The radio horizon extends past the geometric horizon; tropospheric scatter paths run noticeably above their fair-weather budget. Common precursor to ducting.
- $dN/dh < -157 \text{ N-units/km}$: *ducting*. A surface-based or elevated atmospheric duct exists; VHF/UHF signals trapped in the duct can reach hundreds to thousands of kilometres independent of the ionosphere. Operationally this opens 6 m and 2 m DX between stations sharing the duct, particularly across warm-water surfaces (Mediterranean, Sea of Japan, California coast) and under summer subsidence inversions.

Implementation. The dN/dh panel is built from a 24-station basket of UWyo radiosonde sites with global coverage (Europe $\times 10$, North America $\times 7$, SE Asia $\times 3$, Africa $\times 2$, South America $\times 1$, Antarctica $\times 1$; `SONDE_STATIONS` in `functions/_handlers/refractivity.js`). The Cloudflare worker (`functions/_handlers/tropo.js`) fans out over the basket in parallel for the most recent 00 Z or 12 Z slot (with a 12 s per-fetch timeout to bound the worst case), parses each PRE-block sounding, and for each station:

1. Locates the surface sample (lowest row) and the sample nearest $z_{\text{surface}} + 1 \text{ km}$ that is at least 100 m above the surface.
2. Computes N_{surface} and N_{upper} via Eq. 67 from temperature, pressure, and dew-point at each level.
3. Returns $dN/dh = (N_{\text{upper}} - N_{\text{surface}})/\Delta z_{\text{km}}$ and the regime classification.

Stations whose 00 Z / 12 Z release is missing from the upstream archive return `ok: false` with reason `"no data"`, which the ducting panel renders as a placeholder row. The nearest station with valid data is also surfaced as a single dN/dh cell on the VHF band-table, sharing one cell across 6 m and 2 m because the gradient is per-station, not per-band; the cell's tier color uses the same classification (good = ducting, warn = super-refractive, no color = standard) so the band-table verdict and the per-station panel never disagree on what the local atmosphere is doing.

Why this is informational, not a budget term. A duct is only useful if both endpoints sit inside it. The radiosonde network samples vertical structure at fixed sites every 12 h, which is enough to tell an operator at one of those sites whether the local atmosphere is currently trapping signals, but it is not enough to claim a specific TX-to-RX path is ducted: marine-layer ducts can hug a coastline for hundreds of kilometres while inland soundings show standard atmosphere only 50 km away, and elevated subsidence ducts can sit several hundred metres above a surface that a radiosonde column doesn't cleanly resolve. Folding the gradient into a path-level SNR bonus would over-promise the model's spatial coverage; surfacing the per-station classification lets the operator make the geographic inference where the radiosonde basket is dense (Western Europe, the contiguous US) and lets it stay silent where the basket is thin (most of Africa, Central Asia, the open ocean).

7 Ensemble Blend and Self-Calibration

7.1 Verdict Margin

The verdict margin is the physics-budget margin M_{physics} (Eq. 1) plus the per-hop and alternative-mode bonuses, with no further blending or bias correction. Gray-line is additive (it describes a distinct D-region attenuation drop at the terminator), but TEP and scatter recovery both describe F-region irregularity-driven recovery and so are *combined by maximum* rather than summed — on a TEP-eligible 15 m path in the 17–23 LT window above MUF, both fire on the same physical channel and stacking them double-counts the recovery:

$$M_{\text{verdict}} = M_{\text{physics}} + B_{\text{grayline}} + \max(B_{\text{TEP}}, B_{\text{scatter}}) \quad (71)$$

σ **tracks** M_{physics} , **not** M_{verdict} – **caveat.** The reliability formula $R(M, \sigma)$ of Eq. 72 is evaluated with $M = M_{\text{verdict}}$ (the bonus-augmented margin) but σ is computed from M_{physics} inputs (per-band σ_g plus the situational widenings of Section 7.3.1). When the bonuses fire, the bonus terms themselves carry uncertainty that is not folded into σ : $B_{\text{TEP}} = 15 \text{ dB} \pm 5 \text{ dB}$ at the published-experience plateau, and B_{scatter} is calibrated through w_{sc} but the $\pm$ on the underlying spread ($\sigma_{f_{\text{of2}}}$ measurement noise, f/MUF position) is non-trivial. A strictly correct treatment would add the bonus-term variances in quadrature to σ when bonuses are active; under a conservative $\sim 5 \text{ dB}$ bonus-term uncertainty the correction is $\sqrt{\sigma_g^2 + 25} - \sigma_g$, which is $+1.5 \text{ dB}$ on a $\sigma_g = 12 \text{ dB}$ upper-band and $+2.7 \text{ dB}$ on a $\sigma_g = 6 \text{ dB}$ low/mid-band. The current model omits this widening: bonus-active verdicts on upper HF read slightly more confident than they should, and on the low/mid bands the under-counting is $\sim 3 \text{ dB}$. The omission is tracked in Section 10; folding it in requires calibrating per-bonus uncertainty estimates against the per-pair WSPR truth (Section 7.2), which is gated on that mode's wiring.

The TEP plateau ($B_{\text{TEP}} = 15$ dB once the smooth ramps in Section 6.2 have saturated) and the scatter upper bound (15 dB at $w_{\text{sc}} = 1.5$ when both $\sigma_{f_{\text{of2}}} = 1.5$ MHz and $f/\text{MUF} \geq 3$ are simultaneously hit, see Section 6.4) share the same numerical ceiling *at saturation*, but in practice the scatter term rarely saturates: a typical TEP-eligible 15 m cell at 17 LT with $f/\text{MUF} = 1.3$ and $\sigma_{f_{\text{of2}}} = 0.5$ MHz produces $B_{\text{scatter}} = w_{\text{sc}} \cdot \min(1, 0.5/1.5) \cdot \min(2, 1.3 - 1) \cdot 5 = 1.5 \cdot 0.333 \cdot 0.3 \cdot 5 \approx 0.75$ dB, an order of magnitude below the 15 dB TEP plateau. In a TEP-active cell, $\max(B_{\text{TEP}}, B_{\text{scatter}})$ therefore predominantly selects B_{TEP} ; the value of the max versus a sum is to prevent the small-but-non-zero B_{scatter} from stacking on top of the calibrated TEP plateau. The 2026-04-26 fix is a policy correction made on physics grounds rather than an empirical retune: the calibration harness operates on the climatology code path and does not compute either bonus at replay time, so it cannot directly distinguish $B_{\text{TEP}} + B_{\text{scatter}}$ from $\max(B_{\text{TEP}}, B_{\text{scatter}})$. Validation that the change does not overshoot in the other direction, and a per-cell attribution of which mechanism dominates on which TEP-eligible 15 m path (EU–Brazil, NA–Brazil, NA–Argentina, EU–Argentina, EU–South Africa), are deferred to a future harness extension that incorporates the alternative-mode bonuses into `replayMargin` and to VOACAP cross-checks once the fixture set is populated. A previous 0.7 / 0.3 ensemble blend with an N0NBH-style SFI heuristic, plus a per-(band, horizon) bias correction derived from WSPR-tier residuals, were both retired in the 2026-04-25 “D” experiment and the 2026-04-26 P.842 recalibration. Pure physics scored 94.25 % binary accuracy / Brier 0.0386 against 30 d of WSPR aggregates on the production 35-path basket (Section 7.2, “current production” row of Table 10), beating every blended variant tried; and tier verdicts are now defined by ITU-R P.842 reliability buckets in σ -units rather than by a fixed dB-width tier scheme, so the per-band absolute margin bias has no constant width to fit against. The N0NBH-style `heuristicTier` function remains available in `src/physics/physics.js` as an independent reference predictor but is no longer in the verdict path.

7.2 Offline Constant Calibration

Two layers, one set of constants. The ionocast pipeline runs in two distinct contexts. The *online* layer is the browser at runtime: it fetches live data feeds (SWPC, kc2g, GIRO, GFZ, Kyoto, DONKI, wspr.live), evaluates the physics in `src/physics/`, and emits the band-table verdict. The *offline* layer is `scripts/harness.mjs` on a developer workstation: it imports the same `src/physics/` code (no inline copy that can drift), replays the budget over the 35-path basket against 30 d of WSPR aggregates, and writes the calibration baselines (`harness.baseline.json`, `spot-baselines.mjs`) that the online layer ships with. The two layers exchange exactly one thing: the calibrated constants in `src/constants.js` and the WSPR spot baselines in `src/data/spot-baselines.mjs`. Everything else (downstream live data, the ClickHouse query, the per-(path, band) baselines used for regression detection) lives entirely on one side or the other. A formal data-flow diagram is candidate documentation work tracked in Section 10; until then this paragraph is the explicit description.

Several of the budget’s empirical constants — the lumped ionospheric loss L_{iono} , the per-extra-hop defocusing constant D (Section 3.8), and the F2-scatter weight w_{sc} (Section 6.4) — were originally set from published ITU-R decompositions and ham-operator experience rather than measured in place. The per-band σ table (Table 11) is *not* fitted by the harness: σ is anchored to the ITU-R P.533 day-to-day spread (6–12 dB depending on band) and held fixed. The harness scores binary accuracy and Brier against WSPR but does not adjust σ , since under the P.842 reliability framework the tier boundary scales with whatever σ is set, and a machine-fitted σ would chase the WSPR-truth artefact rather than the underlying physics spread.

The current calibration harness is `scripts/harness.mjs`. It imports the production physics directly from `src/physics/` (no inline copy that can drift), replays the SNR budget over a 35-path basket (Appendix C) spanning 1–5 hops and every major continent pair plus polar / transequatorial / NVIS-tail geometries against 30 d of WSPR hourly spot counts (`wspr.live`,

ClickHouse), aligned hour-by-hour with the SWPC K_p history, the $F_{10.7}$ 81-day mean, and pre-fetched 30 d GIRO foF2 / foEs histories from 26 digisonde stations (Appendix D) spanning Europe, North America, the South Atlantic, Australia, the Pacific, and an expanded equatorial-belt set (Ascension and Jicamarca for the Atlantic / Americas trough-and-crest pair — Jicamarca on the dip equator, Ascension at dip lat $\sim -7^\circ$ closer to the trough than to the southern crest peak; Townsville and Niue for the Pacific southern crest). The Pacific / Australia coverage (Townsville, Niue, Perth, Canberra, Hobart) was added 2026-04-25 after a per-station bias diagnostic (`scripts/t1-snapshots-bg.mjs`, `t1-analyze.mjs`) identified consistent under-prediction over the South Pacific. The fetcher chunks queries 3-at-a-time with retry-on-503 because DIDB rate-limits parallel fetches; coordinates are verified against the kc2g station registry by `scripts/verify-station-coords.mjs`. Per-(path, timestamp) cells precompute band-independent state once (climatology, geometry, solar zenith) and the $\sim 25,000$ unique cells are reused across all 10 HF bands per scoring call, keeping a full sweep cycle well under 1 s.

For each candidate config the harness computes the Brier loss of the predicted *open* probability against a binary “alive” floor on the WSPR aggregate. This is a sanity check on the model’s binary verdict, not a calibration target: WSPR-spot activity and link viability for the configured station are correlated but not identical, so binary accuracy is reported with that caveat (see Section 8). An earlier per-band tier-midpoint residual emitter targeted a WSPR-percentile tier truth; that target was retired in favour of ITU-R P.842 reliability buckets defined by the model’s own σ , so per-band absolute-margin biases are no longer fitted. In its place the harness writes a per-(path, band) margin/ σ baseline and flags any cell that drifts more than 2 dB in mean margin or 5 pp in $P(\text{open})$ across runs — a self-consistency regression detector that needs no external truth.

The “alive” floor itself comes in two forms, selectable via `-ground-truth={global,per-path}`:

- **Global** (default, baseline file `harness.baseline.json`): band-level “alive somewhere in the world” floor of ≥ 50 spots/hour, computed from a single bbox-unrestricted WSPR aggregate query. Cheap (one query per harness run, $\sim 7k$ rows for a 30-day window), but every reference path sees the same open-rate truth, so the per-(path, band) drift signal is dominated by the path-independent open-rate bias.
- **Per-path** (baseline file `harness.baseline.perpath.json`): TX bbox + RX bbox aggregate per reference path (bidirectional, default half-width 5°), with a much sparser floor of ≥ 1 spot/hour reflecting the per-pair WSPR density. Adds 35 queries per harness run, but the open-rate truth is path-specific, so the per-(path, band) drift signal directly reflects per-pair behaviour. Used to flag a per-mechanism regression that the global mode averages out — for example, the §4.3 asymmetric-MUF retest, where the bias was concentrated on the minority of cells where kc2g sat above climatology.

Both modes write to separate baseline files so an operator can maintain both anchors simultaneously and compare drift in either register.

A separate driver script, `scripts/tune.mjs`, runs coordinate-descent joint optimisation over $\{L_{\text{iono}}, D, w_{\text{sc}}, w_{\text{NVIS-tail}}, w_{\text{Es-prim}}, \sigma\text{-scale}, \text{fusion-flag}\}$ from three deterministic starting configurations (Appendix E; “baseline”, “fusion-up”, “modes-on”). All three seeds converge to the same basin in ≤ 2 iterations: “same basin” here means the final $\{L_{\text{iono}}, D, w_{\text{sc}}\}$ landed within $\pm 10\%$ across seeds and the final Brier landed within ± 0.001 of the median seed (Brier dominates accBin in the tiebreak because it is more sensitive to per-sample probability shifts). The 2026-04 sweep moved binary accuracy from 91.08% (Brier 0.0625) to 92.35% (Brier 0.0535); the 2026-04-26 / 2026-04-27 polish (climatology rebuild, H-pol Fresnel multi-hop, per-band σ refit, smoothed K_p / HP / cosZ / Bz gates) added another half-point; the 2026-04-28 EIA-trough kernel and per-hop direct-climatology MUF (Section 4.2, Eq. 48) lifted it further to the current 94.25%. Lift was concentrated on 12 m / 10 m, driven entirely by the $w_{\text{sc}} = 1.5$ scatter recovery; the other three parameters held their prior values because no measurable Brier change was observed when sweeping them.

The fixed reference values cited throughout the paper are consolidated in Table 10 so each

percentage can be tied to a specific basket / config / metric.

Table 10: Canonical harness numbers cited in the paper, each tied to its (basket, ground-truth mode, configuration) so a reader can disambiguate. Binary “open” threshold is ≥ 50 spots/h on the global aggregate or ≥ 1 spot/h on the per-path aggregate.

Run	Basket	Global truth		Per-path truth	
		accBin	Brier	accBin	Brier
Pre-R7 baseline (35-path)	35 paths	91.08 %	0.0625	—	—
Post-R7 sweep (2026-04-25)	35 paths	92.35 %	0.0535	—	—
Current production (post-2026-04-28)	35 paths	94.25 %	0.0386	30.97 % [§]	0.6353 [§]

[§] Per-path-truth metric uses TX/RX-bbox-restricted aggregates with the ≥ 1 spot/h floor; the resulting accBin / Brier numbers are *not directly comparable* to the global-truth column because the per-path floor is far stricter (most cells read closed) and because operator-activity sparsity dominates the binary outcome on minority-traffic bands. The headline 94.25 % / 0.0386 is the global-truth metric and inherits the structural “open somewhere” ceiling that the per-path mode was specifically built to expose (Section 10 #9). Treat the global number as a regression-detection signal, not a quality score; the per-path column is for cell-level drift detection against the saved baseline.

The harness does not auto-commit: a human reviews the suggested constants and pastes the winners into `src/constants.js`.

7.3 Circuit Reliability

Given the margin M and spread σ , the circuit reliability is the probability that the achieved SNR meets or exceeds the mode’s required SNR — in other words, the probability the link is usable on a given attempt:

$$R(M, \sigma) = \Phi\left(\frac{M}{\sigma}\right) \quad (72)$$

where Φ is the standard normal CDF. JavaScript has no built-in Φ , so the runtime evaluates it via the Abramowitz & Stegun 26.2.17 rational approximation (`src/physics/tier.js`, `normCdf`); maximum absolute error is $\sim 7.5 \times 10^{-8}$ in the tails, well below the verdict-relevant resolution at the ionocast tier boundaries ($z = \pm 1.2816, +0.2533, -0.3853$, where $\partial R / \partial z = \varphi(z) \approx 0.18, 0.39, 0.37$ respectively). The approximation choice is therefore not a budget-shaping parameter; it is fixed for reproducibility. ionocast applies the ITU-R P.842 [8] circuit-reliability formula with live nowcast inputs — M is the current-hour budget margin (Eq. 71), and σ is anchored to P.842’s day-to-day spread range (6–12 dB, Table 11) augmented in quadrature with the situational widenings of §7.3.1. R drives the tier verdict (Section 8) but the band-table UI surfaces a different percentile alongside the tier label, the *Stability* metric defined in Section 7.3.1.

Strict P.842 vs. instantaneous: an operator-relevant divergence. P.842 as written is a *climatological* predictor: the formula is evaluated with the monthly-median margin at a given hour and the day-to-day spread, so R answers “on what fraction of days does the median link at this hour exceed the requirement?” ionocast evaluates the same algebra with live data — the current-hour observed kc2g MUF, the live $F_{10.7A}$, K_p^{eff} from the past 7 d — and so R answers “what is the probability the link is viable *right now*, given the model’s residual uncertainty?” Both questions use the P.842 formula and a P.842-grade σ , but they are not the same question. An operator deciding whether to call CQ at this minute wants the latter; a frequency-coordination engineer planning circuit allocations across a season wants the former. ionocast surfaces the nowcast interpretation, and the σ widenings of §7.3.1 (storm, terminator, near-MUF) capture the within-hour spread that P.842’s static day-to-day σ does not. The verdict bucket boundaries (Table 13) use the same $R = \Phi(M/\sigma)$ formula as P.842 but apply ionocast-specific reliability

targets (0.90/0.60/0.35/0.10) chosen for operator failure-rate intuition rather than the classical 0.95/0.80/0.50/0.20 schedule; see Table 13.

Tier thresholds (Section 8) are themselves expressed as constant levels of R , so the boundary in dB scales with the band's σ rather than being pinned to a hand-set number. This makes the verdict self-consistent with the model's own uncertainty: a band whose σ is small (low-band, climatology dominated) reaches “excellent” at a smaller dB margin than a band whose σ is large (10 m near MUF, where prediction spread is intrinsic).

What R does and does not measure. R is the probability that the *physics* of the link permits a QSO at the operator's configured rig, conditional on an operator on the other end transmitting and listening at that hour. It does not include operator-availability factors (band-time-of-day social bias, WSPR-spot population density on the destination side, holiday patterns, contests, etc.), which are independent of ionospheric state and not represented in the SNR budget. An empirical calibration against 30 d of WSPR per-path aggregates (TX/RX bbox restricted, floor 1 spot/h; `scripts/wspr-calibration.mjs`) finds that predicted tier ranks correctly — excellent > good > fair > poor > closed in observed open-rate ordering — but the absolute per-path completion rate runs ~ 60 – 65 pp below the predicted reliability across the upper four tiers (the predicted “excellent” bucket sees a $\sim 30\%$ per-path open rate against the WSPR cache; “good” $\sim 14\%$; “fair” $\sim 3\%$; “poor” $\sim 1\%$; “closed” $\sim 0.1\%$). The held-out window (2026-04-26 onwards, post the 2026-04-25 loss-constant calibration) matches the in-sample window within 1 pp on every tier, ruling out a calibration-drift explanation. The gap is a physics-vs-operator-availability semantics issue: R captures “the path's physics permit a QSO with the typical operator's rig” (physics-permitted reliability), while WSPR per-path open-rate captures “a WSPR-active operator pair completed a transmit at this hour” (operator-attempt completion rate), which is uniformly lower because operators do not attempt every available (band, hour, path) cell.

7.3.1 Verdict Stability

The band-table surfaces a per-band *Stability* percentage, defined as

$$S(M, \sigma) = \Phi(\min_{b \in \mathcal{B}} |z - b|), \quad z = M/\sigma, \quad (73)$$

where $\mathcal{B} = \{-1.2816, -0.3853, +0.2533, +1.2816\}$ is the set of tier boundaries in z -space (Table 13). Operator reading: *how likely is the verdict to NOT cross its nearest tier boundary if the true margin moves to its expected value?* The metric is one-sided, it asks about the closest boundary only, not both flanks of the bucket, and bucket-width-independent: a Fair verdict Δz from its nearest boundary reads the same percentage as an Excellent verdict Δz from its boundary.

Reachable values. What S can show depends on the bucket, because the maximum nearest-boundary distance depends on bucket width:

- **Closed** ($z < -1.28$) and **Excellent** ($z \geq 1.28$): open-ended on one side. S rises monotonically the further z moves into the tail: $\sim 84\%$ at 1σ past the boundary, $\sim 97\%$ at 2σ , $\rightarrow 1.0$ deep in the tail.
- **Poor** ($\sim 0.90\sigma$ wide), **Fair** ($\sim 0.64\sigma$ wide), **Good** ($\sim 1.03\sigma$ wide): max nearest-boundary distance is half each bucket's width, so S caps at $\Phi(0.45) \approx 67\%$ for Poor, $\Phi(0.32) \approx 63\%$ for Fair, and $\Phi(0.51) \approx 70\%$ for Good at each bucket centre. $S = 50\%$ on a boundary; the centred-bucket maxima are the most stable a middle-tier verdict in that bucket can read.

Read S on the open buckets against an absolute scale (deep Excellent / Closed approach 100%); read S on the middle buckets relative to their per-bucket centred maxima.

One-sided semantics. S measures the probability that pure- σ noise does not cross the *nearest* boundary; it ignores the far boundary on a finite-width bucket. For a centred Fair verdict

($z = -0.066$, halfway between -0.385 and $+0.253$) the far boundary is also 0.32σ away and is not counted, so $S = 63\%$ rather than the two-sided in-bucket probability $2\Phi(0.32) - 1 \approx 25\%$. The one-sided form is the correct “how locked in is this verdict against the most likely flip” indicator and is what operators typically want; the two-sided form is the literal $P(\text{predicted} = \text{true})$ retained as **tierConfidence** in the source. The earlier band-table column showed the two-sided form, which capped at $\sim 25\text{--}35\%$ for centred middle-tier verdicts and required a “(peak)” annotation; the new S replaces it.

Operator-attempt completion rate. Alongside the physics-permitted tier label, the band-table surfaces an *Expected Completion* (*Comp.*) column that folds the operator-availability gap from the previous paragraph into a single percentage:

$$\text{Comp.}(b, h, M, \sigma) = R(M, \sigma) \cdot a(b, h), \quad (74)$$

where $a(b, h) = \text{baseline}(b, h) / \max_h \text{baseline}(b, h)$ is the per-band operator-availability prior, derived from the 30-day per-(band, UTC-hour) WSPR baselines that already drive the spot-override (§8.1). $a(b, h) = 1$ at the band’s peak activity hour; $a(b, h) \rightarrow 0$ deep in the band’s dead-of-night for that band’s typical user population. *Comp.* reads as “what fraction of attempts at this rig in this hour actually complete a QSO,” which is the question an operator picking a band to call CQ on is asking. Tier and *Comp.* together disambiguate the physics-vs-availability gap: an Excellent tier on 12 m at 0300 UTC reads $\sim 95\%$ physics-permitted (R) and $\sim 5\text{--}10\%$ *Comp.* — the band is open, but nobody is listening. Use Tier when planning around the physics; use *Comp.* when picking a band to actually call CQ on right now.

The factorisation $R \cdot a(b, h)$ assumes physics-permitted reliability and operator-availability are independent. This holds strictly within a band in steady state — $a(b, h)$ is normalised against the band’s own 24-hour maximum, so it encodes *intra-band* hourly activity variation and not inter-band reliability. Across bands the absolute *Comp.* values are not directly comparable (a band with chronically thin WSPR-pair density can never reach *Comp.* $\rightarrow 1$ even when fully open), which is why the band-table surfaces the percentage per row rather than a cross-band ranking. In the limit of a future per-pair availability prior with band-specific ceilings, the factorisation would need refining to account for the fact that bands with high WSPR activity have those operators *because* the band is reliably open — a feedback that the current intra-band normalisation sidesteps but does not eliminate.

Storm-time correlation breaks the independence assumption. During a major geomagnetic storm both R drops (physics) and $a(b, h)$ should drop too (operators on the affected bands stop calling because they observe the band is dead) — i.e. R and a are positively correlated under exactly the disturbance conditions where *Comp.* is most operationally relevant. The current model treats them as independent, which biases *Comp.* *high* during storms (the product of the depressed R with the steady-state a overstates expected completion versus the true $R \cdot a_{\text{actual}}$ where $a_{\text{actual}} < a$). The intra-band normalisation softens but does not remove this: storm-time activity dropouts span all bands roughly together, so the per-band 24-hour maximum baseline still encodes a quiet-day shape that storm hours genuinely depart from. Treat *Comp.* during $K_p^{\text{eff}} \geq 5$ conditions as an upper bound on expected completion, not a point estimate; the storm-aware refinement is tracked in Section 10 and is gated on the per-pair availability data wiring of Section 7.2.

The spread σ is *per-band and condition-dependent*, computed as the RSS of a per-band base σ_g (Table 11; values anchored to the ITU-R P.533 day-to-day spread range 6–12 dB) and five situational penalties.²

²Notation: σ is overloaded across the paper for distinct quantities — the prediction-spread family in this section (σ , σ_g , σ_{MUF} , σ_{storm} , etc.); ground conductivity in Eq. 22 (in S/m); the Gaussian kernel widths σ_{cr} , σ_{tr} in the

- **Near-MUF:** when $f/\text{MUF} > 0.85$, fading and multipath intensify. Adds up to +4 dB at $f = \text{MUF}$.
- **Storm:** $K_p^{\text{eff}} \geq 5$ (the storm-lagged effective K_p from Section 5.3.1, after Bz-forward and Dst bumps) produces ionospheric irregularities. Linear in K_p^{eff} above the activity threshold: $\sigma_{\text{storm}} = 3 + 0.75(K_p^{\text{eff}} - 5)$ dB for $K_p^{\text{eff}} \geq 5$, zero otherwise; that gives +3 dB at $K_p^{\text{eff}} = 5$ and +6 dB at $K_p^{\text{eff}} = 9$. Using the effective rather than live K_p means the storm σ persists into the recovery phase (when the live index has already dropped but the F-region thermosphere is still recovering). σ_{recovery} below fires only when the phase classifier identifies recovery, so the two penalties stack during recovery (covering both the residual lagged-Kp irregularities and the LSTID ripple) and the storm penalty alone applies during main phase. The same K_p^{eff} also drives the auroral amplification described later in this section.
- **Forecast storm:** the SWPC 3-day Kp forecast shows a peak $K_p^{\text{peak}} \geq 5$ within the next 6 h. Sharp electron-density gradients build hours before the index reading, so σ widens ahead of the disturbance. Same magnitude as σ_{storm} above evaluated at K_p^{peak} (+3 dB at $K_p^{\text{peak}} = 5$, +6 dB at $K_p^{\text{peak}} = 9$), attenuated by $0.7\times$ for forecast confidence, then ramped down linearly as the current K_p catches up to the forecast peak:

$$\sigma_{\text{forecast}} = 0.7 \sigma_{\text{storm}}(K_p^{\text{peak}}) \cdot \min(1, \max(0, K_p^{\text{peak}} - K_p^{\text{eff}})) \quad (75)$$

The catch-up ramp is gated on K_p^{eff} rather than the live 3-hour K_p , matching the σ_{storm} branch above and preserving the smooth-ramp philosophy applied at every other gate; using the live index here would step the forecast bump down on each 3-hour tick rather than as the thermosphere actually catches up. Without this gap-based gate, summing in quadrature with the current-Kp σ_{storm} branch would have produced $\sqrt{\sigma_{\text{storm}}^2 + 0.49 \sigma_{\text{storm}}^2} \approx 1.22 \sigma_{\text{storm}}$ at the moment of catch-up (a 22% inflation), instead of the intended hand-off to σ_{storm} alone. The 1 Kp-unit saturation width of the catch-up ramp is empirical: K_p^{eff} typically climbs by ~ 1 unit per hour during onset, so a 1-unit gap maps to roughly an hour of forecast prominence ahead of the live index, which matches the operationally-useful early-warning horizon of the SWPC 3-day forecast. A wider saturation window would extend the forecast bump deeper into peak hours and double-count with σ_{storm} ; a narrower window would cut the forecast bump off before the live index has caught up at slow-onset events.

- **Cross-terminator:** when the path midpoint has $|\cos \chi| \leq 0.15$ (sunrise/sunset line), sharp electron-density gradients increase variance by +3 dB (the variance contribution is 9 dB^2 , added in quadrature to the per-band σ_g^2). Above $|\cos \chi| \geq 0.20$ the contribution is zero; in the $|\cos \chi| \in [0.15, 0.20]$ band the variance contribution ramps linearly from 9 dB^2 down to 0, matching the smooth-ramp philosophy applied at every other gate (auroral $c(\phi)$, Bz bump, twilight $r(\cos \chi_k)$, near-MUF, Es threshold). An earlier formulation used a hard binary at $|\cos \chi| = 0.15$ that produced a tier-confidence cliff at terminator approach; the linear ramp closes that cliff without materially changing the in-window penalty.
- **Storm-recovery TID:** during the recovery phase of a geomagnetic storm (Dst depressed but K_p^{eff} trailing the live K_p , see Section 5.3.1), large-scale travelling ionospheric disturbances ripple through the F-region for hours after the surface-field index settles. The mean margin is no longer suppressed but the path variance is visibly elevated; ionocast adds +4 dB to σ across all paths until the storm phase classifier exits “recovery”.

All penalties are added in quadrature: $\sigma = \sqrt{\sigma_g^2 + \sigma_{\text{MUF}}^2 + \sigma_{\text{storm}}^2 + \sigma_{\text{forecast}}^2 + \sigma_{\text{term}}^2 + \sigma_{\text{recovery}}^2}$.

Why σ_{storm} and σ_{recovery} are path-blind. σ_{MUF} and σ_{term} are evaluated at the path midpoint (so they vary path-by-path across the 10-path basket), but σ_{storm} , σ_{forecast} , and σ_{recovery}

EIA term (Eq. 44, in degrees); the unit-ramp clamp function $\sigma(x) = \text{clamp}(x, 0, 1)$ in the Es persistence model (Section 5.4); and the standard-normal CDF argument in Eq. 72’s $\Phi(M/\sigma)$. The distinct meanings are recoverable from local context (units, subscripts) but the overloading is real.

are global: they fire on the same K_p^{eff} and storm-phase signals regardless of whether any hop on the path actually crosses an auroral CGM latitude. This is intentional rather than a simplification. Storm-time variance arises from at least three mechanisms whose footprints are not confined to high latitude: large-scale travelling ionospheric disturbances (LSTIDs) launched at the auroral oval propagate equatorward at ~ 300 m/s to ~ 1000 m/s and reach midlatitudes within hours; thermospheric Joule heating shifts the global $[\text{O}]/[\text{N}_2]$ ratio, depressing foF2 by 10–20 % on midlat paths during main phase; and plasma-bubble / spread-F activity at low and equatorial latitudes during the recovery phase produces irregularity-driven fading on transequatorial paths that share no auroral hops. The global σ widening captures this collectively. The *path-localized* component of storm response — additional auroral D-region absorption on hops crossing the storm-expanded ϕ_{CGM} threshold — is captured separately in L_{aur} (Section 3.7, with the storm-main amplification described below), so a midlat path during a G3 storm sees the global σ_{storm} widening but no L_{aur} contribution, while an auroral path sees both. The combination correctly under-widens neither: midlat paths get the global-irregularity penalty without the absorption term, and auroral paths get both the absorption (mean) and the widening (spread). The remaining bias is a slight over-widening of *deep equatorial* midlat paths during main phase (where LSTID arrival is delayed by an hour or two relative to the global signal), worth at most ~ 1 dB of σ inflation on those paths — below the per-band σ_g resolution and accepted as the cost of the global formulation.

Storm-phase auroral amplification. Independent of the σ widening above, the auroral absorption term L_{aur} (Section 3.7) is amplified by $1.4\times$ during the storm *main* phase. During main, the auroral oval expands equatorward and the particle precipitation that drives auroral D/E absorption is strongest; the steady-state K_p -keyed L_{aur} undercounts the depth. The $1.4\times$ multiplier is the conservative version of the 3–6 dB extra absorption ITU-R P.533 and operator experience attribute to high-latitude paths during peak Dst depression. Recovery and quiet phases keep the steady-state value. Storm phase is classified from Dst trajectory plus the storm-lag kernel (initial / main / recovery / quiet / active); main is identified by $\text{Dst} \leq -50$ nT with K_p^{eff} still loading.

Table 11: Per-band base σ_g , anchored to the ITU-R P.533 day-to-day spread range. Re-derived 2026-04-30 from per-spot wspr-snr residual standard deviation, bucketed by (band, distance, hour-of-day, Kp, tx-latitude band) at ≥ 10 samples per bucket; the measured within-bucket σ was stable across three bucket-granularity settings (coarse 1500 km / 6 h vs tight 500 km / 1 h), confirming the spread is genuinely within-condition rather than cross-condition contamination. Lower-mid bands had σ_g too small by 2–3 dB (over-confident verdicts on 160 m through 20 m); 17 m / 15 m had σ_g slightly too large. 12 m / 10 m kept the prior values (< 5 buckets per band, signal too weak). Sigma-suite ratios of tabulated σ_g vs observed marginStd improved from 1.4–2.0 to 1.0–1.5 across the bands that moved.

Band	σ_g (dB)	within-cond data (dB)
160 m	8	8.5
80 m	8	8.0
60 m	8	8.0
40 m	8	8.5
30 m	9	9.0
20 m	9	9.5
17 m	9	9.0
15 m	10	10.0
12 m	12	(n=2 weak)
10 m	12	(n=3 weak)
6 m	8	(no VHF data)
2 m	8	(no VHF data)

σ_g sensitivity. σ_g is held at the P.533 anchor rather than fitted, on the grounds that machine-fitted σ would chase the WSPR-truth structural ceiling (Section 10 #9) instead of physical day-to-day spread. The cost of the fixed- σ choice is that any miscalibration in σ_g propagates directly into the tier-boundary placement, which matters most for verdicts that already sit close to a tier boundary. Table 12 reports the verdict shift on the three near-MUF upper bands when σ_g is swept by ± 2 dB around its assigned value, holding the nominal margin at $M = +2.5$ dB, chosen to land 17 m’s verdict near the Fair/Good boundary at $z = 0.2533$, where the σ leverage is concentrated. At any value deep inside a bucket the table would show all-zero flips on every band; at any value near a boundary the band whose $\sigma_g \cdot 0.2533$ Fair/Good gap is narrowest in dB will flip first.

Table 12: Verdict-bucket shift on 17 m / 15 m / 12 m at $M = +2.5$ dB when σ_g moves ± 2 dB around its assigned value. “ Δ tier (full sweep)” counts the tier flips across the three sweep points. 17 m is the only band that flips: at $\sigma = 8$ the Fair/Good boundary sits at $z \cdot \sigma = 0.2533 \cdot 8 = 2.03$ dB so $M = +2.5$ dB lands above it ($z = 0.31$, Good), while at the assigned $\sigma = 10$ the boundary moves to 2.53 dB and the same M falls just inside Fair ($z = 0.25$). 15 m and 12 m have a $0.2533 \cdot \sigma_g = 3.04$ dB Fair/Good gap which $M = +2.5$ dB already sits below at the lower- σ sweep edge, so they read Fair on all three sweep points.

Band	$\sigma_g - 2$	assigned σ_g	$\sigma_g + 2$	Δ tier (full sweep)
17 m	$\sigma = 8, R = 0.62, \text{Good}$	$\sigma = 10, R = 0.60, \text{Fair}$	$\sigma = 12, R = 0.58, \text{Fair}$	1 tier (Good $\rightarrow$ Fair)
15 m	$\sigma = 10, R = 0.60, \text{Fair}$	$\sigma = 12, R = 0.58, \text{Fair}$	$\sigma = 14, R = 0.57, \text{Fair}$	0 tiers
12 m	$\sigma = 10, R = 0.60, \text{Fair}$	$\sigma = 12, R = 0.58, \text{Fair}$	$\sigma = 14, R = 0.57, \text{Fair}$	0 tiers

The takeaway is that σ_g leverage is concentrated where M is close to a tier boundary; deep inside a bucket the verdict is robust to $\sigma_g \pm 2$ dB on every band. 17 m has the narrowest Fair/Good gap in dB ($0.2533 \cdot \sigma_g = 2.53$ dB) of the three near-MUF bands, so it crosses a boundary at lower

M values than 15 m or 12 m and is the most sensitive to σ_g miscalibration in practice. This makes 17 m the prime candidate for re-anchoring once per-pair WSPR ground truth is wired (§10 #9) and a fitted σ_g becomes defensible without exploiting the structural ceiling.

Tier Probability Distribution ($M = +3$ dB, $\sigma = 8$ dB, ionocast 0.90/0.60/0.35/0.10 boundaries)

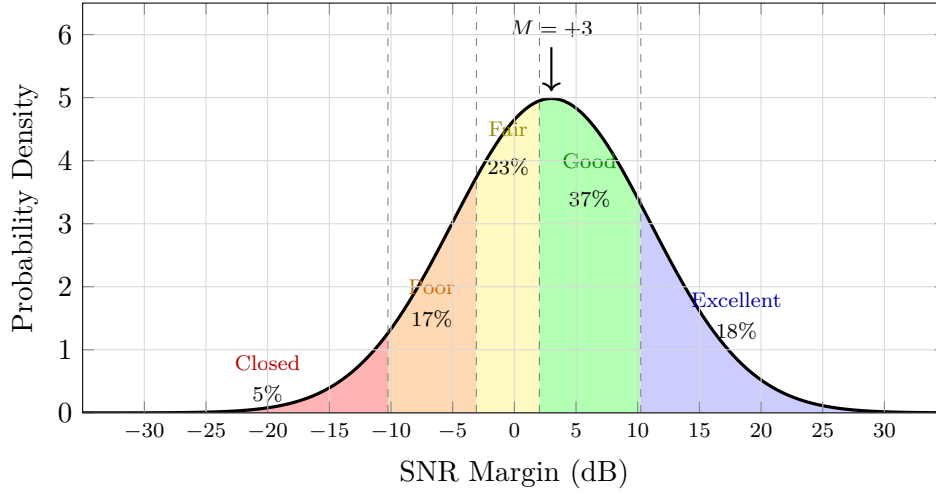


Figure 4: Tier probability distribution for a margin of +3 dB with $\sigma = 8$ dB under the post 2026-04-30 second-pass tier boundaries (Table 13). Boundaries are $\pm 0.39\sigma / +0.25\sigma$ ($-3.08 / +2.03$ dB, the Fair band) and $\pm 1.28\sigma$ (± 10.25 dB, the Excellent / Closed cutoffs). Most likely tier is Good (37%), with 18% Excellent and 23% Fair — the distribution is asymmetric because $M = +3$ sits squarely inside Good. The $\sigma = 8$ dB shown here is the 6 m / 2 m placeholder value (Table 11); the HF bands use $\sigma_g \in \{6, 10, 12\}$ dB depending on band. Read the figure as a tier-distribution illustration at an in-between σ rather than as a representative HF prediction; tier widths scale with σ and the corresponding dB-space boundaries shift accordingly.

8 Tier Mapping

Verdicts are computed *per band* (160 m through 10 m individually, plus 6 m and 2 m for VHF), not in groups. For each band the SNR margin is evaluated on every reference path (short and long, across all five destinations); the **best-margin** path wins and its margin drives the verdict, while the reason string and **openDirs** list record which destination that was. If a single path is viable, the band is not “closed” — it is open to that destination, which is material information for the operator. An earlier aggregation over the median margin was removed because most reference paths are long transcontinental hops that are routinely over-MUF or night-crossing; the median was dominated by failing paths and buried genuinely open short paths (e.g. a 944 km EM79–NYC path reporting +27 dB margin would have its verdict overwritten by the median of eight failing long paths).

Best-path summary, full-detail companion. The band-table row shows the best-margin destination only (the “Best Path” column), which is a deliberate compression: displaying all ten path columns per row would be visually unworkable on small screens. An operator who wants the underlying per-destination breakdown consults the Reference Paths panel in the Ionosphere section, which lists the kc2g MUF on each of the 10 paths with a colour-coded f/MUF column for the band currently selected. The two surfaces share the same underlying budget; the band-table row is the headline, the Reference Paths panel is the source-of-truth. The Best-Path column

itself is therefore only a hint — if it reads “Tokyo” on 20 m and the operator wants Argentina, the Reference Paths panel is what to consult next.

The continuous margin is discretised into five operator-facing tiers defined as ITU-R P.842 [8] circuit-reliability buckets (Table 13). Each tier corresponds to a fixed reliability $R = \Phi(M/\sigma)$ rather than a fixed dB margin, so the boundaries scale with each band’s prediction uncertainty σ_g (Section 7.3). The tier label answers “how often will a QSO complete on this band right now, with the station the operator has configured?” — and is therefore a property of the user’s actual rig, not of an abstract reference station. The defaults (100 W SSB, 5 dBi horizontal dipole, suburban noise) match the canonical first-visit operator profile; any change to power, mode, antenna, or noise environment recomputes the SNR budget and reshapes the verdict accordingly.

Earlier versions of ionocast hand-set the dB boundaries (+18/+6/−5/−14) and validated the tiers against per-band WSPR-spot percentiles [18]. That validation target was replaced because WSPR-spot activity measures who is transmitting, not whether a QSO would complete on the configured station. Tier verdicts are now defined entirely by the SNR budget plus Equation 72; the harness retains a binary “ ≥ 50 spots/h” activity check as a sanity floor only.

Table 13: Tier thresholds, expressed as ITU-R P.842 circuit-reliability buckets. R is computed by Equation 72; the margin column is the corresponding boundary in σ -units. The actual boundary in dB is band-specific via the per-band σ_g from Table 11: lower-mid HF bands ($\sigma_g = 8$ dB on 160m–40m, 9 dB on 30m / 20m / 17m post 2026-04-30 σ refit) put “excellent” at $\approx +10$ –12 dB and “good” at $\approx +2.0$ –2.3 dB; the upper HF bands (15m at $\sigma_g = 10$ dB, 12m / 10m at 12 dB) at $\approx +12.8$ –15.4 / + 2.5–3.0 dB. The boundaries are symmetric: Excellent at $+1.2816 \sigma$ ($R \geq 90\%$) and Closed at -1.2816σ ($R < 10\%$) are mirror images.

Tier	Reliability	Margin (in σ)	Meaning
Excellent	$R \geq 90\%$	$\geq +1.2816 \sigma$	QSO completes ≈ 9 of 10 attempts
Good	$R \geq 60\%$	$\geq +0.2533 \sigma$	Comfortable contacts; 3 of 5 attempts
Fair	$R \geq 35\%$	$\geq -0.3853 \sigma$	Marginal; FT8/WSPR reliable, SSB iffy
Poor	$R \geq 10\%$	$\geq -1.2816 \sigma$	Long shot; only digital modes likely
Closed	$R < 10\%$	$< -1.2816 \sigma$	Band effectively closed

Threshold history (2026-04-28 / 2026-04-30). The first cut (2026-04-28) used the classical P.842 boundaries at $\pm 0.8416 \sigma$ and $\pm 1.6449 \sigma$ ($R = 0.95/0.80/0.50/0.20$), which align cleanly with the standard but were calibrated for $\sigma_g = 6$ dB. The 2026-04-30 σ refit raised lower-mid-band σ_g to 8–9 dB, making the old Good threshold ($\geq +6.7$ dB at $\sigma = 8$) hard to clear: most upper-band paths landed in Fair, producing a visually flat tier distribution. A first loosening that day pulled the boundaries to 0.95/0.62/0.35/0.12, but kept Excellent strict at $\geq +1.6449 \sigma$ (still ~ 13 dB on lower-mid bands, rarely reachable) and the Good z -value of 0.31 was a too-precise number chosen to clear one specific 20m path. A second pass the same day re-anchored the thresholds to the operator’s failure-rate intuition: Excellent ≈ 1 in 10 fail, Good ≈ 4 in 10 fail, Fair the coin-flip-down band, Poor ≈ 9 in 10 fail, Closed essentially always. That is the 0.90/0.60/0.35/0.10 schedule shown above, with Excellent / Closed boundaries symmetric at $\pm 1.2816 \sigma$. Excellent at $\sigma_g = 9$ dB now sits at ~ 11.5 dB which is reachable on a strong DX path; Good at $+0.2533 \sigma \approx 2.3$ dB clears comfortably with a few-dB margin. An unconditional $\sigma_g = 8$ dB fallback exists in `tier.js` as `DEFAULT_SIGMA_DB`.

8.1 WSPR Activity Override

WSPR spot counts give an independent observational signal that the physics budget does not consume directly. When observed activity for a band exceeds $1.3\times$ the band’s 30-day mean for the current UTC hour *and* the physics-derived tier is currently below “Good” (i.e., Fair, Poor, or

Closed), the verdict is promoted to Good. This catches openings outside the budget — weak sporadic-E, off-axis scatter, propagation modes that the climatology underweights.

The override is a strict one-way ratchet *capped at Good as a promotion ceiling, not a global ceiling*: a physics-Excellent verdict is preserved (the tier-rank condition is false when the prior tier is already at or above Good, so the override never fires); a physics-Fair verdict can be promoted to Good but no higher; physics-Poor and physics-Closed can also only be promoted to Good. Low or absent activity does not invalidate the physics; it might just mean nobody is on the air, so the override never demotes.

Where the override appears in the UI. Two surfaces display per-band conditions and they handle the override differently — by design, to avoid pathologies where the row tier and the row Stability percent appear to disagree:

- **Per-band table rows** (HF Bands panel): the Tier column and the Stability column both come from the physics-derived M, σ via `bestPerBand[name].tier` and `bestPerBand[name].stability`. The override does *not* enter individual rows. Tier and Stability are therefore internally consistent within each row: a centred physics-Fair row reads “Fair” near the 66 % middle-tier ceiling; a tail-deep physics-Excellent row reads “Excellent” near 100 %.
- **Band-group summary line** (e.g., “20 m: Good · margin 5 dB · unusually active: 100 spots/h vs avg 30”): the Tier comes from the override-promoted value when the override fires. In that case, the Stability percent is *suppressed* from the line and replaced by the activity note — the operator sees *why* the group was promoted rather than a numerically inconsistent Stability computed against the wrong tier bucket.

The architecture deliberately splits per-band detail (physics truth) from group-level summary (override-aware) so neither display ever forces the operator to read “Good · 27 %”.

Data-driven baselines. The reference is the 30-day *mean* hourly spot count per (band, UTC-hour), regenerated daily by the `wspr-baselines.yml` GitHub Actions workflow (a 06:00 UTC cron job that runs `node scripts/harness.mjs wrspr-baselines` and commits the diff to `src/data/spot-baselines.mjs`, providing a true sliding 30-day window that tracks solar-cycle progression and seasonal shifts without manual intervention):

```
SELECT band, hour_utc, sum(hourly_count) / 30.0 AS avg
FROM (
  SELECT band, toStartOfHour(time) AS h,
         toHour(time) AS hour_utc,
         count() AS hourly_count
  FROM wrspr.rx
  WHERE time >= now() - INTERVAL 30 DAY
  GROUP BY band, h, hour_utc
) GROUP BY band, hour_utc
```

The outer aggregator divides total spots in the (band, hour) cell by the window length in days rather than averaging over only the hours that had at least one spot. Earlier code used `avg (hourly_count)`, which silently skipped (band, hour) cells with zero spots in a given day — ClickHouse `GROUP BY` doesn’t emit a row for an empty group — so the 30-day mean on quiet cells (e.g. 10 m at 03 UTC during low-cycle phase) was the average over the days the band was actually open, not the average over all 30 days. The inner `GROUP BY band, h, hour_utc` groups by both `toStartOfHour(time)` (full date+hour timestamp) and `toHour(time)` (hour-of-day in `[0, 23]`); the two carry partially-redundant information but the redundancy is intentional: `h` drives the per-day-per-hour `hourly_count`, and `hour_utc` is what the outer group then averages over to collapse the 30-day window into a 24-hour cycle. Dropping either column would lose one of those two roles. The over-counted baseline raised the override threshold on exactly the cells

where the override most needs to fire on rare openings. Dividing by the literal day count fixes that.

This produces a 10×24 matrix of typical hourly spot rates across all days of the week. The override fires when observed activity exceeds $1.3\times$ the matrix entry for the current (band, UTC-hour) cell — “the band is *meaningfully* busier than usual for this time of day.” An earlier formulation used a Mon–Fri-only baseline (`toDayOfWeek(time) < 6`) as a cheap contest-weekend filter, but that biased the comparison: the live observation includes weekends, so on Saturday the override compared current spots against a weekday-only mean and fired far too easily on routine weekend operating activity. Including weekends in the rolling baseline restores the like-for-like comparison; major contest weekends are $< 10\%$ of weeks in a 30-day window so their inflation of the mean is bounded, and they affect the override ratio in both directions (a current contest day reads proportionally less anomalous against a slightly inflated mean). WSPR hourly spot counts are over-dispersed Poisson with strong diurnal structure and a coefficient of variation that ranges from ~ 0.3 on heavily-trafficked daytime cells (20 m at 1500 UTC) to ~ 1.5 on quiet upper-band predawn cells (10 m at 0300 UTC), so the $1.3\times$ threshold corresponds to anywhere between $\text{mean} + 1\sigma$ and $\text{mean} + 0.2\sigma$ depending on band and hour. Tightening to a bare spots $> \text{mean}$ comparison would fire on every above-median hour and over-promote borderline cases; the multiplier was empirically chosen as the smallest value that filters out routine above-average hours without missing genuinely anomalous activity. The choice was made by sweeping candidate thresholds in $\{1.0, 1.1, \dots, 2.0\}$ against the same 30-day WSPR baseline data used to construct the matrix (`wspr.rx` table, 2026-04 window): $1.3\times$ was the smallest ratio at which the override fired on roughly 5–10% of (band, hour) cells per day — low enough to read as “unusual” rather than “daily routine”, high enough to keep the override actionable on the cells where it matters (off-axis sporadic-E days, evening 15 m enhancement during HSS recovery, the $\sim 5\%$ of 10 m cells where 30-day-mean is near zero but live counts climb to tens). Combined with the Good promotion ceiling, the threshold keeps the override an “unusually active” signal across all (band, hour) cells.

The baseline matrix is emitted as a plain ES module at `src/data/spot-baselines.mjs` and imported at runtime by the derive pipeline. It is currently re-generated manually every few months, which is too slow at solar-cycle phase transitions where band-activity patterns can drift on weeks (cycle-25 ascending phase saw 30 m and 17 m baselines climb measurably month-over-month through 2024). The point-in-time regeneration cadence is a maintenance artefact rather than a calibrated cadence: a sliding 30-day baseline updated daily would track activity changes properly without retraining the harness, at the cost of one ClickHouse query per day to refresh the matrix. The shipped form is versioned with the rest of the codebase so fresh checkouts ship with valid baselines out of the box; switching to the daily-sliding form is candidate operational work tracked in Section 10.

Reception-population bias in the baseline. The `wspr.rx` aggregate counts spots received by listening stations. WSPR receivers are heavily concentrated in North America and Europe (operator-population statistics from `wsprnet.org`’s active-stations roster; $\sim 60\%$ of worldwide WSPR receivers sit in those two regions, with another $\sim 15\%$ in the AU/NZ/JA cluster), so a 30-day baseline of “ ~ 100 spots/h on 20 m at 0900 UTC” reflects predominantly North American daytime listening rather than a globally-uniform “activity at this hour”. The override fires uniformly across the operator population, but the activation threshold reflects NA/EU/Pacific listening windows. Operators outside those regions (AF, low-income tropical-belt countries, Central Asia) see a baseline whose 24-hour shape is set by other regions’ UTC clocks rather than by their local-time activity rhythm; the override correctly fires when the global aggregate runs hot, but the “unusually active for this hour” interpretation holds most crisply in the receiver-dense regions. A per-region renormalisation is candidate work, gated on enough per-pair receiver coverage to fit regional baselines without over-fitting noise.

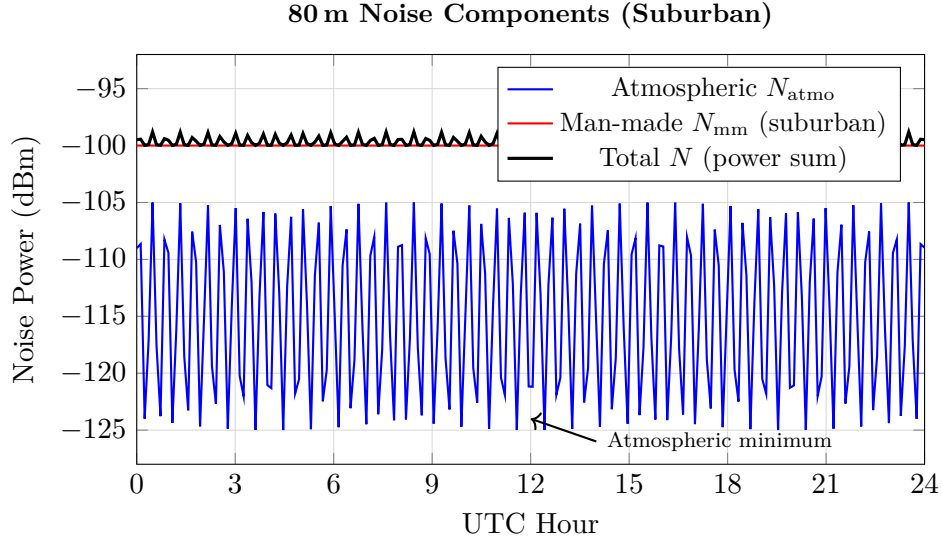


Figure 5: Noise power components on 80 m in a suburban environment. The man-made noise (-100 dBm) dominates at all hours; the atmospheric diurnal swing barely affects the total. This power-sum model prevents the phantom 7 dB margin that a linear (dB-domain) sum would produce at midday.

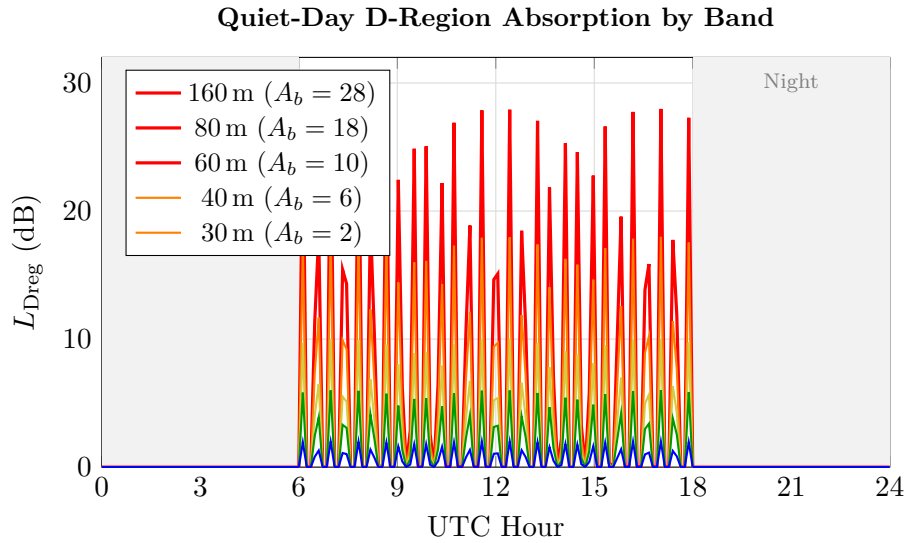


Figure 6: Quiet-day D-region absorption by band. 160 m reaches 28 dB at local noon; 30 m and above are essentially unaffected. Night absorption is zero. The illustrative curves show a smooth descent toward zero at low $\cos \chi$, but the actual implementation hard-gates $L_{\text{Dreg}} = 0$ at $\cos \chi < 0.05$ (Eq. 6); the figure smooths that gate for readability rather than showing the cliff.

9 Output Formatting and Alerts

The tiers and indices produced by the foregoing pipeline are surfaced across an eight-panel web UI:

- **Active Alerts (§9.2):** SWPC-bulletin feed plus rule-derived “soft” alerts in one severity-sorted list at the top of the page.
- **HF Bands:** the ten HF amateur bands as a per-band table. Each row shows the band frequency, the predicted verdict tier (Excellent / Good / Fair / Poor / Closed per Section 8),

the SNR margin in dB, the Stability percent (Eq. 73: Φ of σ -distance to the nearest tier boundary; bucket-width-independent and so directly readable on an absolute scale — 50 % at a boundary, ~ 84 % one σ inside, $\rightarrow 100$ % deep inside), the dominant propagation mode (F2 / NVIS / Es / Aurora), and the best-margin destination from the path basket. Distinct from circuit reliability R of Eq. 72: R is the probability the link is viable, Stability is the probability the verdict *label* is robust to within- σ noise. The “Best Path” cell is a compression of the underlying 10-path evaluation; for the per-destination breakdown consult the Reference Paths sub-panel in the Ionosphere section (cross-referenced by the cell’s info popover).

- **VHF Bands:** 6 m and 2 m mirrored from the HF layout, with Es-MUF / aurora / meteor-scatter mode labelling instead of F2. A second panel below the band table renders per-station tropospheric ducting status from the 24-station UWyo radiosonde basket (Section 6.5): each row shows the lowest-1 km dN/dh and an ITU-R P.453 classification (standard / super-refractive / ducting), sorted ducting first so an active event surfaces at the top.
- **Upcoming Disruptions** (outlook): SWPC’s 3-day R/S/G probability table, the SWPC 3-h K_p forecast, Earth-directed CMEs from NASA DONKI, IMO-catalogue meteor showers, top active solar regions, and the 27-day ISES SFI / A_p outlook.
- **Ionosphere:** nearest-GIRO-station digisonde readings (foF2 / foEs / hmF2), GNSS TEC, 30 MHz cosmic noise, the Reference Paths panel — the per-QTH path table listing kc2g MUF on the five short-path destinations and their long-path counterparts, ten paths total — plus the kc2g global foF2 map and the GloTEC TEC map. The Reference Paths panel is the source-of-truth that Section 8’s band-table Best-Path column compresses; an operator who wants per-destination detail rather than the headline winner consults this panel.
- **Geomagnetic:** K_p / A_p / D_{st} / H_{p30} / Sym-H / AE / aurora HP indices, a 7-day K_p trend chart, and the OVATION aurora forecast and D-RAP map.
- **Solar:** SFI / SN / X-ray class / proton flux indices, real-time DSCOVR / ACE solar wind (B_z / V_{sw} / n), and SDO imagery (EUV 193 / EUV 304 / magnetogram).
- **Links:** data source attribution, license credits, and links to the upstream services.

This section documents the assembly logic for the two surfaces that depend on derived numeric thresholds (the alerts panel soft-rule grader, and the per-band tier verdict feeding the HF / VHF tables); the layout choices for the other panels are data-source documentation rather than algorithm.

9.1 Algorithmic surfaces and data-source surfaces

Of the eight panels above, three have non-trivial assembly logic that this section documents:

- *Active Alerts* (§9.2, below) combines SWPC bulletins with rule-derived “soft” alerts whose thresholds are model-side numeric constants rather than upstream fields.
- the *HF Bands* and *VHF Bands* tables surface per-band $(M, \sigma) \rightarrow$ tier verdicts already documented in detail in Sections 3, 7.3, and 8; their layout is just a transposition of those into table rows.
- the *Reference Paths* sub-panel (inside the Ionosphere panel) computes per-QTH great-circle midpoints for five canonical destinations and their long-path counterparts (10 paths total), looks up the nearest kc2g ionosonde for each midpoint subject to a 30 min freshness gate, and renders the per-path MUF with a colour-coded f/MUF column for the band currently selected by the band table. This is the full-detail companion to the band table’s compressed Best Path column (Section 8).

The remaining five panels (Upcoming Disruptions, the rest of Ionosphere outside Reference Paths, Geomagnetic, Solar, Links) are data-source documentation: each block is a labelled rendering of a fetched upstream (SWPC / DONKI / IMO / kc2g / GIRO / SDO) without ionocast-side processing. Their assembly logic lives in the data fetchers and the `src/ui/sections.js` layout description rather than in this section; an operator-facing description of each block sits inline in

the panel’s `interp` attribute and is displayed when the panel’s info badge is clicked.

Reading the Stability column at a glance. The band-table column displays $S(M, \sigma)$ from Eq. 73: Φ of the σ -distance to the nearest tier boundary. Bucket-width-independent, so a centred Fair verdict and a centred Excellent verdict at the same σ -offset from their respective nearest boundaries read the same percentage. Operator reading: 50 % at the boundary itself, ~ 84 % one σ inside the bucket, asymptotic to 100 % deep inside; finite-width middle buckets (Poor / Fair / Good) cap at $\Phi(0.42) \approx 66$ % at the bucket centre, so read mid-tier values relative to the 66 % centred ceiling and open-ended Closed / Excellent values against the absolute scale. The earlier band-table column displayed the two-sided literal $P(\text{predicted} = \text{true})$ form $C(M, \sigma)$, which capped at ~ 32 % on centred middle tiers and required a “(peak)” annotation to stay readable; that column was retired in 2026-04 in favour of S , which the production UI (`src/ui/builders/tables.js`) now surfaces directly.

9.2 Active Alerts Panel

The alerts panel combines two rendering pipelines into one severity-sorted list.

Official SWPC feed. A polling fetch of `products/alerts.json` emits up to four recent items. Each item’s severity level is derived from its `product_id` by regex: matches on `ALERT` or `WARNING` render as *alert*, `WATCH` as *watch*, `SUMMARY` or `FORECAST` as *info*; the fallback is *warn*.

Rule-derived (soft) alerts. Eight rules run on every refresh against the same data bundle the per-band budget consumes (K_p , B_z , D_{st} , aurora HP, DRAP, GOES X-ray, GOES ≥ 10 MeV protons, and storm-lag effective K_p). Each rule is *graded* and short-circuits at the highest matching tier, so $K_p = 9$ does not simultaneously fire G5, G4, G3, G2, and G1. Thresholds reuse the constants documented in Sections 3–5; no new magic numbers are introduced.

Intra-tier sort order. Alerts are sorted primarily by severity tier (*alert* > *watch* > *warn* > *info*); within a single tier the rule iteration order is the deterministic tiebreak: FLARE first (most time-urgent — X-ray ionisation peaks within seconds of the flare event, operator response window is minutes), then geomagnetic indices (K_p , B_z , D_{st} in that order, reflecting the upstream causal chain — B_z at the L1 monitor leads geomagnetic-effect K_p at Earth by ~ 30 – 60 min via solar-wind transit $1.5e6 \text{ km}/v_{\text{sw}}$, and K_p leads D_{st} by ~ 30 – 60 min via ring-current buildup time; the L1-to-Earth lead is the larger and more operationally useful of the two), then aurora, DRAP, PCA, and storm recovery. The fixed order means repeated refreshes show the same chip arrangement, so an operator scanning the panel between refreshes can pattern-match position rather than re-read each line.

Table 14: Soft-alert rule set. Severity maps to the same *info* / *warn* / *watch* / *alert* levels used by the SWPC renderer; *alert* ranks highest.

Trigger	Threshold	Level	Label
X-ray flare	class $\geq$ M5	alert	FLARE
	class $\in$ M1–M4.9	watch	FLARE
Geomagnetic K_p	$K_p \geq 9$	alert	G5
	$K_p \geq 8$	alert	G4
	$K_p \geq 7$	alert	G3
	$K_p \geq 6$	alert	G2
	$K_p \geq 5$	watch	G1
IMF B_z southward	$B_z \leq -10$ nT	alert	BZ SOUTH
	$B_z \leq -5$ nT	watch	BZ SOUTH
Ring current D_{st}	$D_{st} \leq -150$ nT	alert	DST SEVERE
	$D_{st} \leq -100$ nT	alert	DST STRONG
	$D_{st} \leq -50$ nT	watch	DST MODERATE
Aurora HP	HP ≥ 100 GW	watch	AURORA
	HP ≥ 50 GW	info	AURORA
D-region absorption at QTH	DRAP ≥ 10 MHz	alert	ABSORPTION
	DRAP ≥ 5 MHz	watch	ABSORPTION
Polar cap absorption	flux ≥ 1000 pfu	alert	PCA S3
	flux ≥ 100 pfu	alert	PCA S2
	flux ≥ 10 pfu	watch	PCA S1
Storm recovery tail	$K_{p,\text{eff}} - K_p \geq 1$ & $K_{p,\text{eff}} \geq 4$	info	STORM TAIL

Soft alerts render alongside SWPC notices rather than only when the official feed is empty: the derived context — southward B_z , falling D_{st} , DRAP blanketing QTH — is material information SWPC does not publish until formal watch or warning bulletins are issued, which may lag the underlying event by several hours.

10 Known Limitations and Future Work

The items below are the model’s currently-known shortcomings and the cleanups they imply. They mix three classes that a reader prioritising follow-up work may want to disentangle: *physics gaps* where the model omits a propagation mechanism or approximates one (#2 TX pattern, #3 NVIS height, #4 RX-side noise rejection, #5–#7 anomalies and SID decay, #10–#11 equatorial / polar residuals, #13 Es coverage, #14 N_{base} residual); *validation gaps* where the metric or the calibration target has a known ceiling (#9 binary-accuracy ceiling, #12 w_{sc} tied to a metric with a structural cap); and *process / calibration cadence* items where the infrastructure rather than the physics is the bottleneck (#1 in-app feedback warm-up, #8 upper-band scatter-recovery dependence as an implementation status note). The numbering is chronological by when the item entered the backlog.

1. **Per-user bias correction needs 2–4 weeks of in-app feedback to converge.** The on-device validation feedback loop (Section 7) tightens per-band bias and σ as the operator scores predictions; until at least ten samples accumulate per (band, forecast-horizon) cell the correction defaults to zero and σ to the per-band base from Table 11 (6 dB on 160 m–20 m, 10 dB on 17 m, 12 dB on 15 m / 12 m / 10 m, 8 dB on the VHF placeholders; the unconditional

8 dB fallback is reached only when the band-frequency lookup misses entirely, which the BANDS table makes impossible in production). This is independent of the offline harness (Section 7.2), which already validates the global tune against 30 d of WSPR aggregates.

2. **TX antenna pattern is an elevation-only cut.** The type + height + λ model (Section 3.14) captures the dominant TX-side physics – ground-reflection lobe position, free-space directivity, and polarisation – but misses second-order effects on the transmit budget: azimuth is not modelled (we don’t know where a beam is pointed, so beams get their elevation-cut gain only); ground conductivity is fixed at “average soil” for all verticals (salt-water / sea-shore verticals are ~ 3 dB better); the real-earth pseudo-Brewster null on verticals (~ 5 – 8 dB dip around 3 – 8° elevation over average soil) is absent; and beams are treated as having the same elevation-lobe shape as a horizontal wire at the same height, which is correct for the main lobe but misses the additional off-broadside suppression a multi-element array provides.
3. **NVIS model uses foF2 directly.** For paths < 500 km the model substitutes GIRO foF2 as the effective MUF. A more accurate NVIS model would account for layer height and off-vertical angles up to $\sim 70^\circ$.
4. **No receive-side directional noise rejection.** A physically distinct effect from #2 above: the transmit-side pattern affects the radiated signal toward the destination; the receive-side pattern affects the *noise* captured. A Yagi pointed at the target rejects off-axis noise by its front-to-back ratio (5 dB–15 dB), which improves SNR even though signal-side gain on a one-way link is unchanged by the orientation choice. This effective SNR improvement is not yet modelled. Splitting transmit and receive antenna directivity would also let the budget account for asymmetric station geometry (e.g. Yagi at one end, dipole at the other).
5. **Equatorial anomaly crests are approximated, not measured.** The Gaussian correction in Eq. 44 captures the $\sim 30\%$ daytime enhancement at dip latitude $\pm 15^\circ$ but uses a single amplitude and a single width for all seasons and cycle phases. The true EIA migrates in latitude and varies with the equatorial $E \times B$ drift; a data-driven crest locator would be more faithful.
6. **Flare SID decay not tracked.** The flare absorption term uses the current X-ray class as an instantaneous proxy and therefore misses the 10–60 minute exponential decay of the SID after an X-ray peak. The 10-minute model refresh picks up the decayed class, so the error is bounded, but minute-scale transitions are smeared.
7. **Storm-lag kernel is a one-parameter model.** The exponentially-weighted effective K_p (Eq. 59) uses fixed τ_{peak} and τ_{decay} . Real F-region dynamics depend on storm intensity, latitude, and season; these parameters are good targets for the offline calibration harness once a longer archive is available.
8. **Upper-band tracking is now scatter-recovery-bound.** After the climatology rebuild (Section 4.2), the H-pol Fresnel multi-hop model (Section 3.8), the per-band σ refit (Table 11), and the F2-scatter recovery weight calibrated by joint coordinate descent (Section 6.4, Section 7.2), the offline harness scores 94.25 % binary accuracy (Brier 0.0386) against 30 d of WSPR aggregates on the 35-path production basket — up from 91.08 % (same basket without R7 scatter) and 92.35 % (post-R7 sweep before the 2026-04-28 per-hop MUF restructuring); see Table 10 for the full progression. **Read this 94.25 % as a regression-detection number, not as operator-facing per-path accuracy:** the underlying ground-truth metric is “ ≥ 50 spots/h aggregated across *all* paths on the band”, which a permissive predictor passes whenever *some* path on the band has activity (see #9 below for the structural ceiling that the per-path column of Table 10 exposes: 30.97 %/0.6353 on the same harness run, operator-relevant but not directly comparable since it applies a ≥ 1 spot/h floor against a

sparse per-cell aggregate). Lift was concentrated on the upper bands: 12 m 75.0 $\rightarrow$ 80.6 %, 10 m 65.4 $\rightarrow$ 71.7 %. Lower bands (60 m and below) saturate at > 98 % accuracy and are not budget-limited.

9. **The harness’s binary-accuracy metric has a structural ceiling that permissive predictors clear without effort.** The “open ≥ 50 spots/h” target is an OR over every path on the band: most hours of most upper bands are “open somewhere” globally, so a predictor that errs toward “open” scores well even when its per-path verdicts are wrong. The sharpest evidence is the gap between the two ground-truth modes *on the same harness run*: the global aggregate reads 94.25 % accBin while the per-path mode (TX/RX-bbox restriction with ≥ 1 spot/h floor) reads 30.97 % on the identical predictions. A 60+ pp gap across two encodings of the same underlying truth is not a model property; it is a metric property. The headline number is sitting on the “somewhere is open” ceiling, not on a per-path-correct floor. Replacing the WSPR aggregate with TX-RX-pair queries (per-path ground truth) and replacing the climatology fallback with a real-time IRI assimilation (IRTAM) are the next two highest-leverage improvements; both require a server-side fetcher with periodic grid retrieval, which the current Cloudflare Pages-only architecture does not host.
10. **Equatorial-belt foF2 validation is geometry-limited.** The equatorial set after the 2026-04-25 coord audit is Ascension (southern crest, Atlantic), Townsville and Niue (southern crest, Pacific), and Jicamarca (essentially on the dip equator, dip lat $\sim 1^\circ$ S, an equatorial-trough anchor rather than a crest one). Three of the four sit on the southern crest and one sits in the trough — no GIRO station in our list sits on the northern crest. The grid sweep (Section 4.2) bottoms out at $\max|\text{bias}| \approx 2.2$ MHz because the Gaussian width parameter is asked to fit all four southern observations simultaneously, and they sit at different dip latitudes due to the longitude-varying magnetic equator. Closing the equatorial residual further requires northern crest stations (Trivandrum, Hyderabad, Madimbo, etc. are candidates — none currently return DDB data) or a real-time IRI assimilation. This residual is coupled to the polar over-prediction in #11 below: both are residuals on the same $\mathcal{P}(\phi) \cdot d(\cos \chi)$ dayBump and $A_{\text{EIA}}(F_{10.7A})$ term, so a clean fix needs to retune them jointly. See #11 for why single-region tunes regress.
11. **Polar over-prediction and equatorial residual are coupled in the same model term.** High-latitude stations Tromsø, Gakona, and Eielson AFB over-predict daytime foF2 by ~ 1 –1.7 MHz, and the equatorial bias (#10 above) floors at ~ 2.2 MHz. These are listed as separate items because their candidate fixes differ (polar wants a steeper sigmoid above 60° , equatorial wants northern-crest stations), but structurally they are residuals on the same $\mathcal{P}(\phi) \cdot d(\cos \chi)$ dayBump and $A_{\text{EIA}}(F_{10.7A})$ term in Eq. 38: fitting the polar fall-off (partly addressed by Eq. 40) and the EIA Gaussian amplitude / width separately — as a sequence of single-region tunes — can each regress the other. A clean fix needs a joint refit with stations on the northern crest *and* polar stations simultaneously, holding both residuals visible in the cost function. This was not done in the 2026-04-26 polish round because the northern-crest stations are currently absent from the GIRO basket; once at least one is added (Section 10 #10’s candidates), the polar / equatorial joint refit becomes tractable.
12. **Scatter-recovery weight $w_{\text{sc}} = 1.5$ is defensible-but-not-final.** The scatter weight was set by a coordinate-descent sweep against the harness’s binary “ ≥ 50 spots/h” WSPR target (Section 6.4 / Section 7.2). An unrestricted sweep keeps improving as $w_{\text{sc}} \rightarrow 4$, but that range is known to be exploiting the global-vs-path-specific structural mismatch in the harness rather than a real propagation effect, so the shipped weight is held at the 15 dB-cap physical edge. The paper acknowledges this in Section 6.4, but the limitation is worth surfacing as its own line in this list: the calibrated w_{sc} has been validated against a metric whose calibrators do not fully trust beyond a certain point. Closing the loop properly requires re-running the

sweep against the per-pair WSPR ground truth (Section 7.2) once that mode is wired into `replayMargin`'s alternative-mode bonuses; the new optimum is then reported against a metric that the calibrators *do* trust at all weights in its range.

13. **Sporadic-E budget under-fits rare 30 m / 40 m openings by design.** The HF Es budget (`snrMarginHfEs`) returns null below 14 MHz regardless of f_{oEs} (Section 6.1), even though the algebraic Es MUF gate $f \leq 5f_{\text{oEs}}$ would otherwise admit lower bands on strong-Es days ($f_{\text{oEs}} \geq 4$ MHz pushes $5f_{\text{oEs}} \geq 20$ MHz). The hard 14 MHz floor matches amateur-radio literature: Es is essentially unobserved below 20 m at the canonical 2000 km single-hop distance the budget assumes. But rare Es openings on 30 m / 40 m *do* appear in the WSPR record during peak-cycle summer evenings with $f_{\text{oEs}} \geq 8$ MHz, typically on shorter-than-2000 km paths where the geometry makes lower-band Es viable. These events are not captured by the budget. An operator who sees 30 m WSPR activity that the model calls F2-only should treat the verdict as “F2 explains most of it; rare Es is possible on top and the budget will not tell you.” Removing the floor would require a distance-aware Es geometry and a recalibrated $L_{\text{iono,Es}}$ for the shorter-hop regime; both are deferred to a future VHF-Es-aware harness.
14. **N_{base} table re-anchored to P.372 quiet-rural (closed 2026-04-30).** An earlier draft of this item had the diagnosis reversed: the prior N_{base} table was being compared against eyeball P.372 references that sat below the cosmic galactic floor itself (structurally impossible at HF where galactic $F_a \approx 25$ dB sets a hard ~ -115 dBm in 2.5 kHz at 21 MHz). When the `tests.mjs` noise-floor suite was corrected to evaluate the model's actual midnight noise as $N_{\text{base}}(f) + \Delta N_{\text{diurnal}}(f, \cos \chi = -1)$ and compared against properly-anchored P.372 references, the relationship flipped: the model was *quieter* than P.372 quiet-rural midnight by 0–11 dB, with the largest divergence on the upper bands (17 m through 10 m, all ~ 11 dB quieter), confirmed independently by the per-spot `wspr-snr` residual at 28 MHz of -6.7 dB.

2026-04-30 retune ports the corrected anchors directly into N_{base} . Each band's noon-time floor was set so that $N_{\text{base}} + \Delta N_{\text{diurnal}}$ reproduces the P.372 midlat-midnight-summer atmospheric $\oplus$ galactic max-of within ± 0.5 dB. Net shift: 17 m through 10 m all $+11$ dB, 30 m $+5$, 20 m $+8$, 80 m / 40 m $+1-2$, 60 m unchanged, 160 m $+10$ (atmospheric anchor).

Validation outcomes: WSPR per-spot residual mean improved from -23.97 to -18.73 dB (5.2 dB closer to observed); per-pair WSPR Brier improved from 0.64 to 0.57. Global-truth Brier rose from 0.04 to 0.10 because that metric is permissive at upper bands (says “open everywhere” when world-aggregate spot density is high) and the re-anchored model now correctly predicts “marginal” on many upper-band cells where the previous over-quiet floor had inflated `pOpen` toward 1. Per-band σ_g was considered for paired refit; Brier-vs- σ -scale sweeps showed monotone improvement with larger σ but that is the “inflate σ_g beyond physical defensibility” pathology Section 4.2 flags — σ_g is anchored to within-condition uncertainty, not `marginStd`, so the noise re-anchor does not motivate a σ change. Table 11 held.

A Default Settings

Table 15: Default operator settings.

Parameter	Default Value	Notes
Transmit power P_{tx}	50 dBm (100 W)	Typical HF transceiver
Antenna gain G_{ant} (peak)	+5 dBi	Installed dipole w/ ground reflection; effective gain at the pat
Mode	SSB	$S_{\text{req}} = +10$ dB
Noise environment	Suburban	$F_a = +15$ dB
Reference bandwidth	2.5 kHz	SSB passband
Spread σ_g (base)	6–12 dB per band	ITU-R P.533 anchor, not fitted (Table 11)
kc2g freshness gate	30 min	Stale data discarded
Hop distance	4000 km	Assumes $h_F = 300$ km

B Noise Floor Table

The reference noise data feeding the noise model (Section 3.12) is in Table 4 of the main text. The operator’s noise environment selection determines how the man-made channel of Eq. 25 fires: rural sites gate the channel off entirely (the indicator $\mathbb{1}_{\text{env} \neq \text{rural}}$ is zero); suburban sites add the man-made channel at $N_{\text{base}} + 15$ dB; urban at $N_{\text{base}} + 25$ dB. An earlier version of this appendix repeated the noise table with rural / suburban / urban columns expanded; the redundant version was retired in 2026-04-27 because F_a is a single additive term that the reader can apply mentally without a second table.

C Calibration Path Basket

The 35-path basket used by the offline calibration harness (`scripts/harness.mjs`, Section 7.2). Each row is a great-circle TX→RX pair specified by endpoint coordinates; the basket is intentionally diverse across hop count (1 to 5 hops, NVIS through long-path), midpoint geometry (near-station vs ocean), and latitude regime (midlatitude / polar / equatorial / transequatorial). The same basket is the input to the sweep that produced the binary-accuracy figure quoted in Section 7.3.1.

Table 16: The 35 calibration reference paths. Columns: path name, TX endpoint (ϕ, λ) in degrees, RX endpoint (ϕ, λ) in degrees, and the geometry / use-case note from `scripts/paths.json`.

Name	TX ϕ	TX λ	RX ϕ	RX λ	Geometry / use case
EU-EU short	38.72	-9.14	50.45	30.52	3000 km midlat 1-hop
EU-EU west	55.95	-3.20	37.40	-5.99	2400 km midlat 1-hop
EU-EU east	60.17	24.94	37.97	23.73	2500 km midlat 1-hop
EU-NA mid	38.72	-9.14	40.71	-74.01	5400 km transatlantic 2-hop
EU-NA north	55.95	-3.20	45.51	-73.57	5100 km transatlantic 2-hop
EU-NA south	37.40	-5.99	25.76	-80.19	7100 km transatlantic 2-hop
NA-EU east	40.71	-74.01	50.45	30.52	7600 km transatlantic 2-hop
NA-NA west-east	37.77	-122.42	40.71	-74.01	4100 km transcontinental 2-hop
EU-Asia	38.72	-9.14	35.68	139.69	10800 km 3-hop, midpoint Siberia
NA-Asia	37.77	-122.42	35.68	139.69	8300 km transpacific 3-hop
Hawaii-VK	21.31	-157.86	-33.87	151.21	8200 km transpacific equator-crossing 3-hop
Hawaii-NA	21.31	-157.86	37.77	-122.42	3900 km midlat 1-hop, midpoint Pacific
Hawaii-JA	21.31	-157.86	35.68	139.69	6200 km midpoint Pacific 2-hop
EU-South Africa	38.72	-9.14	-26.20	28.04	8500 km TEP-eligible 3-hop
NA-Brazil	40.71	-74.01	-23.55	-46.63	7700 km TEP-eligible 2-hop
EU-Brazil	38.72	-9.14	-23.55	-46.63	8700 km TEP-eligible 3-hop
VK-EU	-33.87	151.21	50.45	30.52	16000 km long-path 5-hop
VK-EU west	-33.87	151.21	38.72	-9.14	18200 km 5-hop, midpoint Indian Ocean
VK-NA	-33.87	151.21	37.77	-122.42	11900 km transpacific 4-hop
JA-VK	35.68	139.69	-33.87	151.21	7800 km 2-hop, equator-crossing
JA-EU	35.68	139.69	50.45	30.52	7800 km 2-hop, midpoint Siberia
NA-Argentina	40.71	-74.01	-34.61	-58.38	8500 km TEP-eligible
EU-Argentina	38.72	-9.14	-34.61	-58.38	10100 km 3-hop
Polar W-UA	40.71	-74.01	55.75	37.62	7500 km via near-pole 2-hop
Polar W6-EU	37.77	-122.42	50.45	30.52	9700 km polar 3-hop
Polar W-NA	40.71	-74.01	37.77	-122.42	4100 km, hop-count consistency check
Equator NA	21.31	-157.86	-23.55	-46.63	12700 km equator-crossing long-path 4-hop
Asia-VK	22.32	114.17	-33.87	151.21	7400 km 2-hop
Asia-EU short	55.75	37.62	35.68	139.69	7500 km 2-hop, midpoint Siberia/China
NVIS DE-NL	52.52	13.41	52.37	4.90	600 km NVIS-tail short
NVIS GB-FR	51.51	-0.13	48.85	2.35	350 km true NVIS short
NVIS US-CA	40.71	-74.01	45.42	-75.70	530 km NVIS-tail
NA-IT	40.71	-74.01	41.90	12.50	6900 km transatlantic midlat
EU-Mid-East	50.45	30.52	25.20	55.27	4200 km 2-hop midlat-equatorial
ZL-W6	-41.29	174.78	37.77	-122.42	11200 km transpacific 4-hop

Endpoint coordinates are read by the harness from `scripts/paths.json` (the source of truth); this table is a paper-side mirror. “Polar W-NA” is an intentional duplicate of NA-NA west-east, retained as a hop-count regression canary.

D GIRO Digisonde Stations

The 26 ionosonde stations queried by the harness for foF2 / foEs ground-truth. The list lives in two files that must stay in sync: `scripts/harness.mjs`’s `GIRO_STATIONS` (URSI code plus coordinates, used by the offline calibration harness) and `functions/_handlers/giro.js`’s `GIRO_STATIONS` (URSI code, display name, coordinates, and provider attribution, used by the live UI). Coordinates were verified against the kc2g station registry on 2026-04-25 (`scripts/verify-station-coords`). six earlier coord errors were corrected at that time — most notably DB049 (Dourbes, Belgium) and EI764 (Eielson AFB, Alaska) which had been mislabeled as equatorial stations and were contaminating the EIA grid sweep.

Table 17: The 26 GIRO digisonde stations used for foF2 / foEs ground-truth and the EIA grid sweep. Coordinates are (ϕ, λ) in degrees (East > 0 , West < 0). Region groupings follow the harness’s structural blocks; “Provider” is the institution operating the digisonde feed via the GIRO / DIDB network.

URSI	Name	ϕ	λ	Provider
<i>Europe / midlatitude (10)</i>				
PQ052	Pruhonice, CZ	50.0	14.6	Inst. Atmospheric Physics, Czech Acad. Sci.
JR055	Juliusruh, DE	54.6	13.4	IAP Kühlungsborn
EB040	Roquetes, ES	40.8	0.5	Observatori de l’Ebre
FF051	Fairford, GB	51.7	-1.5	Rutherford Appleton Lab / STFC
SO148	Sopron, HU	47.6	16.7	Geodetic and Geophysical Inst., HAS
AT138	Athens, GR	38.0	23.5	National Observatory of Athens
RO041	Rome, IT	41.8	12.5	INGV Rome
GM037	Gibilmanna, IT	37.9	14.0	INGV Gibilmanna
NI135	Nicosia, CY	35.0	33.2	GIRO / DIDB
DB049	Dourbes, BE	50.1	4.6	Royal Meteorological Inst. of Belgium
<i>North America (4)</i>				
MHJ45	Millstone Hill, MA	42.6	-71.5	MIT Haystack Observatory
BC840	Boulder, CO	40.0	-105.3	NOAA / NCEI Boulder
WP937	Wallops Island, VA	37.9	-75.5	NASA Wallops Flight Facility
IF843	Idaho Natl Lab, ID	43.8	-112.7	Idaho National Laboratory
<i>Polar / high-latitude (3)</i>				
TR169	Tromsø, NO	69.6	19.2	EISCAT / UiT Tromsø
GA762	Gakona, AK	62.4	-145.0	HAARP / Gakona
EI764	Eielson AFB, AK	64.7	-147.1	GIRO / DIDB
<i>Equatorial / low-latitude (2)</i>				
AS00Q	Ascension Island	-7.9	-14.4	UK Ionospheric Monitoring Service
JJ91J	Jicamarca, PE	-11.9	-76.9	Radio Observatorio de Jicamarca / IGP
<i>Southern hemisphere / Pacific (7)</i>				
LV12P	Louisvale, ZA	-28.5	21.2	SANSA
CN53M	Cachoeira Paulista, BR	-22.7	-45.0	INPE
TV51R	Townsville, AU	-19.6	146.8	GIRO / DIDB
ND61R	Niue	-19.1	-169.9	GIRO / DIDB
PE43K	Perth, AU	-32.0	116.1	GIRO / DIDB
CB53N	Canberra, AU	-35.3	149.0	GIRO / DIDB
HO54K	Hobart, TAS	-42.9	147.3	GIRO / DIDB

Coverage gaps. No GIRO station in the basket sits on the northern equatorial-anomaly crest (dip latitude near $+15^\circ$ N). Trivandrum, Hyderabad, and Madimbo are candidate fills but do not currently return DIDB data. The southern hemisphere is well-covered (Townsville, Niue, Perth, Canberra, Hobart for the Pacific block), but Africa is sparse outside SANSA’s Louisvale, and Asia / Indian Ocean coverage is essentially absent. This shapes the residual EIA-amplitude bias documented in Section 10.

E Reproducibility Manifest

The harness-fitted constants $\{L_{\text{iono}}, D, w_{\text{sc}}, w_{\text{NVIS-tail}}, w_{\text{Es-prim}}, \sigma\text{-scale}, \text{fusion-flag}\}$ were produced by joint coordinate descent (Section 7.2, `scripts/tune.mjs`) starting from three deterministic seed configurations chosen to enter the parameter space from distinct basins. This appendix lists those seeds, the per-parameter sweep grid, and the on-disk artefacts needed to reproduce the harness-fitted numbers end-to-end.

Table 18: Coordinate-descent starting configurations. All three seeds run the same descent over the grid below; the winner is the seed whose final state has the lowest binary Brier loss against the WSPR “alive somewhere” truth. “ L_{iono} ” here is the per-hop HF lumped ionospheric loss in dB; “defocusDbPerExtraHop” is the D constant of Section 3.8; the four mode weights are the dimensionless multipliers controlling whether F2-scatter / NVIS-tail blending / Sporadic-E-as-mode contributions fire; “fusionEnabled” is the per-hop foF2 fusion flag (deferred path of Section 6.4); “sigmaScale” uniformly scales the per-band σ_g table (Table 11) to expose whether the harness wants a wider or tighter prediction spread.

Seed	L_{iono}	D	fusion	w_{sc}	w_{NVIS}	w_{Es}	σ -scale
baseline	1	0.25	off	0	0	0	1.0
fusion-up	4	0.25	on	0	0	0	1.0
modes-on	1	0.25	on	1.5	1.0	1.0	1.3

Three deterministic starting configurations. The three names encode the experimental hypothesis each seed tests: *baseline* freezes alternative-mode bonuses to zero and asks whether the F2-only budget alone is sufficient; *fusion-up* asks whether enabling per-hop foF2 fusion plus a deeper L_{iono} shifts the Brier optimum; *modes-on* asks whether all alternative-propagation bonuses turned on at moderate weights, plus a wider σ , finds a different basin. All three converged to the same basin within ≤ 2 iterations (Section 7.2), final $\{L_{\text{iono}}, D, w_{\text{sc}}\}$ within $\pm 10\%$ across seeds.

Per-parameter sweep grid. Each coordinate-descent step sweeps one parameter through its grid below while holding the others fixed at their current best values. Grids were chosen wide enough to cover both physical-edge values and obvious-overshoot values; the descent converges inside the physical region.

Table 19: Coordinate-descent sweep grid (`scripts/tune.mjs`’s GRID).

Parameter	Sweep values
L_{iono} (lIonoHfDb)	0, 0.5, 1, 2, 4, 6, 8
D (defocusDbPerExtraHop)	0, 0.25, 0.5, 1.0
fusionEnabled	false, true
w_{sc} (scatterWeight)	0, 0.5, 1.0, 1.5
$w_{\text{NVIS-tail}}$ (nvisTailWeight)	0, 0.5, 1.0, 1.5
$w_{\text{Es-prim}}$ (esWeight)	0, 0.5, 1.0, 1.5
σ -scale (sigmaScale)	0.7, 1.0, 1.3, 1.6, 2.0

On-disk calibration artefacts. The harness writes two baseline files and one report file; together with the source configurations above they specify the calibration completely.

scripts/harness.baseline.json. Global “open somewhere” baseline computed from a single bbox-unrestricted WSPR aggregate query (cheap, $\sim 7k$ rows for a 30-day window). Used as the default truth for the Brier / accBin numbers in the headline metrics (Table 10).

scripts/harness.baseline.perpath.json. Per-path baseline using TX/RX bounding-box restricted aggregates (35 queries per harness run, ≥ 1 spot/h floor). Adds a path-specific drift signal that the global mode averages out.

scripts/harness.report.json. Latest harness run output. Regenerated each node `scripts/harness.mjs` invocation; not committed to source control.

src/data/spot-baselines.mjs. Per-(band, UTC-hour) 30-day WSPR spot-count baseline used by the spot-override surface (Section 8.1) and the operator Comp. availability prior of Eq. 74. Regenerated by a separate ClickHouse query on a slower cadence (currently once per few months; see m-tier item m-5h regarding cadence improvement).

What this paper documents and what it does not. Every constant in the SNR budget (Section 3) and MUF estimation (Section 4) is given inline. The σ_g table is in Table 11; the noise-floor table in Table 4; the full path basket in Appendix C; the GIRO station list in Appendix D; the Φ approximation in Section 7.3; the IGRF pole and EIA / winter / equatorial / polar amplitudes inline at their respective equations. What this paper does *not* reproduce is (a) the WSPR ground-truth aggregates themselves — those are queried live from `wspr.live` at harness-run time and their values shift week-to-week as activity rebalances; and (b) the SWPC, GIRO, GFZ, and Kyoto data feeds, which are also live. Reproducing the headline accBin / Brier numbers in Table 10 therefore requires running the harness against contemporaneous data from those feeds, which means the absolute numbers will differ slightly run-to-run; the regression detection signal (drift > 2 dB mean margin or 5 pp in $P(\text{open})$ across runs) is the reproducibility guarantee, not bit-exact metric values.

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